Optimal Control of SOEC's and sensitivity towards degradation

Analysis of the impact of degradation on the optimal control of SOEC's, using finite volume method and sensitivity analysis

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Contents

1 Abstract

The abstract is given below. A reference [15]

Home pc setup ready

2 Introduction

This project aims to do the following:

The thesis studies the sensitivity of the optimal control for a solid oxide electrolysis cell (SOEC) over its lifetime, with the goal of minimizing degradation to prolong lifetime.

Starting by defining the optimal control problem. In loose terms, the objective maximizes revenue while penalizing degradation. A rough formulation is:

$$\max_u \left\{ \alpha(t) \int_0^T \text{revenue}\, \mathrm{d}t - (1-\alpha(t))\text{degredation} \right\} \tag{2.1}$$

The weighting function $\alpha(t)$ balances the importance of production versus degradation, and can be directly linked to real market prices. The details of how the overall optimization problem is formulated are provided later; at this stage, this serves as a broad overview. The core focus is to examine the optimal control $u^*(t)$ (the set of control choices that maximize the objective) and to see what happens as we give the controller more freedom, and lesser fixation. Before diving into the technical setup, the basic idea of giving more freedom will be demonstrated using a simple example.

2.1 A Simple Example of Control Freedom

Consider a simple example: a water gun at a fixed spot that shoots with constant muzzle speed v_0. The only thing we can control is the angle α. The angle sets how far the water goes; the maximum range is at 45°, so restrict α to $[0^\circ, 45^\circ]$ and call the range at 45° simply $L_{\max}$ [20].

The setup is illustrated below:

Thomas (a co-supervisor) stands somewhere on the field at L_T. The goal is to hit him, or get as close as possible. In symbols, something like $\min_\alpha |L(\alpha) - L_T|$.

If L_T is within range, pick the angle that lands on him. If he is too far away, shoot at 45° and accept the miss. The optimal angle as a function of where Thomas stands can be sketched as in Figure 2.2.

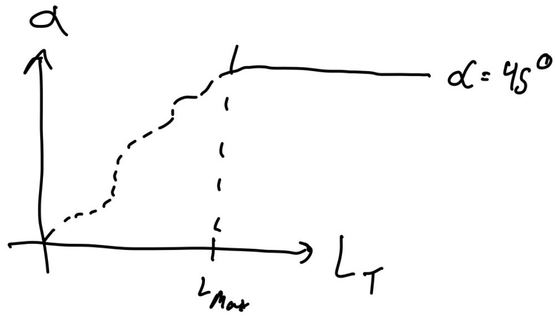


Figure 2.2: Optimal angle when only α can be chosen and $v = v_0$ is fixed.

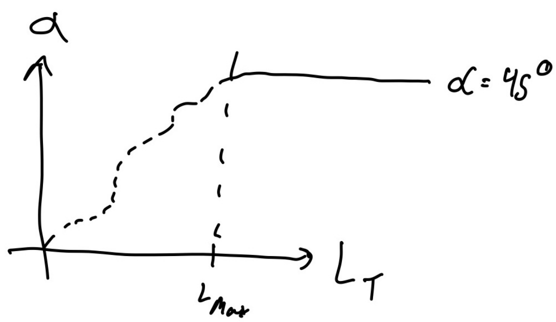


Figure 2.3: Optimal aim when v is fixed (angle varies) versus when v can also be chosen (angle stays at 45°, speed varies).

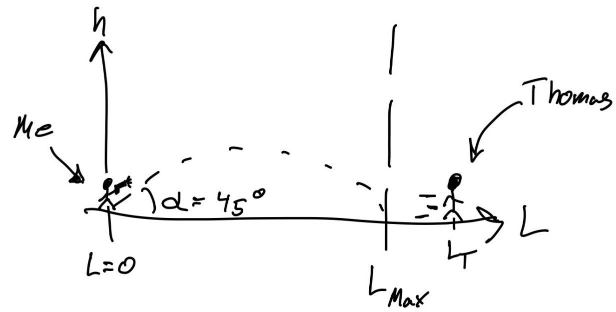


Figure 2.1: Sketch of the optimization problem: the aim is to hit as close to Thomas as possible.

So far so obvious. The point of the example is what happens when we add another knob.

Suppose we can also choose the muzzle speed v, not only the angle. Then the picture flips: instead of carefully tuning α for every L_T, it is natural to leave the gun at 45° (maximum range for a given speed) and adjust v until the water reaches Thomas. Targets that were out of range before can be reached by turning up the pressure; targets that were already in range can still be hit, but the optimizer no longer needs to lean on the angle the way it did when v was fixed.

Figure 2.3 sketches the contrast: one curve for the old case (angle does the work), one for the new case (angle stays near 45°, speed does the work).

The takeaway is loose but important: giving the optimizer more freedom does not just add options — it can change what the first control variable is asked to do.

The same idea carries over to the cell problem in Equation 2.1. If only the cell voltage U is free and the inlet temperature T_{in} is fixed, the optimizer works mainly through U. Once T_{in} is also a decision variable (as in Equation 5.1), the optimal voltage profile can look quite different — just as the water gun stops fiddling with the angle once the pump pressure is on the table. The full setup is spelled out later in the optimal-control chapter.

3 Nomenclature and Assumptions

3.1 List of Assumptions

3.1.1 Geometry, flow, and thermodynamics

1. **Constant channel area.** The cross-sectional area A_{int} is uniform along the channel.
2. **One-dimensional flow.** Properties vary only in the axial direction z; width w and height d enter as geometric scalars.
3. **Ideal gas.** Gas mixtures obey the ideal-gas law.
4. **Negligible concentration overpotential.** Mass-transport limitations at the electrodes are not resolved; molar fractions on each electrode side are taken as uniform (Equation 4.8).
5. **Negligible heat conduction.** The conductive term $\nabla \cdot (\lambda_T \nabla T)$ is dropped in the energy balance (Equation 4.16).
6. **Constant pressure.** $\partial_t p \approx 0$ and $h \approx c_p T$ in the enthalpy formulation.
7. **Prescribed inlet flows.** Fuel and air volumetric flow rates are inputs, not optimized.

3.1.2 Electrochemistry and degradation

8. **Butler–Volmer kinetics.** Anodic and fuel-side (cathode) activation overpotentials follow Equation 4.12 with parameters from [14].
9. **Two dynamic ohmic degradation mechanisms.** Nickel migration and interconnect oxidation evolve via Table 4.2; other ageing paths are not modelled dynamically.
10. **Nickel agglomeration at steady state.** Because agglomeration develops on a short timescale relative to the optimization horizon, the cathode multiplier λ_c is fixed at ≈ 0.55 [5] rather than solved as an additional state.
11. **Spatially resolved degradation.** Slow states are stored per FVM cell ($\mathbf{d}_{\text{Ni-mi},i}, \mathbf{s}_{\text{IC-ox},i} \in \mathbb{R}^{N_{\text{cell}}}$); interconnect oxidation is advanced in squared-thickness form (Equation 5.18).

3.1.3 Quasi-steady coupling and numerical solution

12. **Quasi-steady cell physics.** At each time step the FVM steady-state problem (molar fractions and temperature along z) is solved for the current controls and degradation state before the slow ODEs are updated.
13. **Piecewise-constant controls.** U and T_{in} are constant on each of the N_t control intervals (Section 5.2.6).

14. **Implicit Euler in time.** Degradation states are advanced with the implicit Euler scheme on a uniform grid (Equation 5.16).
15. **Recoverable outlet enthalpy.** Auxiliary heating power P_{heat} credits recovery of outlet sensible heat toward the inlet stream, minus a fixed loss T_{loss} (Equation 5.2).

3.1.4 Economic boundary conditions (scenario-dependent)

16. **Finite rolling horizon.** The objective maximizes revenue and salvation over a chosen horizon T, not the full stochastic stack lifetime (Figure 5.1).
17. **Tank scenarios (when active).** Buffer storage is unlimited; contractual delivery $\dot{n}_{\text{H}_2,\text{agreement}}$ is constant; shortfalls are covered by inventory or spot purchase at $\kappa_{\text{buy}} c_{\text{H}_2}$ (Section 5.1.4).

3.2 List of Symbols

Symbol	Unit	Description
Geometry, discretization, and time		
A_{cell}	m^2	Active electrode area
A_{int}	m^2	Flow-channel cross-sectional area
d	m	Cell thickness (channel height)
L	m	Channel length
w	m	Channel width
z	m	Axial coordinate along the channel
N	—	Number of axial channel segments
N_{cell}	—	Number of FVM control volumes along z
N_t	—	Number of control / time segments in the NLP
Δz	m	FVM mesh spacing
z_k	m	Centre of FVM cell k
Ω_k, V_k	m^3	FVM cell k and its volume
t	s	Time
T	s or days	Optimization horizon
Δt_k	s	Duration of control interval k

Table 3.1: Roman symbols (1/5): geometry and discretization.

Symbol	Unit	Description
Species, flow, and thermodynamics		
$\mathcal{S}$	—	Gas species $\{H_2O, H_2, O_2\}$
$\mathcal{S}_\zeta$	—	Species on side $\zeta \in \{a, f\}$ (anode / fuel)
$\mathcal{M}$	—	Species index set for mass fractions y_k
c_i, μ_i	mol/m^3	Molar concentration; μ_i along channel (state)
x_i, y_k	—	Molar / mass fraction of species i or k
N_i, $N_{c,i}$	$\text{mol/(m}^2\cdot\text{s)}$	Molar flux; electrochemical source flux
$\dot{n}_i$, $\dot{V}_\zeta$	mol/s, m^3/s	Molar flow rate; volumetric flow on side ζ
v_{fuel}, v_{air}	m/s	Gas velocity on fuel / air side
p_i, P	Pa	Partial / total pressure
p_{ref}, p_{air}	Pa	Reference and air-side pressures
ρ	kg/m^3	Gas density
$c_{p,i}$, c_p, C_p	J/(mol·K), $\text{J/(K}\cdot\text{m}^3)$	Species, mixture, and volumetric heat capacity
$h_i(T)$, Δh	J/mol	Species enthalpy; reaction enthalpy change
Q_T	W/m^3	Volumetric heat source

Table 3.2: Roman symbols (2/5): species, flow, and thermodynamics.

Symbol	Unit	Description
Electrochemistry		
F	C/mol	Faraday constant
n_e	—	Electrons per reaction ($n_e = 2$)
ν_i	—	Stoichiometric coefficient of species i
i	$\mathrm{A/m^2}$	Current density
I, I_{cell}	A	Total cell current
U	V	Applied cell voltage (control)
U_{th}	V	Thermal (reversible) cell voltage
η_{cell}	V	Cell overpotential ($\eta_{\text{cell}} = U$)
η_{nernst}, η_{act}, η_{ohm}	V	Nernst, activation, and ohmic overpotentials
$\eta_{\text{act}}^{\zeta}$	V	Activation overpotential on side ζ
i_0^{ζ}	$\mathrm{A/m^2}$	Exchange current density on side ζ
R_{deg}, R_{ohm}	$\Omega{\cdot}\mathrm{m^2}$	Degradation and total ohmic resistance
R_{contact}	$\Omega{\cdot}\mathrm{m^2}$	Contact resistance
B_{ohm}	$\Omega{\cdot}\mathrm{m^2/K}$	Base factor in intrinsic ohmic term
$E_{\text{act}}^{\text{ohm}}$	J/mol	Activation energy in intrinsic ohmic term
ΔG	J/mol	Gibbs free energy change

Table 3.3: Roman symbols (3/5): electrochemistry.

Symbol	Unit	Description
Degradation		
$\mathcal{D}$	—	Set of slow degradation mechanisms
$\mathcal{D}_{\text{ohm}}$	—	Subset in R_{ohm} (Ni-mi, IC-ox)
$\mathbf{d}$, d_i	—	Degradation state vector and component i
$h_{\text{Ni-mi}}$, $h_{\text{IC-ox}}$	m	Ni-migration gap; interconnect oxide thickness
$s_{\text{IC-ox}}$	m^2	Squared IC-oxide thickness (ODE state)
$d_{i,\text{critical}}$	m	End-of-life limit for mechanism i
$R_{\text{Ni-mi}}$, $R_{\text{IC-ox}}$	$\Omega{\cdot}\text{m}^2$	Ohmic contributions from Ni-mi and IC-ox
σ_{elec}, $\sigma_{\text{elec},0}$	S/m	Electrode conductivity and pre-factor
$\sigma_{\text{IC-ox}}$	S/m	Oxide-layer conductivity
$E_{a,\text{elec}}$, $E_{a,\text{ox}}$, E_{el}	J/mol	Activation energies (electrode, IC ox., IC cond.)
$c_1, \dots, c_4$	various	Fitted Ni-mi growth-rate coefficients
k_{mg}, k_{ox}	various	IC-oxidation rate and conductivity prefactors
ε, τ	—	Electrode porosity and tortuosity
Optimal control and economics		
$\mathbf{u}(t), \mathcal{U}$	—	Control vector (U, T_{in}); admissible set
T_{in}, T_{out}, T_{loss}	K	Inlet (control), outlet, and thermal recovery loss
$R(\mathbf{u}, \mathbf{d})$, R_{tank}	EUR/s	Revenue rate; contract rate with tank
P_{cell}, P_{heat}	W	Stack power; auxiliary sensible heating
φ, $\Psi(\mathbf{d})$, Ψ_{tank}	EUR	Integrated revenue; salvation; tank liquidation
c_{H_2}, c_P, $c_{P,k}$	EUR/kg, EUR/kWh	Hydrogen and power prices
$\bar{c}_P$, a	—	Mean diurnal power price and relative swing
c_{cell}	EUR/m^2	Cell capital cost in Ψ
$\alpha(t)$, α_Ψ	—	Schematic weight; salvation exponent
κ_{buy}, κ_{sell}	—	Spot-purchase penalty; liquidation multiplier
m_{tank}, $n_{\text{purchase},k}$	mol, kg	Tank inventory; shortfall purchase
$\dot{n}_{H_2,\text{agreement}}$, $\dot{n}_{\text{prod}}$, $\dot{n}_{\text{purchase}}$	mol/s	Delivery, production, and purchase rates
$\Delta\dot{n}_{H_2}$	mol/s	Net H_2 production in R
PDE / FVM notation		
u, Q_u, F_u	various	Generic FVM field; source and flux terms
m_k	kg	Gas mass in FVM volume k

Table 3.4: Roman symbols (4/5): degradation, optimal control, and numerics.

Symbol	Unit	Description
R	J/(mol·K)	Universal gas constant
α_ζ	—	Butler–Volmer symmetry factor; $\alpha_a = 0.65$, $\alpha_f = 0.59$ [@Leonide2009]
γ_ζ	A/m^2	Exchange-current pre-factor; $\gamma_a = 1.515\times10^8$, $\gamma_f = 7.666\times10^5$
E_{act}^ζ	J/mol	Electrode activation energy; $E_{\text{act}}^a = 1.40\times10^5$, $E_{\text{act}}^f = 1.05\times10^5$
m_i	—	Species exponent in i_0^ζ; $m_{\mathrm{H_2O}} = 0.33$, $m_{\mathrm{H_2}} = -0.1$, $m_{\mathrm{O_2}} = 0.22$
λ_a, λ_c	—	Exchange-current multipliers ($\lambda_a = 1$; $\lambda_c \approx 0.55$)
λ_T	W/(m·K)	Thermal conductivity (neglected)
Ω	m	Spatial domain
x_0	—	Reference molar-fraction scale ($p_{\text{ref}}/p_{\text{air}}$)

Table 3.5: Greek symbols (5/5). Parameter values follow Table 4.1 and Table 4.2 unless stated otherwise.

4 Physical Modelling

4.1 Modelling SOEC

4.1.1 Introduction

In this chapter the aspects of solid oxide electrolysis cell's (SEOC's) will be introduced. Briefly visiting different scales of operation, before a more detailed description of how the mole flow is modelled, rounding it up with the energy balance that determines the behavior of temperature flow.

The basics are presented so that readers with a background in applied mathematics can follow the system. Derivations throughout the chapter therefore assume familiarity with standard intermediate steps; lengthier or less routine derivations are given in the main text or in an appendix. Readers who wish to proceed directly to the governing equations will find them summarized in Table 4.1.

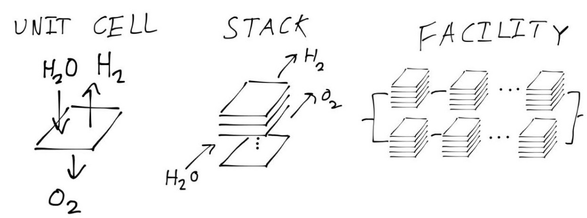


Figure 4.1: A sketch of the scales: the three different scales mentioned in this paper from smallest (left) to largest (right)

The three different scales of SOEC discussed in this paper are defined as follows. The first and smallest scale is the unit cell, where the focus is only on the dynamics occurring within a single SOEC cell. We will spend the majority of the Thesis on this scale.

Next is the stack. At this level, multiple unit cells are stacked together, which naturally introduces more complexity to the system. Figure 4.1 is a quick visualization of the different scales.

Finally, the largest scale considered is at facility scale. At this level, multiple SOEC stacks are connected together; the facility operates much like an electrical circuit, with each stack behaving as its own component with distinct characteristics. However as stated, we will be focusing on the cell scale. And might be relating this to the operation of other scales.

To draw a parallel with the earlier water gun example: in this chapter, we begin constructing the "water gun" of our model, step by step, laying out its fundamental mechanics and underlying principles.

4.1.2 Basics of SOEC

The “water gun” also known as the unit cell, can in simple terms be described by Figure 4.2, here it is visible what happens at the small scales.

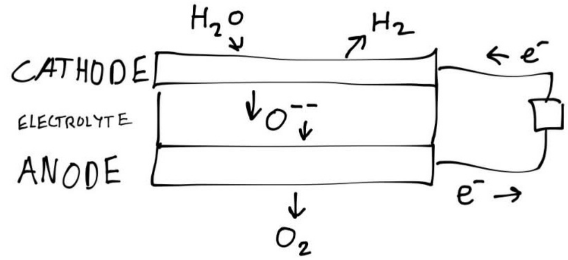


Figure 4.2: A sketch of a unit cell: the basic principles of an SOEC cell are illustrated. The cathode pulls apart the hydrogen and oxygen in the water. The oxygen is transferred through the electrolyte, then formed to dioxygen in the anode, all due to an applied current.

The unit cell is composed of three main components: the **cathode**, the **electrolyte**, and the **anode**. At the cathode, water molecules are split into dihydrogen (H_2) and oxide ions (O^{--}). This process is only possible due to the presence of electrons provided by an external electric current: these electrons participate directly in the reaction at the cathode, facilitating the reduction of water.

The newly-formed oxide ions then migrate through the solid **electrolyte**, propelled by the electric field generated by the applied voltage, highlighting the central role of electrons and electrical energy in driving the overall process. Once the oxide ions reach the **anode**, they release their excess electrons and recombine to form dioxygen (O_2). The electrons that are released here travel through the external circuit, closing the electrical loop.

Summing up, the operation of the SOEC is fundamentally tied to the movement of electrons, both in the chemical reactions at the electrodes and in sustaining the ionic and electronic currents through the device. This is captured in the overall reaction, $2H_2O + 4e^- \rightarrow 2H_2 + 2O^{--} \rightarrow 2H_2 + O_2 + 4e^-$, where the electrons facilitate water splitting at the cathode and are recovered at the anode. For modeling purposes, and since the focus will generally not be on the internal mechanisms, the model simplifies to the absolute global equation (eq. 4.1).

$$H_2O \rightarrow 2H_2 + O_2 \tag{4.1}$$

To clarify some terminology. The upper side of the unit cell—where water is supplied and dihydrogen (H_2) is produced—is called the **fuel side**. This is because water acts as the “fuel” for the process, supplying the reactant that gets split. The lower side—where dioxygen (O_2) is released—is called the **air side**, since atmospheric air is usually introduced here as the inlet. These terms, “fuel side” and “air side,” are commonly used throughout SOEC literature to describe the two compartments of the cell.

4.1.2.1 The chemical flow in a unit cell

The equation in 4.1 treats the reaction as strictly one-way, but in reality, some produced hydrogen and oxygen can recombine to form water again, especially as the system approaches equilibrium based on its electrical state. This reverse reaction is rare at the start but becomes more significant near equilibrium. Yielding equation 4.2.

$$2H_2O \underset{p_r}{\overset{p_l}{\leftrightharpoons}} 2H_2 + O_2 \tag{4.2}$$

Here, p_l stands for the transition probability that two hydrogen and one oxygen molecule will combine and turn back into water (going from right state $\rightarrow$ left state), while p_r is the probability for splitting water into hydrogen and oxygen (left state $\rightarrow$ right state).

From these probabilities, the production rate of H_2O at every point in the cell can be determined (assuming concentrations and probabilities are known everywhere). The difference between how much H_2O is produced versus how much is consumed is taken. Keeping in mind that this is calculated per unit volume of the cell, it has the unit $\frac{\mathrm{mol}}{\mathrm{s \cdot m^3}}$:

$$R_{\mathrm{H_2O}} = p_r c_{\mathrm{H_2O}}^2 - p_l c_{\mathrm{H_2}}^2 c_{\mathrm{O_2}}$$

Where c_x is the concentration $\left(\frac{\mathrm{mol}}{\mathrm{s \cdot m^3}}\right)$ of molycule x

To obtain the production rate for the whole cell (the global rate), this quantity is integrated over the thickness (height) of the cell. Which gives the total rate $N_{c,\mathrm{H_2O}}$ (later this will be treated as a local source, but for now it is kept global):

$$N_{c,\mathrm{H_2O}} = \int_0^d R_{\mathrm{H_2O}}$$

This gives the average production rate per area of the cell, which is a useful parameter to work with. But realistically, calculating this from scratch could get a bit complicated. Thankfully, Faraday already did the heavy lifting here. By Faraday's law [18], the rate per cell area can be written as in equation 4.3:

$$N_{c,\mathrm{H_2O}} = \frac{i}{2F} \tag{4.3}$$

Where i is the current, F is Faraday's constant, and the 2 comes from the number of electrons involved in the chemical equation.

For now, equation 4.3 is kept as it is, since the formulation is still at the global level; hence the current in the equation is the global current. Later, when making all this local, the current must be computed using the Nernst equation. More about that later. First, a form that describes the system locally is introduced.

4.1.2.2 Behavior through a channel over the cell

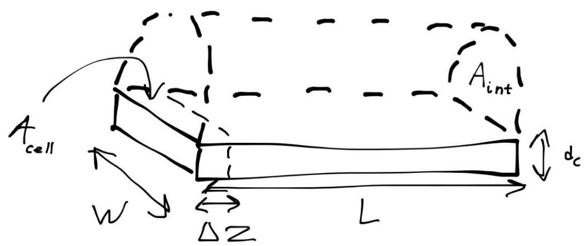


Figure 4.3: Schematic illustration of channel discretization for modeling local behavior along the fuel side of a SOEC. The channel is divided into N segments of length Δz, showing the flow of steam over the cathode.

To describe the system locally, figure 4.3 is used. The fuel side is modeled as a channel (the upper section in the figure), where the steam (H_2O) flows over the cathode. The width of the channel is w and the length is L, with the channel extending from the inlet at $z = 0$ to the outlet at $z = L$ (where L is the length of the cell). The gas flows through a constant cross-sectional area A_{int}.

For any control volume within this channel, mass conservation dictates that the accumulation of water over time equals the flow in plus the production of the cell reaction minus the flow out. Following the formulation by Primdahl and Mogensen [15] for a continuously stirred tank reactor (CSTR), the mass balance for the water molar fraction $x_{\mathrm{H_2O}}$ is:

$$\frac{PV}{RT}\frac{\partial x_{\mathrm{H_2O}}}{\partial t} = A_{int}N_{in,\mathrm{H_2O}} - A_{int}N_{out,\mathrm{H_2O}} + A_{cell}N_{c,\mathrm{H_2O}} \tag{4.4}$$

Where V is the volume of the gas in the reactor, N_i is the molar flux for species i, and A_{cell} is the active electrode area.

To obtain the local behavior along the channel, discretization along the channel length is required, meaning the length is split into sections. This is also illustrated in figure 4.3. Hence $z_i - z_{i-1} = \Delta z$ for all $i = 1, \dots, N$, with $z_0 = 0$ and $z_N = L$, giving a sequence:

$$z_0 < z_1 < \cdots < z_N$$

Each section $z \in (z_{i-1}, z_i)$ can then be viewed as its own cell, where the inlet flow at z_{i-1} is the outlet flow of the previous interval. Letting $\Delta z \to 0$ and inserting the definition of $N_{c,\mathrm{H_2O}}$ (equation 4.3), and the PDE in equation 4.5 pops right out. (Appendix A1)

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{int}}N_{c,\mathrm{H_2O}} - \frac{\partial N_{\mathrm{H_2O}}}{\partial z} \tag{4.5}$$

Where $N_{\mathrm{H_2O}} = v_{fuel}\,\mu_{\mathrm{H_2O}}$ and $N_{c,\mathrm{H_2O}} = \nu_{H_2O}\frac{i}{n_e F}$, whith n_e the number of electrones in the reaction being 2.

By similar computations for H_2 and O_2 the following equations will appear:

$$\frac{\partial \mu_{\mathrm{H_2}}}{\partial t} = \frac{w}{A_{int}}N_{c,\mathrm{H_2}} - \frac{\partial N_{\mathrm{H_2}}}{\partial z} \tag{4.6}$$

$$\frac{\partial \mu_{O_2}}{\partial t} = \frac{w}{A_{int}} N_{c,O_2} - \frac{\partial N_{O_2}}{\partial z} \tag{4.7}$$

Where $N_{H_2} = v_{fuel}\,\mu_{H_2}$ and $N_{O_2} = v_{air}\,\mu_{O_2}$, with ν_i the stochiometric coefficient for species i, μ_i the molar concentration of species i, and $v_{fuel/air}$ the corresponding velocity. And $N_{c,i} = \nu_i \frac{i}{n_e F}$.

From the chemical equation of the reaction:

$$2H_2O \leftrightharpoons 2H_2 + O_2$$

The stoichiometric coefficients are determined as $\nu_{H_2O} = -\nu_{H_2} = -2\nu_{O_2} = 1$.

This equation describes the local evolution of the species concentration. As noted earlier, the current i is no longer a global current. To determine it, the Nernst equation and more is required; this will be explained shortly.

4.1.2.3 Introducing the overpotentials

From the equation stated in 4.5 only the current density i remains to be determined. The control variable is the voltage U, which equals the overpotential of the whole cell (η_{cell}) or cell voltage.

The cell voltage can be written as the sum of all overpotentials over the cell as in equation 4.8. Concentration overpotentials that appear when accounting for diffusion are neglected here, and the concentration is assumed to be the same everywhere on each side of the cell. (Section 3.1)

$$\eta_{cell} = \eta_{nernst} + \eta_{act} + \eta_{ohm} \tag{4.8}$$

$$\eta_{nernst} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log\left(\frac{x_{H_2}\sqrt{x_{O_2}}}{x_{H_2O}\sqrt{x_0}}\right), \tag{4.9}$$

Where $x_0 = p_{ref}/p_{air} \approx 1$, ΔG is Gibbs free energy approximated by the polynomial $\Delta G = -0.0031\,T^2 - 49.119\,T + 244778$. [5]

$$\eta_{ohm} = iR_{\text{deg}} \tag{4.10}$$

Where R_{deg} is the resistince induce from degradation mechanisms.

$$\eta_{act} = \eta_{act}^a + \eta_{act}^c \tag{4.11}$$

Where η_{nernst} is the Nernst overpotential, representing the voltage generated by the electrochemical reaction, and η_{act} is the activation overpotential, which accounts for the energy barrier that must be overcome for the reaction to proceed, η_{ohm} is the Ohmic resistence, which accounts for resistance that is active due to degredative mechanisms. [5]

The Ohmic overpotential is essentially Ohm's law with some resistance; but it is in this resistance we get the degradation, or in terms of the water gun problem, we find where Thomas is standing within the ohmic resistance. The Nernst overpotential describes the willingness of the reaction to proceed. When there is a lot of H_2O and very little H_2 and O_2, the Nernst overpotential is large in absolute value but negative. This drives the reaction away from the high concentration in H_2O and closer to equilibrium. Away from the Gibbs term and the other overpotentials, the *equilibrium* occurs where the fraction is 1. The Nernst overpotential can therefore be viewed as the *force* that pulls the concentrations towards equilibrium.

The activation overpotential is probably the most difficult part to compute. The governing equations are listed first; their meaning is explained afterwards.

The overpotentials for the anode and cathode arise by solving Equation 4.12, also known as the Butler-Volmer equation.

$$i = i_0^\zeta \left(\exp\left[\alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta\right] - \exp\left[-(1-\alpha_\zeta)\frac{n_e F}{RT}\eta_{act}^\zeta\right] \right), \qquad \zeta \in \{a, f\}. \tag{4.12}$$

Where α_ζ is the symmetry factor for the corresponding side, and i_0^ζ is the exchange current for the respective side, with $\alpha_a = 0.65$ and $\alpha_c = 0.59$. [14]

The exchange current is computed as in Equation 4.13.

$$i_0^\zeta = \lambda_\zeta \gamma_\zeta \exp\left[-\frac{E_{act}^\zeta}{RT}\right] \prod_{i \in \text{Species}_\zeta} \left(\frac{x_i}{x_0}\right)^{m_i}, \qquad \zeta \in \{a, f\}. \tag{4.13}$$

Where Species$_a$ $= \{O_2\}$ and Species$_c$ $= \{H_2O, H_2\}$. $\lambda_a = 1$ (in principle it does not appear in the equation) and λ_c is also determined by degradation mechanisms, discussed later. γ_ζ is the pre-exponential factor for the respective side with values $\gamma_a = 1.515 \cdot 10^8$ and $\gamma_f = 7.666 \cdot 10^5$. E_{act}^ζ is the activation energy, with $E_{act}^a = 1.40 \cdot 10^5$ and $E_{act}^f = 1.05 \cdot 10^5$. And $m_{H_2O} = 0.33$, $m_{H_2} = -0.1$ and $m_{O_2} = 0.22$. [14]

This describes the electrical energy required to overcome the energy barrier for the reaction. Somewhat similar to how water does not necessarily freeze at 0 C°. There is some discussion in the field about whether the Butler-Volmer equations are adequate for this phenomenon; but for this project the Butler-Volmer model is taken as a sufficient approximation.

4.1.2.4 Computing the current i

With the overpotentials introduced, it is clear that some of them depend on the current i, which is the missing piece in 4.3 needed to complete the coupled differential equations 4.5, 4.6 and 4.7.

The current is obtained by solving Equation 4.8, with $\eta_{cell} = U$, and i defined as in Equation 4.14

$$i = \underset{i}{\text{Solve}} \left[U - \eta_{nernst} = \eta_{ohm}(i) + \eta_{act}(i)\right] \tag{4.14}$$

Using various simplifications this equation can be solved in different ways; for now it is solved numerically, using a root solver from the python library Scipy [19].

With the equations for mole transport in place, the temperature behavior is treated next, before all equations are summarized in a table.

4.1.3 Energy Balance

To derive the energy balances, the derivation starts at the thermal energy equation given by Equation 4.15 [12]. A number of simplifications are then applied to obtain Equation 4.16.

$$\rho\frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla\cdot(\lambda_T \nabla T) - \sum_{i\in\mathcal{S}} \nabla\cdot(h_i N_i) + Q_T \tag{4.15}$$

where ρ is density, h specific enthalpy, λ_T thermal conductivity, h_k and N_k the specific enthalpy and flux of species k, and Q_T the volumetric heat source (e.g. electrochemical heating).

Starting with Equation 4.15, essentially stating how much energy a volume accumulates, in enthalpy per unit volume (left-hand side). Is equal to the compression (The change in pressure) plus the energy added by heat conduction (Second order differential term in temperature) and the transportation of enthalpy by species (H_2O, H_2, O_2), and lastly adding the reaction heat Φ.

The first simplifications is to neglect the energy accumulation due to heat conduction ($\nabla\cdot(\lambda\nabla T) = 0$). Also assuming constant pressure, $h \approx c_p T$ and $\partial_t p = 0$, which yields:

$$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i\in\mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + Q_T \tag{4.16}$$

Where $c_p = \sum_{e\in\mathcal{M}} c_{p,k} y_k$, with y_k being the mass fractions (Not the molar fraction), for further information look at Appendix A2

It remains to define the source term, which follows from the power expression in Equation 4.17. Power is the energy per unit time, hence the energy added to the system.

$$Q_T = \frac{i\,w}{A_{int}} U \tag{4.17}$$

4.1.3.1 At steady state

At steady state the expression for the molar progression can be used to simplify the equations further, yielding Equation 4.18.

$$0 = i\frac{w}{A_{int}}(U - U_{th}) - C_p\frac{\partial T}{\partial z} \tag{4.18}$$

Where $U_{th} = \frac{\Delta h}{n_e F}$ with $\Delta h = h_{\mathrm{H_2O}} - h_{\mathrm{H_2}} - \frac{h_{\mathrm{O_2}}}{2}$, (Appendix A2).

With the energy balance, or temperature differential equation, in place, the cell model is complete. The following section takes a deep dive into “Thomas’s position” and “how he moves” also known as the degradation mechanisms.

Description	**Equation**	**Ref.**
Faraday's law	$N_{c,i} = \nu_i \frac{i}{2F}$	Equation 4.3
Molar concentration DEs	$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{int}} N_{c,i} - \frac{\partial N_i}{\partial z}$	Eqs. 4.5, 4.6, 4.7
Cell voltage	$\eta_{cell} = \eta_{nernst} + \eta_{ohm} + \eta_{act}^a + \eta_{act}^c$	Equation 4.8
Ohmic overpotential	$\eta_{ohm} = iR_{deg}$	Equation 4.10
Nernst overpotential	$\eta_{nernst} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log\left(\frac{x_{\mathrm{H_2}}\sqrt{x_{\mathrm{O_2}}}}{x_{\mathrm{H_2O}}\sqrt{x_0}}\right)$	Equation 4.9
Butler–Volmer	$i = i_0^{a/f}\big(\exp[\alpha_{a/f} \frac{n_e F}{RT} \eta_{act}^{a/f}] - \exp[-(1-\alpha_{a/f}) \frac{n_e F}{RT} \eta_{act}^{a/f}]\big)$	Equation 4.12
Exchange current	$i_0^{a/f} = \lambda_{a/f} \gamma_{a/f} T \exp\big[-\frac{E_{act}^{a/f}}{RT}\big] \prod_{i \in \mathcal{S}_{a/f}} (x_i/x_0)^{m_i}$	Equation 4.13
Gibbs free energy	$\Delta G \approx -0.0031T^2 - 49.119T + 244778$	
Energy balance	$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i \in \mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + \frac{i\,w}{A_{int}} U$	Eqs. 4.16, 4.17

Table 4.1: Overview of model equations (Here without degradation which is in Section 4.2.1).

4.2 Modelling degredation

4.2.1 Introduction

This section examines the two components R_{ohm} that appears in Equation 4.10 and λ_c which appears in Equation 4.13. The latter is more complicated, so the Ohmic resistance is treated first. Below a equation is set up, which describes how the degradation plays a role on the ohmic resistance.

$$R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp\left\{\frac{E_{act}^{\text{ohm}}}{RT}\right\} + R_{contact} + \sum_{d \in \mathcal{D}_{\text{ohm}}} R_d$$

Here, the first two terms are time-invariant and depend only on the present conditions of the cell. In contrast, the degradation terms, represented by the summation, evolve and increase as the cell ages. The first degradation mechanism considered is the resistance arising from nickel migration.

4.2.2 Nickel migration

Nickel migration is a simple phenomenon to understand. Essentially the electrode is saturated with nickel, but as time goes by the saturated part of the cell will shrink, hence we will have a gab in the electrode where there are no nickel. Since the nickel plays a huge part in transporting the oxygen ions fast and smoothly, the absence of nickel will induce an added travel resistance for the ions. Now the resistance can be represented by Equation 4.19.

$$R_{\text{Ni-mi}} = \frac{h_{\text{Ni-mi}}}{(\frac{\varepsilon}{\tau}\sigma_{\text{elec}})} \tag{4.19}$$

Where $d_{\text{Ni-mi}}$ is the thickness of nickel absence layer, which develops over time, σ_{elec} is the electrode conductivity, ϵ is the zirconia fraction and τ is the tortuosity of the zirconia fraction [5]

The sketch in Figure 4.4 illustrates how the nickel migration develops over time, and the impact it has.

The conductivity of the electrode is given by Equation 4.20. This conductivity is in reality the reduction in conductivity due to the creep of the nickel part of the electrode.

$$\sigma_{\text{elec}} = \frac{\sigma_{\text{elec,0}}}{T} \exp\left(-\frac{E_{a,\text{elec}}}{RT}\right) \tag{4.20}$$

Javid Beyrami et al. [5] describe the development of nickel migration by the following differential equation, where c_1 and c_2 are negative and c_3 and c_4 are positive. For high values of current and temperature, migration increases; if either T or i is too small or too large, migration also increases.

$$\frac{\partial h_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 i T + c_0$$

Where c_k is fitted to experimental data.

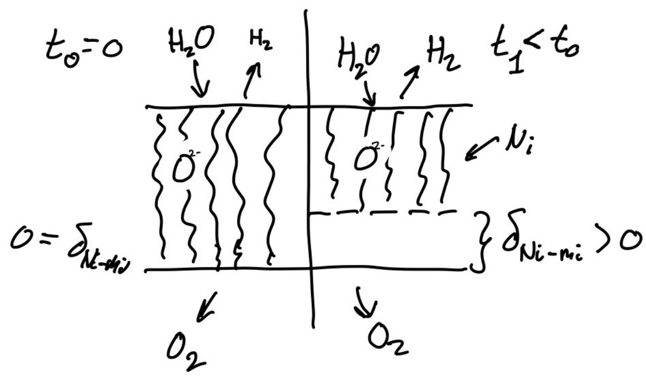


Figure 4.4: For a cell without degradation (Left), nickel migration has not occurred and $d_{\text{Ni-mi}} = 0$ at the layer between the nickel-filled electrode and the electrolyte. Over time the nickel-filled part of the electrode creeps away from the electrolyte (Right), giving rise to resistance induced by the raw electrode on the traveling species.

4.2.3 Interconnect Oxidation

Another important degradation mechanism to consider is interconnect oxidation. This process also contributes to the overall ohmic resistance, and is relatively straightforward to understand: it occurs when the metallic interconnect undergoes oxidation, in other words, when the metal effectively "rusts."

The resistance resulting from interconnect oxidation is in structure much like that of nickel migration. Specifically, it is determined by the thickness of the oxidized layer divided by its (reduced) electrical conductivity, as shown in Equation 4.21 [5], [13], [6].

$$R_{\text{IC-ox}} = \frac{h_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}} \tag{4.21}$$

Where the oxidized interconnect conductivity is given as a function of temperature in Equation 4.22.

$$\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp\left\{-\frac{E_{\text{el}}}{RT}\right\} \tag{4.22}$$

Beyrami et al. [5] describe the growth of the oxide layer by the following rate law.

$$\frac{dh^2_{\text{IC-ox}}}{dt} = k_{\text{mg}} \exp\left\{-\frac{E_{a,\text{ox}}}{RT}\right\} \tag{4.23}$$

The squared-thickness form in Equation 4.23 reflects parabolic oxidation kinetics: layer growth follows a square-root-of-time law, and the rate depends on local temperature alone through the Arrhenius term. Unlike nickel migration, interconnect oxidation is therefore not sensitive to current density. As $h_{\text{IC-ox}}$ increases, the contribution $R_{\text{IC-ox}}$ in Equation 4.21 rises and adds to the degradation summation in the total ohmic resistance. The constants k_{mg} and $E_{a,\text{ox}}$ are fitted to experimental data [5], [6]. Together with nickel migration, this completes the ohmic degradation terms; the remaining mechanism acts on the cathode exchange current and is discussed next.

4.2.4 Nickel Agglomeration

A further degradation mechanism affecting the cathode exchange current is nickel agglomeration. Unlike the ohmic contributions discussed above, it is typically modelled not as a resistance but as a multiplier on λ_c, the normalized triple phase boundary [5]. Whether the underlying process is nickel migration or another phenomenon remains debated, and the mechanistic details are therefore not pursued here. Because agglomeration develops on a relatively short timescale, λ_c is assumed to have reached its asymptotic value of approximately 0.55 [5].

Description	**Equation**	**Ref.**
Total ohmic resistance	$R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp\left\{\frac{E_{\text{act}}^{\text{ohm}}}{RT}\right\} + R_{\text{contact}} + \sum_{k \in \mathcal{D}_{\text{ohm}}} R_k$	
Nickel-migration resistance	$R_{\text{Ni-mi}} = \frac{d_{\text{Ni-mi}}}{(\frac{\varepsilon}{\tau}\sigma_{\text{elec}})}$	Equation 4.19
Electrode conductivity	$\sigma_{\text{elec}} = \frac{\sigma_{\text{elec,o}}}{T} \exp\left(-\frac{E_{a,\text{elec}}}{RT}\right)$	Equation 4.20
Nickel thickness rate	$\frac{\partial d_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 i T + c_4$	
Interconnect oxidation resistance	$R_{\text{IC-ox}} = \frac{d_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}}$	Equation 4.21
IC oxide conductivity	$\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp\left\{-\frac{E_{\text{el}}}{RT}\right\}$	Equation 4.22
IC oxidation rate	$\frac{dd^2_{\text{IC-ox}}}{dt} = k_{\text{mg}} \exp\left\{-\frac{E_{a,\text{ox}}}{RT}\right\}$	Equation 4.23
Normalized triple phase boundary	$\lambda_c = 0.55$	[@BEYRAMI2024100653]

Table 4.2: Overview of degradation mechanisms (ohmic contributions and slow-state dynamics). The base cell equations are summarized in Table 4.1.

5 Optimal Control and Solution Method

5.1 Optimal Control Problem

5.1.1 Introduction

With the cell model in place, and a nice degradation setup. Operation can be simulated for any admissible inlet conditions and cell voltage (within a reasonable range). What remains is to specify an economic objective and formulate the corresponding optimal control problem. Once again, related to the water gun problem, in this chapter, the objective of “hitting Thomas” is formulated.

Following discussions with Dynelectro and DTU Energy, and a review of how the objective is set in most literature [10],[4],[7], the general structure adopted is to maximize running revenue (hydrogen production while minimizing production cost) and to minimize degradation.

$$
\begin{aligned}
&\max_{\mathbf{u}\in\mathcal{U}} \int_0^T R\,\mathrm{d}t + \Psi \\
&\text{s.t.} \quad \frac{\partial n_i}{\partial t} \text{ satisfies Table 4.1,} \qquad \text{for } i \in \mathcal{S} \\
&\qquad\quad \frac{\partial d_i}{\partial t} \text{ satisfies Section 4.2.1,} \quad \text{for } i \in \mathcal{D}
\end{aligned}
\tag{5.1}
$$

Here $\mathbf{u}(t)$ collects the control inputs, that is the applied voltage $U(t)$ and the inlet temperature $T_{\text{in}}(t)$ that can be chosen as functions of time, and $\mathbf{d}$ collects the degradation state variables;

$$
\mathbf{d} = \Big[d_i(t)\Big]_{i\in\mathcal{D}}\,.
$$

Looking at Equation 5.1, the integrand $R(\mathbf{u}, \mathbf{d})$ is the **revenue rate**; maximizing its time integral corresponds to maximizing operating income over $[0, T]$. The term $\Psi(\mathbf{d})$ is the **salvation** contribution: a scalar that rewards the optimizer for how much useful life remains in the cell at the end of the horizon.

It is worth noting that minimizing degradation is in its core, a part of minimizing production cost: when a cell degrades the resistance rises, leading to higher power for the same hydrogen output, and in the worst case degradation leads to a malfunctioning cell, requiring replacement of the cell or, at facility scale, a stack. It should also be noted that using terminal degradation is a modeling choice; using the running degradation rate inside the integral would have been equally valid and may be simpler to implement.

5.1.2 Revenue

The revenue rate is taken in the explicit form shown in Equation 5.2.

$$
\begin{aligned}
R &= c_{\mathrm{H}_2}\Delta\dot{n}_{\mathrm{H}_2} - c_P P \\
\Delta\frac{\partial n_{\mathrm{H}_2}}{\partial t} &= A_{int}N_{out,\mathrm{H}_2} - \frac{\partial n_{in,\mathrm{H}_2}}{\partial t} \\
P &= P_{cell} + P_{heat}
\end{aligned}
\tag{5.2}
$$

Here $\frac{\partial n_{in,\mathrm{H}_2}}{\partial t}$ is specified as an input (part of **u***), P is the total electrical power, P_{cell} the power supplied to the stack, and P_{heat} the auxiliary sensible heating power.*

Thus, R is the value of net hydrogen production (outlet minus inlet, in appropriate units) minus the cost of power. All quantities in Equation 5.2 follow from the cell model summarized in Table 4.1.

From the model date the power of the cell is computed by $P_{cell} = I_{cell}U$ with:

$$
I_{cell} \approx \frac{A_{cell}}{N_{cell}} \sum_{k=1}^{N_{cell}} i_k.
$$

The heating power is the **sensible** auxiliary preheat duty: for each stream, the enthalpy difference between the **inlet** composition at the target inlet temperature T_{in} and the same composition at the recovered outlet temperature $T_{out} - T_{loss}$. Composition enthalpy of the electrochemical conversion is excluded from P_{heat} and is accounted for via P_{cell}. Positive P_{heat} means external preheat is required ($T_{in} > T_{out} - T_{loss}$); negative values are recovery credits.

$$
P_{\text{heat}} = \sum_{\zeta\in\{a,f\}} \dot{V}_\zeta\, A_{int} \left(\sum_{i\in\mathcal{S}_\zeta} \mu_{i,\text{in}}\, h_i(T_{in}) - \sum_{i\in\mathcal{S}_\zeta} \mu_{i,\text{in}}\, h_i(T_{out} - T_{loss}) \right)
\tag{5.3}
$$

Where outlet thermal energy is assumed recoverable toward the inlet stream with an absolute loss T_{loss}, and $\mu_{i,\text{in}}$ denotes inlet molar concentrations.

With that the running revenue or integrant is settled, hence we can look at the salvation.

5.1.3 Salvation

After operating up to time T, the running profit captures cash flow from hydrogen sales net of power cost. A salvation term is included, with the objective in mind, that the cell shouldn't be overused leading to, a cell which cannot gain its own cost in revenue. The idea is to measure, in a simple aggregate way, how close the cell is to end-of-life limits across degradation mechanisms that can cause failure.

Equal weight is assigned to each index in $\mathcal{D}$, leading to

$$
\Psi = \frac{c_{cell}}{|\mathcal{D}|} \sum_{i\in\mathcal{D}} \left(1 - \frac{d_i}{d_{i,\text{critical}}}\right)^{\alpha_\Psi}
\tag{5.4}
$$

Here c_{cell} scales the reward to the capital cost of the cell, and $\mathcal{D}$ is the set of degradation. Initially $\alpha_\Psi = 1$

For each i, the factor $\max\{0, 1 - d_i/d_{i,\text{critical}}\}$ is the unused fraction of the margin to the critical level at stop time T. Averaging over $\mathcal{D}_{\text{fail}}$ and scaling by c_{cell} yields a terminal value that increases when the cell stops farther from those limits.

5.1.3.1 Extra note on Salvation

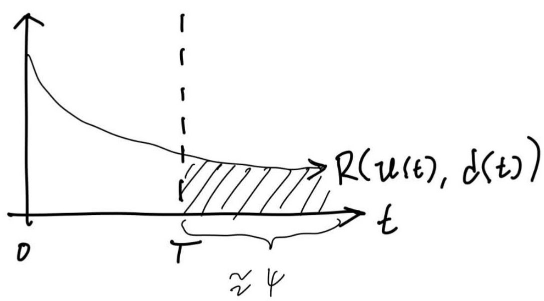


Figure 5.1: CAPTION

The ideal aim would be to extract the most value from a cell over its lifetime, which would correspond to optimizing the integral in Equation 5.1 with $\Psi = 0$ and T a conditioned variable equal to the lifetime. That horizon is not practical, since degradation and prices cannot be predicted that far ahead. Instead, the objective optimizes value over a short future horizon while retaining potential revenue. This is illustrated in Figure 5.1.

The objective is essentially the area under the revenue-rate curve. The formulation used here approximates maximization of that area over the full lifetime, but only up to the finite horizon T.

5.1.4 $\mathrm{H_2}$ storage and delivery agreement

Beyond the simple revenue model in Equation 5.2, a contract-style extension adds buffer storage and a fixed hydrogen delivery obligation. The cell and degradation equations are unchanged; only the economic layer is extended. The contractual offtake is represented by a constant delivery rate $\dot{n}_{\mathrm{H_2},\text{agreement}}$. In practice, delivery may occur in batches; here it is approximated as a continuous flow over $[0, T]$. Shortfalls relative to production are met from inventory or, if the tank is empty, from spot purchases at rate $\dot{n}_{\text{purchase}}$, penalized in the revenue at $\kappa_{\text{buy}}\, c_{\mathrm{H_2}}$. The resulting revenue rate is given in Equation 5.5.

$$R_{\text{tank}} = c_{\mathrm{H_2}}\, \dot{n}_{\mathrm{H_2},\text{agreement}} - c_P P - \kappa_{\text{buy}}\, c_{\mathrm{H_2}}\, \dot{n}_{\text{purchase}}. \tag{5.5}$$

Inventory is modelled as a lumped storage tank. The net production surplus $\Delta\dot{n}$ is the difference between stack output and the agreement rate. Inventory increases only when production exceeds delivery; spot purchases activate only when production falls short. This gives Equation 5.6 and Equation 5.7.

$$\Delta\dot{n} = \dot{n}_{\text{prod}} - \dot{n}_{\mathrm{H_2},\text{agreement}}$$
$$\dot{m}_{\text{tank}} = \max(0,\ \Delta\dot{n}) \tag{5.6}$$
$$\dot{n}_{\text{purchase}} = \max(0,\ -\Delta\dot{n})\,. \tag{5.7}$$

At the end of the horizon the tank may still hold inventory. Two terminal treatments are used in the results.

In the first, remaining hydrogen is valued as liquidation on the open market, giving the additional term in Equation 5.8.

$$\Psi_{\text{tank}} = \kappa_{\text{sell}}\, c_{\mathrm{H_2}}\, m_{\text{tank}}(T), \tag{5.8}$$

where $\kappa_{\text{sell}} \approx \frac{1}{8}$ when end-of-horizon stock must compete with *black* hydrogen. In the second, a cyclic inventory constraint replaces terminal liquidation:

$$m_{\text{tank}}(0) = m_{\text{tank}}(T), \tag{5.9}$$

which is natural when the operating window is meant to repeat periodically.

Under Equation 5.9, the objective is invariant to a uniform upward shift of the inventory trajectory: if $m_{\text{tank}}(0) = m_{\text{tank}}(T) = m_0$ is feasible, so is $m_{\text{tank}}(0) = m_{\text{tank}}(T) = m_0 + a$ for any $a > 0$, provided the shifted trajectory remains admissible. To select the trajectory with the smallest inventory, a penalty term is subtracted from the objective:

$$-\min_t m_{\text{tank}}(t).$$

When the inventory stays strictly positive over $[0, T]$, this term equals the minimum inventory level and removes the shift degeneracy.

5.1.4.1 Full objective (tank mode)

When tank mode is active, the optimizer solves

$$\begin{aligned} \max_u \quad & \int_0^T R_{\text{tank}}\, \mathrm{d}t + \Psi(\mathbf{d}(T)) + \Psi_{\text{tank}}(m_{\text{tank}}(T)) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = 0 \end{aligned} \tag{5.10}$$

or

$$\begin{aligned} \max_u \quad & \int_0^T R_{\text{tank}}\, \mathrm{d}t + \Psi(\mathbf{d}(T)) - \min_t m_{\text{tank}}(t) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = m_{\text{tank}}(T) \end{aligned} \tag{5.11}$$

instead of Equation 5.1 with Equation 5.2. Fixed delivery revenue decouples cash flow from instantaneous production: the optimizer minimizes power cost and spot purchases while building inventory when production is favourable, subject to the usual salvation trade-off on degradation.

An overview of all relavant equations for the optimal control problem is stated in Table 5.1.

Description	Equation	Ref.
General objective function	$\int_0^T R, \mathrm{d}t + \Psi$	Equation 5.1
Revenue	$R = c_{\mathrm{H}_2} \Delta \dot{n}_{\mathrm{H}_2} - c_P P$	Equation 5.2
Salvation	$\Psi = \frac{c_{cell}}{\lvert\mathcal{D}\rvert} \sum_{i \in \mathcal{D}} \left(1 - \frac{d_i}{d_{i,\text{critical}}}\right)^{\alpha_\Psi}$	Equation 5.4
Tank balance	$\dot{m}_{\text{tank}} = \max\bigl(0,\, \dot{n}_{\text{prod}} - \dot{n}_{\mathrm{H}_2,\text{agreement}}\bigr)$	Equation 5.6
Spot purchase	$\dot{n}_{\text{purchase}} = \max\bigl(0,\, \dot{n}_{\mathrm{H}_2,\text{agreement}} - \dot{n}_{\text{prod}}\bigr)$	Equation 5.7
Contract revenue	$R_{\text{tank}} = c_{\mathrm{H}_2} \dot{n}_{\mathrm{H}_2,\text{agreement}} - c_P P - \kappa_{\text{buy}}\, c_{\mathrm{H}_2}\, \dot{n}_{\text{purchase}}$	Equation 5.5
Terminal liquidation (alternative A)	$\Psi_{\text{tank}} = \kappa_{\text{sell}}\, c_{\mathrm{H}_2}\, m_{\text{tank}}(T)$	Equation 5.8
Cyclic inventory constraint (alternative B)	$m_{\text{tank}}(0) = m_{\text{tank}}(T)$	Equation 5.9
Minimum-inventory penalty (with alternative B)	$-\min_t m_{\text{tank}}(t)$	Equation 5.11
Cell Power	$P_{cell} = I_{cell} U$	
Heating Power	$P_{\text{heat}} = \sum_{\zeta \in \{a,f\}} \dot{n}_\zeta \left(\sum_{i \in \mathcal{S}_\zeta} x_i h_i \left(T_{in}\right) - \sum_{i \in \mathcal{S}_\zeta} x_i h_i \left(T_{out} - T_{loss}\right)\right)$	

Table 5.1: Overview of the optimal control formulation (objective, revenue, terminal value, and power bookkeeping). The underlying cell equations are summarized in Table 4.1; degradation is treated in Section 4.2.1. Optional tank economics (Section 5.1.4) add contract revenue and inventory dynamics; end-of-horizon inventory is closed either by terminal liquidation (alternative A) or by a cyclic constraint with a minimum-inventory penalty (alternative B).

5.2 Building the numerical solver

5.2.1 Introduction

This chapter presents the construction of numerical tools required to determine the optimal control $\mathbf{u}^*(t)$ for the system. Unlike simple textbook examples, where a Wikipedia lookup was sufficient, the problem considered here is more complex and necessitates a structured, step-by-step methodology. The central question addressed is: What tools and methods are needed to solve the optimal control problem outlined in Section 5.1.1?

The overall strategy is as follows:

1. **Steady-state cell equations:**
   As the focus is on long-term behavior, it is assumed that the dynamics described by the differential equations in Table 4.1 occur much faster than the overall optimization timescale, treating them as instant. This allows for replacement with a set of steady-state equations, which are simpler to solve. The Finite Volume Method (FVM) is used to solve these equations numerically.

2. **Degradation dynamics:**
   With the steady-state cell solver established, its outputs serve as algebraic inputs to the degradation model (Table 4.2). This model captures the system's degradation over time through differential equations, which are solved numerically using the implicit Euler method.

3. **Objective and optimal control:**
   The optimal control objective (Table 5.1) is addressed by discretizing the control input with the single shooting method. In this approach, both the revenue and degradation equations are integrated forward in time as a system of differential equations.

Structuring the problem in these three parts (steady-state solution, degradation modeling, and control optimization) enables a systematic approach to the full optimal control challenge.

5.2.2 Finite volume method (FVM)

With the descriptive equations in place (Table 4.1, Section 4.2.1, and Section 5.1.1), a numerical solution method is required. All equations can be written in the general form

$$\frac{\partial u}{\partial t} = Q_u - \frac{\partial F_u}{\partial z} \tag{5.12}$$

where Q_u denotes the source term and F_u the flow term; both can be computed directly from the state. For now only the steady-state solution is of interest, which simplifies the problem considerably and yields the following *boundary value problem*:

$$0 = Q_u + \frac{\partial F_u}{\partial z}$$

The *finite volume method* is used to solve this steady-state BVP [9].

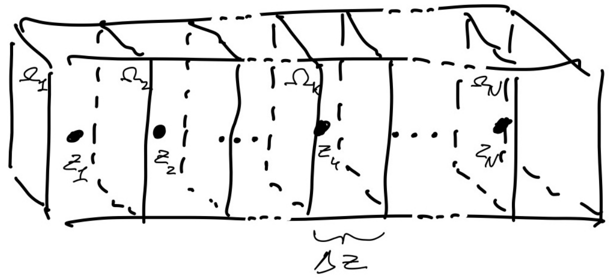


Figure 5.2: A sketch of the finite volume method. A large volume, corresponding to the air and gas channel, is split into smaller volumes. The greater volume is divided into N volumes, each represented by the midpoint, yielding a 1D system instead of 3D. Integration over the densities in each small volume gives the mass in that volume. The figure also introduces z_k (midpoint), Ω_k (small volume), and Δz (length of the volume).

As shown in Figure 5.2, the domain is split into smaller subdomains coupled by flux from one volume into the next. Integration of the density u over a small volume Ω_k of volume V_k proceeds as follows.

$$\int_{\Omega_k} \frac{\partial u}{\partial t} = \int_{\Omega_k} Q_u \, dV - \int_{\Omega_k} \frac{\partial F_u}{\partial z} \, dV$$

The source term is approximated by a rectangle (midpoint rule): $\int_{\Omega_k} Q_u \approx Q_u(z_k) A_{int} \Delta z$. For the flux term, the fundamental theorem of calculus is used to evaluate the antiderivative in the z-direction, with the assumption that the concentration is constant perpendicular to z, giving:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \, dV = A_{int} F_u \Big|_{\partial \Omega_k} = A_{int} \Big(F_u(z_{k+\frac{1}{2}}) - F_u(z_{k-\frac{1}{2}}) \Big)$$

The boundary values $F_u(z_{k-\frac{1}{2}})$ are approximated by the value in the previous cell, consistent with the flow direction of the system (in cell k there is no information from cell $k+1$, but there is from cell $k-1$). The integral is therefore approximated as:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \approx \Big(F_u(z_k) - F_u(z_{k-1}) \Big) A_{int}$$

Combining the source and flux approximations gives the discrete equation Equation 5.13.

$$\frac{\partial}{\partial t} \int_{\Omega_k} u \, dV = Q_u(z_k) A_{int} \Delta z - \Big(F_u(z_k) - F_u(z_{k-1}) \Big) A_{int} \tag{5.13}$$

or at steady state:

$$0 = Q_u(z_k) A_{int} \Delta z - \Big(F_u(z_k) - F_u(z_{k-1}) \Big) A_{int}$$

5.2.2.1 Applying boundary conditions

The problem has two boundary conditions. The first is a Dirichlet condition at the inlet, imposed by defining the first flux as:

$$F_u(z_{-1}) = F_u(z_{inlet})$$

For the Neumann condition at the outlet, no change is made to the equations, since the upstream flux in the FVM makes it possible to neglect a Neumann boundary term at the end.

5.2.2.2 FVM on specific equations

Applying the FVM to the governing equations yields the following discretization; see Appendix A3 for derivations.

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) \approx w N_{c,i} \Delta z - A_{int}(N_i(z_k) - N_i(z_{k-1})), \tag{5.14}$$

And

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T A_{int} \Delta z - A_{int} \sum_{i \in \mathcal{S}} v_i c_{p,i} \Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1}) \Big) \tag{5.15}$$

5.2.3 FVM model validation and Grid independence

5.2.3.1 IV-curve validation

After implementing all the equations given in Table 4.1, the model was validated against an IV-curve from experimental data [17]. The error is below 10^{-2} except at quite high voltage/current, where the error is larger. Figure 5.3 compares the model with the data.

This level of agreement is sufficient for the intended operating range, as very high voltages are not relevant for typical operation. The main variable of interest is the change in moles across the cell, which exhibits similar trends; at high voltage or current, the error in molar change decreases further, since the molar fractions approach their extrema (0 or 1), making relative errors negligible.

5.2.3.2 Grid independence

Grid refinement was used to assess how much the states change when the mesh is refined, and thereby to estimate the resolution required for reliable simulations later. The steady-state solver was run while doubling the number of grid points each time. Figure 5.4 indicates that if a relative error target of approximately 10^{-3} was the target. The amount of volumes needed would be around 200. But for this thesis, the amount of volumes will be 50, meaning that a relative error of around 10^{-2} will have to suffice.

But nonetheless the Steady-state equations can be solved to a rather fine precision. Leading to the next step: Solving the ODE's using the outputs of the output of the FVM.

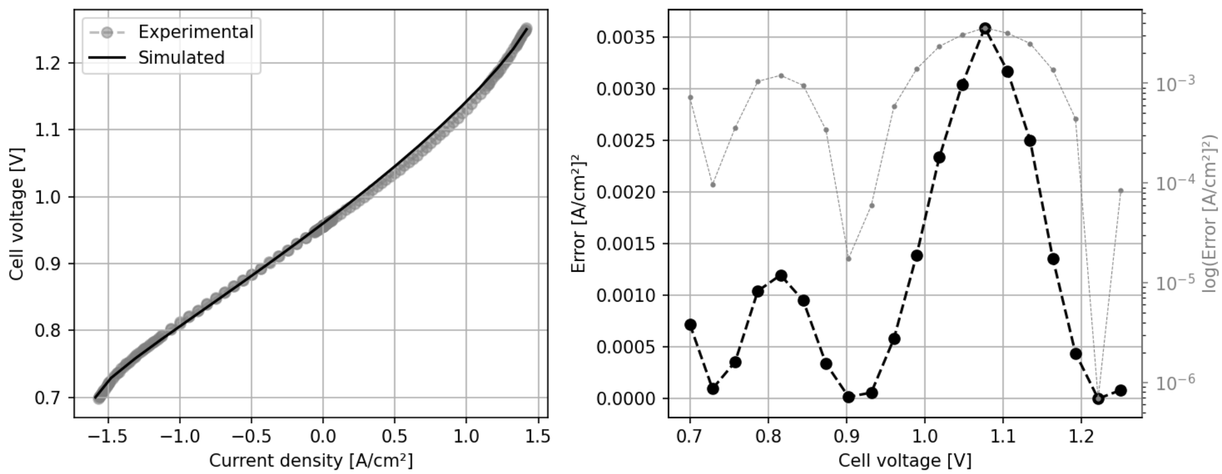


Figure 5.3: Model validation by comparing to an IV-curve from experimental data. The simulated current–voltage curve is compared to the experimental curve on the left. On the right, the error in current is shown as a function of voltage, on both logarithmic and linear scales. The experimental data is from [14], Fig. 2.

5.2.4 Implicit Euler method (ODE solver)

As stated in the introduction to this chapter, the FVM gives the algebraic outputs needed to solve the differential equations that define degradation (Table 4.2). Those ODEs are advanced with the implicit Euler method on a uniform time grid [11].

5.2.4.1 General form

Take the ODE: $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$. Discretize time on $t \in [0, T]$ into N_t steps with $\mathbf{x}_i = \mathbf{x}(t_i)$, $t_i = t_{i-1} + \Delta t_i$, $t_0 = 0$, and $t_{N_t} = T$. Given an initial state $\mathbf{x}_0$, the implicit Euler update is

$$\mathbf{x}_i = \mathbf{x}_{i-1} + \Delta t_i\, \mathbf{f}(t_i, \mathbf{x}_i), \qquad i = 1, \dots, N_t. \tag{5.16}$$

Because $\mathbf{x}_i$ appears on both sides, Equation 5.16 is a fixed-point equation at each step. Damped fixed-point iteration is therefore a natural solver. The update is also Markovian: step i depends only on $\mathbf{x}_{i-1}$ and the control on $[t_{i-1}, t_i)$, so the integration can proceed one time step at a time.

5.2.4.2 Specific ODE setup

For this thesis the vector field $\mathbf{f}$ is expensive: evaluating $\dot{\mathbf{d}}$ requires converged FVM fields (temperature and current density along the stack), which themselves depend on the degradation state. The degradation state advanced at each control interval is

$$\mathbf{d}_i = \begin{bmatrix} \mathbf{d}_{\text{Ni-mi,i}} \\ \mathbf{s}_{\text{IC-ox,i}} \end{bmatrix}, \tag{5.17}$$

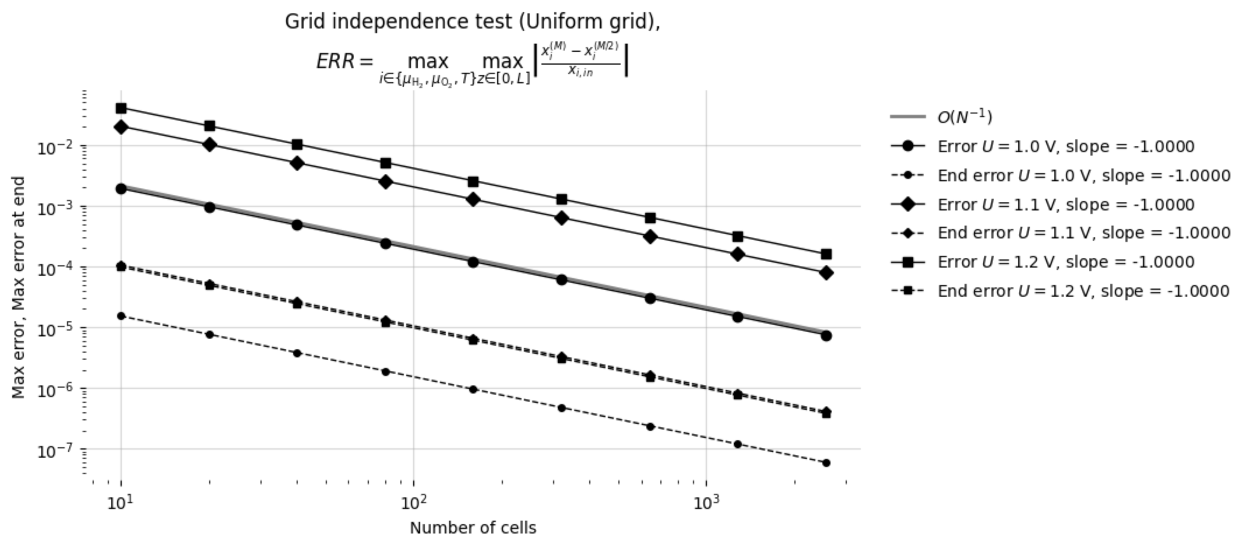


Figure 5.4: Grid Independence test

where $\mathbf{d}_{\mathrm{Ni-mi,i}}, \mathbf{s}_{\mathrm{IC-ox,i}} \in \mathbb{R}^{N_{\mathrm{cell}}}$ *collect the per-cell slow states at time* t_i*, with* $\mathbf{s}_{\mathrm{IC-ox,i}} = \mathbf{d}^2_{\mathrm{IC-ox,i}}$*. Nickel migration stores layer thickness* $d_{\mathrm{Ni-mi}}$ *directly; interconnect oxidation stores squared oxide thickness* $s_{\mathrm{IC-ox}} = d^2_{\mathrm{IC-ox}}$*, consistent with Equation 4.23 and* $R_{\mathrm{IC-ox}} = d_{\mathrm{IC-ox}}/\sigma_{\mathrm{IC-ox}}$ *with* $d_{\mathrm{IC-ox}} = \sqrt{s_{\mathrm{IC-ox}}}$*.*

With piecewise-constant controls $\mathbf{u}_i$ on $[t_{i-1}, t_i)$, implicit Euler gives

$$\begin{aligned} \mathbf{d}_{\mathrm{Ni-mi,i}} &= \mathbf{d}_{\mathrm{Ni-mi,i-1}} + \Delta t_i\, \dot{\mathbf{d}}_{\mathrm{Ni-mi}}(\mathbf{d}_i, \mathbf{u}_i), \\ \mathbf{s}_{\mathrm{IC-ox,i}} &= \mathbf{s}_{\mathrm{IC-ox,i-1}} + \Delta t_i\, \dot{\mathbf{s}}_{\mathrm{IC-ox}}(\mathbf{d}_i, \mathbf{u}_i), \end{aligned} \qquad i = 1, \dots, N_t, \tag{5.18}$$

where $\mathbf{d}_i$ *stacks all per-cell degradation components at time* t_i*. The rates follow Table 4.2:* $\dot{d}_{\mathrm{Ni-mi,k}} = c_1 T_k + c_2 i_k + c_3 T_k i_k + c_4$ *depends on local temperature* T_k *and current density* i_k*, while* $\dot{s}_{\mathrm{IC-ox,k}} = k_{\mathrm{mg}} \exp\{-E_{a,\mathrm{ox}}/(RT_k)\}$ *depends on* T_k *only.*

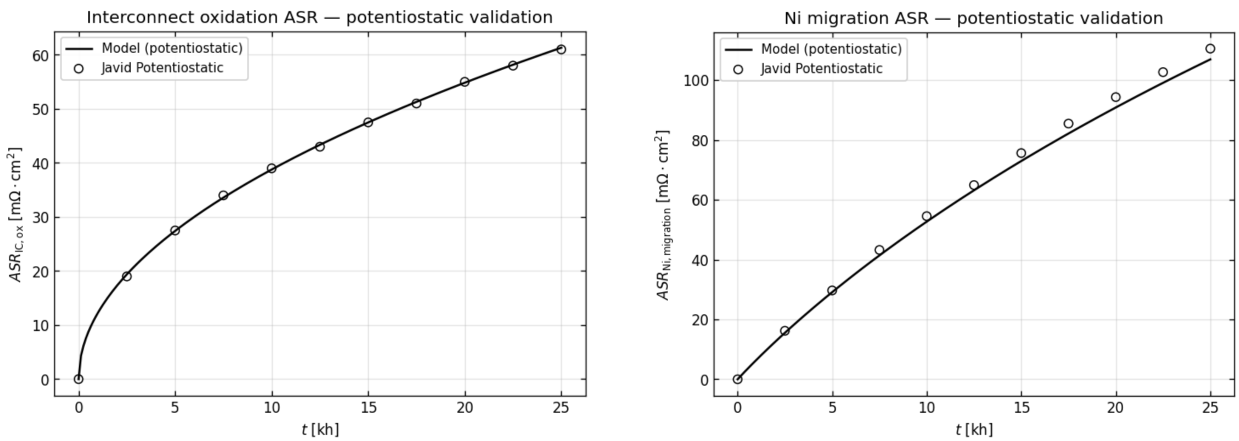
Those FVM outputs are not explicit inputs to the ODE; they are obtained by running the steady-state FVM solver with control $\mathbf{u}_i$ and degradation state $\mathbf{d}_i$. Because $\mathbf{d}_i$ appears inside that solve, each time step is a coupled fixed-point problem: guess $\mathbf{d}_i$, run FVM, evaluate the rates in Equation 5.18, update $\mathbf{d}_i$, and iterate until convergence. Giving us the heavy computation of solving the steady state for each fixed-point iteration, leading to a large computational time.

In general, the problem can be viewed on a two-dimensional grid: each spatial row (FVM cell) must be resolved repeatedly within a time step before advancing to the next column (time index).

5.2.5 Validation of the ODE solver

With the steady-state FVM validated in the previous section, the next step is to check that the coupled degradation integrator in Equation 5.18 reproduces expected long-term behaviour.

The reference case follows the potentiostatic traces reported by Beyrami et al. [5] (Fig. 8): fixed cell voltage $U = 1.285\,\mathrm{V}$, inlet temperature $T_{\mathrm{in}} = 750\,^\circ\mathrm{C}$, and fuel composition $10\,\%$ H_2 / $90\,\%$ H_2O. At each time



(a) Interconnect oxidation. At $t = 25\,\mathrm{kh}$: model $61.3\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$, reference $61.0\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$. RMSE $= 0.26\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$.

(b) Nickel migration. At $t = 25\,\mathrm{kh}$: model $106.9\,\mathrm{m}\Omega{\cdot}\mathrm{cm}^2$, reference $110.5\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$. RMSE $= 2.45\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$.

Figure 5.5: Potentiostatic validation of ASR for (a) interconnect oxidation and (b) nickel migration. Black line: implicit-Euler simulation. Open markers: digitized reference data from [2].

knot the FVM steady-state problem is solved, the degradation rates from Table 4.2 are evaluated, and the states are updated with implicit Euler (Equation 5.16) until $t = 25\,\mathrm{kh}$. This produces a history of how the degradation mechanisms develop over time. From these quantities, the area-specific resistances (ASR) can be computed and compared with the traces from Beyrami et al. in Figure 5.5.

The interconnect-oxidation agreement is very good over the full 25 kh horizon. Nickel migration follows the same trend but sits slightly below the reference curve. Given the digitization uncertainty and the simplified inlet specification relative to the original study, this level of agreement is considered sufficient for the degradation model to be used in the optimal-control setup in Section 5.2.6.

5.2.6 Solving the Optimal Control Problem

The optimal control problem (Equation 5.1) is solved using a nonlinear programming (NLP) solver. The problem is set up for single shooting so that an NLP solver can find the optimum. The control function $\mathbf{u}$ is taken to be piecewise constant (as would also be the case in real applications).

$$\mathbf{u}^{(k)}(t) = \begin{cases} \mathbf{u}_i^{(k)} & t \in [t_i, t_{i+1}) \end{cases}$$

The problem is split into N_t subproblems, each with constant control input. From Equation 5.1 the objective φ can be rewritten as:

$$\begin{aligned} \varphi &= \int_0^T R(\mathbf{u}, \mathbf{d})\, \mathrm{d}t + \Psi(\mathbf{d}) \\ \Rightarrow \dot{\varphi} &= R(\mathbf{u}, \mathbf{d}) \end{aligned} \tag{5.19}$$

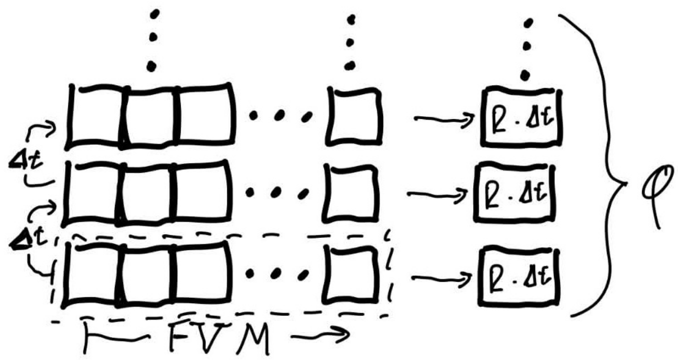


Figure 5.6: CAPTION

Thus φ is obtained by solving the auxiliary differential equation with initial condition $\varphi(0) = 0$ (numerical integration), concurrently with the state equations, and adding $\Psi(\mathbf{d})$ at the final time when the degradation at $t = T$ is known.

For the constraints in Equation 5.1, which are equality constraints, they are used directly to compute degradation and molar flows, leaving only $\mathbf{u}$ as a decision variable for the optimizer. The NLP solver also requires the gradient $\nabla_{\mathbf{u}}\varphi$; adjoint gradient methods are used [16], [8]. Further details are in Appendix A5; here it suffices that $\nabla_{\mathbf{u}}\varphi$ is available.

5.2.6.1 The full setup

The discretized problem solved is:

$$
\begin{aligned}
&\max_{\mathbf{u}} \quad \varphi_N + \Psi(\mathbf{d}_N) \\
\text{s.t.} \quad & \varphi_{i+1} = \varphi_i + \dot{\varphi}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \dots N-1 \\
& \mathbf{d}_{i+1} = \mathbf{d}_i + \dot{\mathbf{d}}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \dots N-1 \\
& \varphi_1 = 0, \\
& \mathbf{d}_1 = \mathbf{0}.
\end{aligned} \tag{5.20}
$$

as a single shooting problem, with $\mathbf{d}$ computed from the constraint equations. A convex programming algorithm is used to find the optimal u; the problem is generally convex. For this project the trust-constr method from SciPy is used [19], a trust-region method that employs the computed gradient of φ.

6 Results and Findings

6.1 Introduction

With the full machinery in place, it was time to put it to work. The preceding chapters set up the model, the objective, and the solver; what was still missing before any results could be discussed was a single list of the **numerical settings** actually used in the optimization runs. Those defaults are collected here. Later sections vary individual knobs; this section is the common baseline they all share unless stated otherwise.

The objective itself is unchanged from Equation 5.1: maximize integrated revenue plus terminal salvation over a finite horizon. The prices and break levels are the concrete numbers plugged into that formulation. Unless stated otherwise, the runs use the settings in Table 6.1. Literature values for stack geometry and flows [17], end-of-life limits [3], cell cost [4], and energy prices [1] are indicated in the reference column; entries marked *Chosen* are project defaults.

Several entries in Table 6.1 are marked as chosen rather than taken from the literature. The voltage and inlet-temperature bounds keep the optimizer within a range of physically plausible operating conditions for the stack model. The number of FVM cells N_{cell} and the control segments N_t were set by the computational budget of the project: finer discretizations were possible in principle, but the selected values keep each forward solve and outer iteration within tractable run times on the available hardware.

The 200-day horizon is chosen for the same practical reason. It is long enough that a change in the optimal control profile can show up within the optimization window, without extending T so far that run times and degradation forecasts dominate the workflow.

The parameter α_Ψ in Equation 5.4 controls the sharpness of the terminal penalty as the cell approaches its end-of-life degradation threshold. A value of $\alpha_\Psi = 0.5$ is chosen here, making the penalty function relatively gentle when the stack is healthy, but causing it to increase steeply as degradation nears its critical limit. This ensures the optimizer is sensitive to imminent failure while still allowing for some degradation earlier in the lifetime.

6.2 Baseline optimization result

As announced in Section 6.1, the first result is a baseline in which both controls are held constant over the full horizon: a single cell voltage U and a single inlet temperature T_{in}, broadcast to all $N_t = 5$ segments. This is the most restricted operating mode considered (U and T_{in} each collapse to one scalar decision variable), analogous to fixing both aim and pump pressure in the water-gun example from Section 2.1.

Setting	Baseline value	Unit	Reference
Horizon T	200	days	Chosen
Control segments N_t	5	—	Chosen
U bounds	$[0.8,\ 1.5]$	V	Chosen
T_{in} bounds	$[700,\ 900]$	°C	Chosen
FVM cells N_{cell}	20	—	Chosen
Fuel / air flows	24 / 50	NL/h	[14]
Initial $h_{\text{Ni-mi}}$ / $h_{\text{IC-ox}}$	0.75 / 1.5	μm	Values after 200 hours simulation
Break $h_{\text{Ni-mi}}$ / $h_{\text{IC-ox}}$	5 / 10	μm	[18]
c_{H_2}	7.44	EUR/kg	[19]
c_P	0.079	EUR/kWh	[19]
c_{cell}	1000	EUR/m^2	[14], [1]
α_Ψ	0.5	—	Chosen
Optimization solver	SLSQP/Trust-Constr	—	—

Table 6.1: Baseline numerical settings for the optimization results in this chapter (unless a subsection states otherwise).

All other settings follow Table 6.1. The NLP returns

$$U^* \approx 1.29\,\text{V}, \qquad T_{\text{in}}^* \approx 758\,^\circ\text{C}\ (1031\,\text{K}).$$

Figure 6.2 shows the replayed trajectory at these controls. Nickel-migration thickness grows from $0.75\ \mu$m to about $3.7\ \mu$m; interconnect oxidation from $1.5\ \mu$m to about $3.1\ \mu$m. Running revenue accumulates over the 200-day window while degradation approaches the break limits enforced in the solver.

The accumulated revenue is 18.985 EUR and the terminal salvation term is 1.452 EUR, meaning the cell has lost 0.548 EUR of salvage value over the horizon. Due to the initial conditions, 0.089 EUR had already been lost at $t = 0$, leaving a loss within the control domain of 0.459 EUR and a net economy of 18.526 EUR from a $20\,\text{cm}^2$ cell over 200 days of operation.

The optimal voltage lies near the thermoneutral point, where electrolytic efficiency is highest. At this operating condition, heat generation and losses balance, so the temperature profile along the cell remains nearly uniform. Both nickel-migration and interconnect-oxidation rates depend on local temperature; with a flat temperature field, these rates are therefore similar at every spatial location. Degradation accumulates evenly along the cell rather than being concentrated in a local hot spot that would accelerate failure at one point.

6.2.1 Allowing more control freedom

The natural next step is to let the optimizer choose inlet temperature and voltage freely in each control interval. In this case, however, the outcome is unremarkable: even with full temporal freedom, the best operating mode remains potentiostatic-isothermal operation. This baseline does not impose a required hydrogen delivery rate. As reported in the literature, this mode is among the most efficient ways to operate the cell, and potentiostatic control is gentler than galvanostatic operation. The result therefore serves as a sanity check for the full model and confirms qualitatively expected behavior. Before exploring further variants, the setup

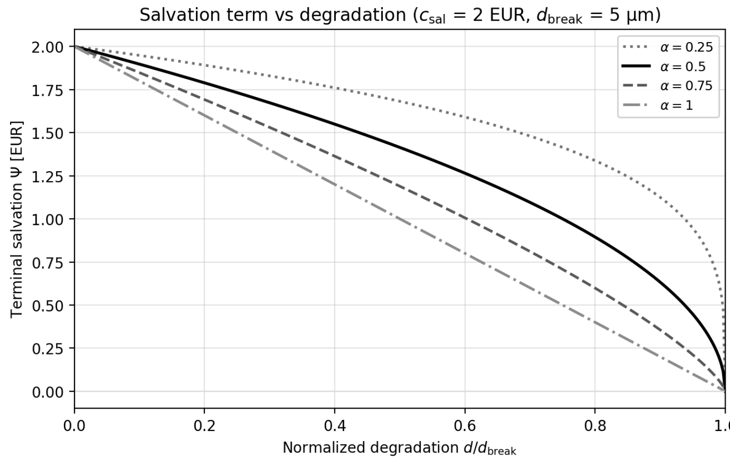


Figure 6.1: WRITE

Setting	**Value**	**Unit**	**Reference**
Contractual delivery $\dot{n}_{H_2,\text{agreement}}$	8.25	NL/h	Chosen
Initial tank inventory $m_{\text{tank},0}$	0	kg	Chosen
Spot purchase penalty κ_{buy}	1.5	—	Chosen
Terminal liquidation credit κ_{sell}	$\frac{1}{8}$	—	Chosen
Updated Variables	**Value**	**Unit**	**Reference**
Control segments N_t	20	—	Chosen

Table 6.2: Tank and delivery-agreement settings for the constant-delivery results (in addition to Table 6.1). Updating N_t to be finer to allow for better galvaniostatic controll

is extended to a production setting with a fixed contractual delivery rate, as described in Section 5.1.4.

6.3 Constant delivery rate

The next result uses the storage-tank implementation from Section 5.1.4 with a fixed contractual delivery rate. All other settings follow Table 6.1; the additional tank parameters are listed in Table 6.2.

Meaning that we have a contract that we need to deliver what would average to 7.5 normal liters an hour for one cell. Which would be equivalent to a productional setup of two stacks each having about 150 cells each. These two stack should then produce 5kg of hydrogen combined each day to live up to the customer agreement. Then κ_{buy}, means that if we don't live up to the agreement with the customer. Means that a producer is fined by 50% of the hydrogen price for the hydrogen not delivered. And lastly κ_{sell}, means that if the hydrogen storage tank have additional hydrogen left in the tank at the end of the period, this is assumed to be sellable to 12.5% of the agreement price on an open marked, assuming that it has to compete with "black" hydrogen.

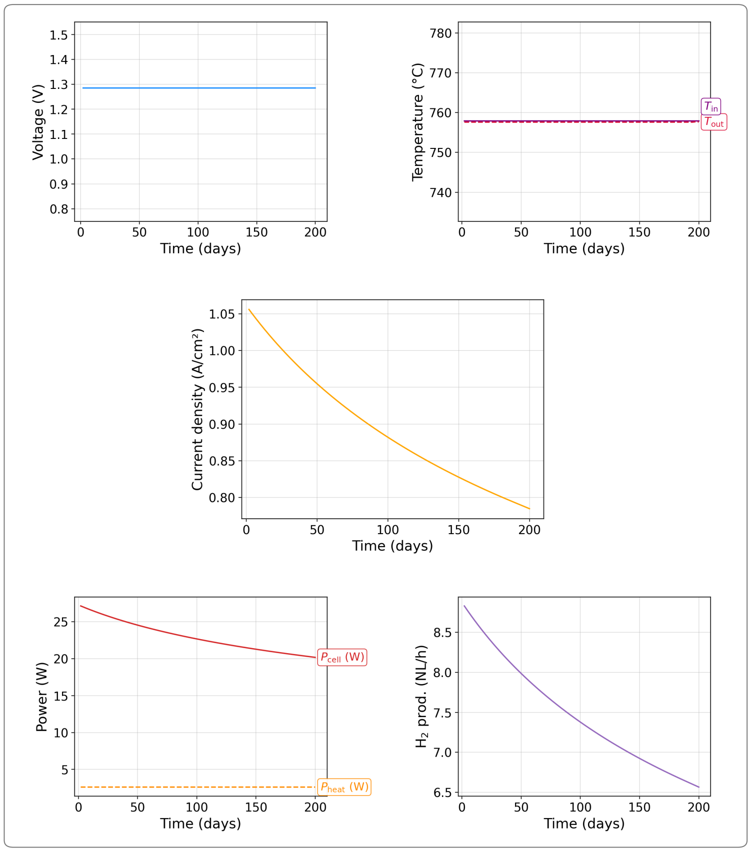


Figure 6.2: Baseline potentiostatic–isothermal optimization: constant U and T_{in} over 200 days, with segment-resolved degradation and economics.

6.3.1 Freedom on all controls

Now for the most interesting case we look at what happens if we let both voltage and temperature vary through time. Here we end up having that we keep at the optimal control conditions found in Section 6.2. But then we end up two local minima, that have the same objective value, if we take the numerical precision into account. One where once we hit the agreement rate we start tuning the temperature to hit galvanotactic operation. Once the optimizer hit a point where it is possible to switch back to isothermal and potentiostatic operation and the tank will have enough content to make up for the difference between production rate and agreement rate. Once it hit this point it will prefer this to save the cell rather than sell on the free marked, this operation mode can be seen in Figure 6.3.

The other option is to run at isothermal and potentiostatic until the tank runs dry, and the switch to galvanotactic operation mode for the rest of the operation window, and can be seen in Figure 6.4.

We must assume that these two minima are the two extreme points on a continuum of minima. Although the modes along this continuum are equivalent with regard to the implemented objective function, one is clearly more favorable than the other, since in production it would be much nicer to run with a spare reserve; hence the best practical solution is the first. Another benefit is that we continuously control both temperature and current, whereas in the second control option we have a large jump in both when the tank runs dry. When piecewise controls are optimized, abrupt segment-to-segment changes can be discouraged by the optional

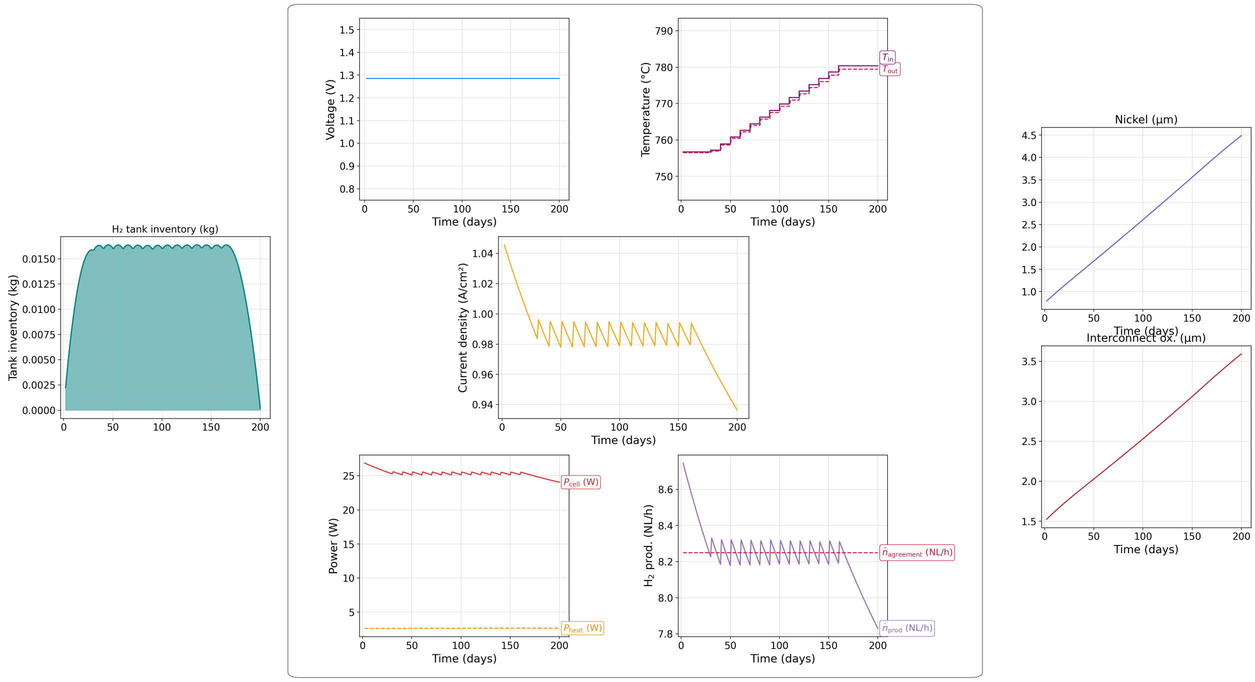


Figure 6.3: WRITE

squared continuity soft penalty defined in Appendix A6; it was not activated in the constant-delivery runs reported here ($w_U = w_T = 0$).

In both cases the running revenue is the same add 15.87 EUR. And the final salvation is 1.21 EUR, leading to a loss in cell-value of 0.7 EUR. And in both cases the tank has as little hydrogen left, such that the influence on the final objective can be neglected. But And the final net-revenue ends up at 15.10 EUR.

6.4 24-hour cycle with varying power price

The final scenario shortens the horizon to one day and replaces the flat power price with a time-varying profile. The cosine form in Table 6.3 is a stylized stand-in for solar-driven day–night spreads: electricity is cheapest around midday and most expensive around midnight. Together with the contractual delivery setup from Section 6.3, this case tests whether the optimizer shifts production toward low-price periods while still meeting the agreement.

Terminal liquidation is disabled ($\kappa_{\text{sell}} = 0$), and the cyclic inventory constraint Equation 5.9 is enforced so that $m_{\text{tank}}(0) = m_{\text{tank}}(T)$. The day can therefore be studied as a repeating operating window without end-of-horizon sell-off. The updated settings are listed in Table 6.3.

Voltage is held potentiostatic and a large weight w_U discourages segment-to-segment voltage changes; temperature remains the primary free control. The optimal trajectory is shown in Figure 6.5. Production is highest when the power price is low and reduced when prices rise at night. The response is qualitatively as expected for load shifting, but the profile is simple: the optimizer modulates inlet temperature rather than exploiting both voltage and temperature.

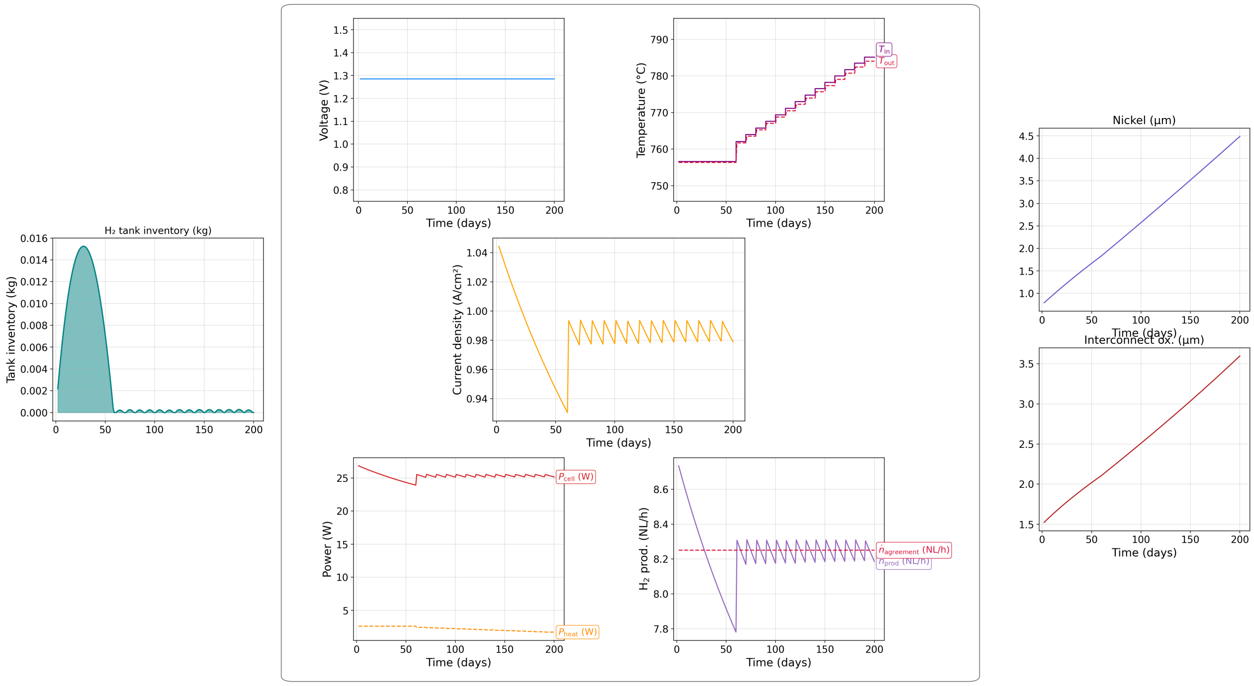


Figure 6.4: WRITE

The economic decomposition of the optimized run is as follows. Integrated running revenue over the 24,h window is 0.074 EUR and the terminal salvation term is 1.91 EUR, giving a total objective of 1.98 EUR. Degradation over a single day is negligible (salvage loss $\lesssim 0.01$ EUR), so the net economy from operation is effectively 0.07 EUR for the $20\,\text{cm}^2$ cell.

Two modelling choices shape this outcome strongly. The cyclic tank constraint removes any terminal credit for leftover inventory and, together with the delivery obligation, forces production and storage to balance over the day rather than deferring output indefinitely. The continuity penalties bias the solution toward smooth temperature profiles and, through $w_U \gg w_T$, toward an effectively fixed voltage. In exploratory runs, increasing w_T pushes the optimizer back toward near-isothermal operation, whereas smaller values allow sharper concentration of production in the cheapest price windows.

6.4.1 Summary

This 24-hour case closes the results sequence begun in Section 6.2 and Section 6.3. Where the long-horizon baseline favours steady potentiostatic–isothermal operation and the delivery case exposes degenerate optima tied to tank use, the 24-hour cyclic run shows that time-varying electricity prices alone can induce deliberate scheduling—here through inlet temperature—while contractual delivery and periodic inventory are satisfied. Economically, almost all of the reported objective value sits in terminal salvation rather than the one-day cash flow (1.98 EUR total versus 0.07 EUR net operating economy), which is expected for a short horizon with little degradation.

The pattern is economically sensible but not surprising: shift electrolysis toward cheap power and lean on the tank to bridge expensive intervals. That simplicity is still informative, because it confirms that the revenue, storage, and price models are coupled as intended. At the same time, the result should not be over-

Setting	Value	Unit	Reference
Temperature squared continuity penalty w_T	0.1	—	Chosen (Appendix A6)
Voltage squared continuity penalty w_U	1000	—	Chosen (Appendix A6)
Updated Variables	**Value**	**Unit**	**Reference**
Horizon T	24	h	Chosen
Control segments N_t	12	—	Chosen
Terminal liquidation credit κ_{sell}	0	—	Chosen
Initial tank inventory $m_{\text{tank},0}$	$m_{\text{tank},T}$	—	Periodic boundary
Power price $c_{P,k}$	$\bar{c}_P\left(1 - a\cos\left(2\pi\dfrac{h_k - 12\,\text{h}}{24\,\text{h}}\right)\right)$	EUR/(W·s)	Chosen
Mean power price $\bar{c}_P$	0.079	EUR/kWh	[19]
Relative price swing a	0.5	—	Chosen

Table 6.3: Updated model settings for the 24-hour cyclic scenario. No terminal sale of hydrogen is allowed ($\kappa_{\text{sell}} = 0$), and the initial tank content is enforced as a periodic boundary. Twelve control segments are used for the day. The piecewise-constant power price on segment k follows a single cosine cycle over 24 h, with h_k the hour-of-day at the segment midpoint; the minimum $c_{P,k} = \bar{c}_P(1 - a)$ occurs at midday ($h_k = 12\,\text{h}$) and the maximum $\bar{c}_P(1 + a)$ at midnight, modelling lower prices when solar supply is high. The continuity weights w_U and w_T enter the soft penalty Appendix A6; with potentiostatic voltage only w_T is effective.

interpreted: a single-harmonic price curve, fixed continuity weights, and the cyclic boundary are strong structural assumptions. Real markets exhibit ramps, negative prices, and correlation with renewable output that this stylized setup does not capture; the sensitivity to w_T, w_U, and the tank closure rule noted above underlines that operational conclusions from this scenario depend on those choices. A natural next step would be to test alternative terminal treatments, finer temporal resolution, or measured price series before drawing firm scheduling recommendations.

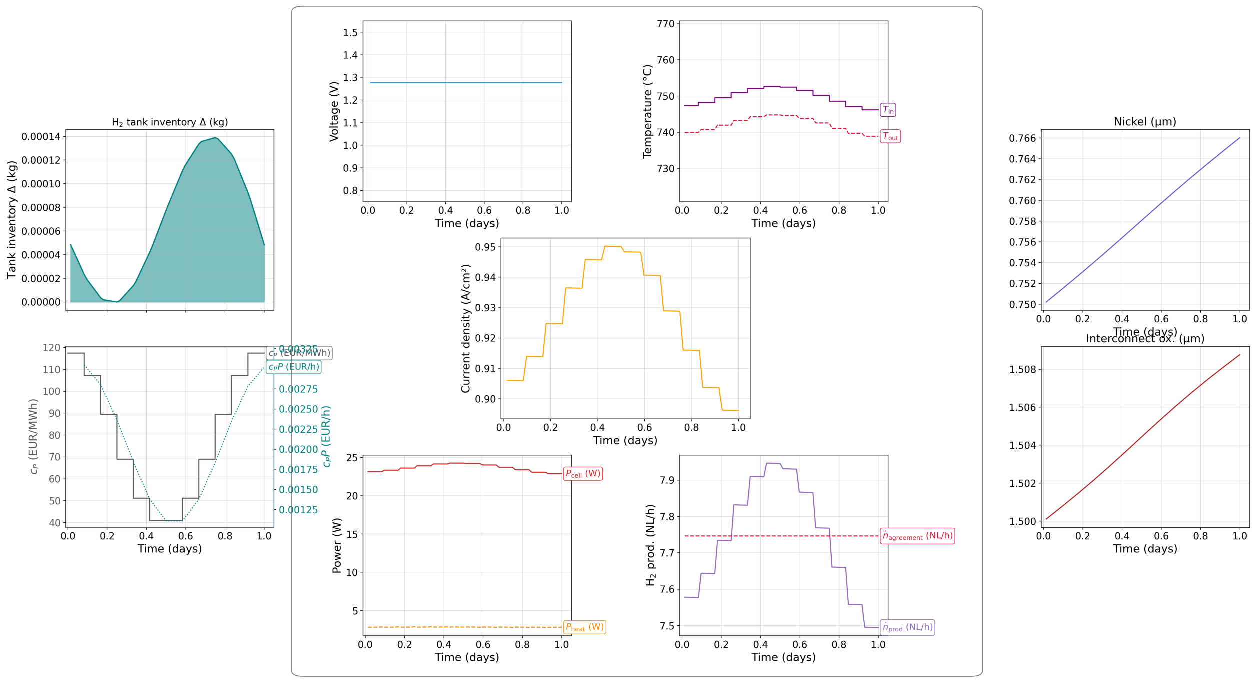


Figure 6.5: Optimal temperature control over a 24-hour cycle with varying power prices. The controller modulates temperature to take advantage of the lower power price during midday and reduces hydrogen production at night when prices are higher. The cyclic boundary condition ensures the tank inventory is the same at the start and end of the day.

7 Discussion and Conclussion

7.1 Introduction

The results found in Section 6.1

7.2 Interpretation of control freedom

7.2.1 Baseline without delivery obligations

7.2.2 Effect of a fixed delivery contract

7.2.3 Time-variyng electricity prices

7.3 Physical model limitations

7.3.1 Notes

The optimal-control conclusions rest on the modelling assumptions collected in Section 3.1 and on the degradation and cell equations summarized in Table 4.1 and Table 4.2.

Spatial and transport simplifications. The stack is modelled as one-dimensional flow with constant channel area. Concentration overpotential is neglected, which is reasonable when reactant utilisation remains moderate but may underestimate losses at high current. Heat conduction is neglected; in reality, lateral and axial conduction can smooth temperature peaks that would otherwise locally accelerate degradation rates.

Degradation mechanisms. Only nickel migration and interconnect oxidation are resolved dynamically. Nickel agglomeration enters indirectly through a fixed cathode exchange-current multiplier $\lambda_c \approx 0.55$ (Section 3.1), justified by its short timescale relative to the 200-day horizon. A time-varying λ_c could alter the trade-off between current density and temperature over life, especially in scenarios with extended galvanostatic operation.

Validation and spatial resolution. IV-curve validation (Figure 5.3) shows agreement within roughly 10^{-2} in the intended operating range. Grid-independence tests (Figure 5.4) indicate that $\mathcal{O}(10^{-3})$ relative error in the steady-state fields would require on the order of 200 FVM cells, whereas the optimization runs use $N_{\text{cell}} = 20$ (Table 6.1). The coarser mesh is a deliberate computational compromise; degradation trajectories and optimal controls should therefore be interpreted as qualitatively reliable rather than grid-converged to high precision.

Unit scale. All results are computed for a single $20\,\mathrm{cm}^2$ laboratory-scale cell. Stack-level effects—temperature maldistribution, flow allocation, and cell-to-cell variability—are not represented. Scaling arguments (for example, the industrial stack count mentioned in Section 6.3) illustrate production context but do not change the per-cell optimal-control structure derived here.

7.4 Economic and boundary-condition limitations

7.4.1 Notes

The objective combines running revenue, terminal salvation, and—when the tank is active—contract revenue and optional liquidation (Table 5.1). Several economic parameters are marked *Chosen* in Table 6.1 and Table 6.2 rather than taken from market data. The qualitative conclusions are therefore conditional on those numbers.

Salvation weighting. The terminal penalty Ψ in Equation 5.4 with $\alpha_\Psi = 0.5$ rises steeply only when degradation approaches critical thicknesses. A smaller α_Ψ would make the optimizer more willing to accept degradation early in life; a larger value would push operation further from break limits, potentially at the expense of production revenue.

Tank penalties and credits. The spot-purchase multiplier κ_{buy} and liquidation credit κ_{sell} set how severely shortfalls and end-of-horizon inventory are priced. In the constant-delivery results, $\kappa_{\text{buy}} = 1.5$ and $\kappa_{\text{sell}} = 1/8$ were used (Table 6.2), whereas the tank formulation in Section 5.1.4 quotes $\kappa_{\text{sell}} = 0.9$ as a default. Sensitivity to these coefficients was not explored systematically; they should be aligned and, where possible, justified from offtake-contract structures or market studies [4, 10].

Delivery rate and horizon. The contractual rate $\dot{n}_{\mathrm{H_2},\text{agreement}}$ and the 200-day horizon jointly determine how often the tank forces galvanostatic operation. A longer horizon, a higher delivery obligation, or a stricter purchase penalty would strengthen the incentive for early inventory build-up and may reduce the set of economically equivalent trajectories observed in Section 6.3.

Diurnal pricing. The cosine power-price profile in Table 6.3 is a stylized stand-in for solar-driven day–night spreads. Real price series include ramps, negative prices, and correlation with renewable availability that a single-harmonic model cannot capture. Conclusions about load shifting under diurnal prices should be revisited once results for that scenario are finalized.

7.5 Solver and numerical limitations

7.5.1 Notes

The optimal-control problem is transcribed to a single-shooting NLP as described in Section 5.2.6. Piecewise-constant controls over N_t segments are appropriate for implementable set-points, but they bound how finely the optimizer can track fast price or load variations. The baseline used $N_t = 5$; the delivery case used

$N_t = 20$; the diurnal case uses $N_t = 12$ over 24 hours (Table 6.1, Table 6.2, Table 6.3). Refining N_t increases the decision-space dimension and the cost of each adjoint gradient evaluation.

The NLP solvers (SLSQP and trust-region constrained methods) find local optima. The pair of equivalent delivery solutions in Figure 6.3 and Figure 6.4 is consistent with a flat or weakly curved region of the objective landscape: many inventory-management policies deliver the same contractual hydrogen while incurring similar power and degradation costs. Without additional regularization—for example, penalties on control jumps or a minimum tank reserve constraint—the solver has no reason to prefer one member of that family over another.

Gradients were verified for the forward model and economic terms (Appendix, Appendix A5 and gradient-verification figures). Remaining numerical uncertainty comes mainly from the FVM spatial discretization, the implicit time stepping over long horizons, and finite convergence tolerances in the outer optimizer. Reported objective values should be read to the precision stated in the results chapter; differences below that level are not physically meaningful.

7.6 Alternative objective formulations

7.6.1 Notes

The optimization results in Section 6.2–Section 6.3 all use the objective implemented in Equation 5.1: maximize integrated running revenue plus a terminal salvation term. This section contrasts that choice with the introductory formulation in Equation 2.1 and with objectives common in the SOEC optimal-control literature.

7.6.2 Weighted revenue–degradation form

At the outset of the thesis, the problem was described schematically in Equation 2.1:

$$\max_u \left\{ \alpha(t) \int_0^T \text{revenue}\, \mathrm{d}t - (1-\alpha(t))\, \text{degradation} \right\} \tag{7.1}$$

where $\alpha(t) \in [0,1]$ weights instantaneous production against degradation. The motivation was that $\alpha(t)$ could be tied to market conditions—for example, favouring production when hydrogen prices are high and penalizing degradation more when replacement cost dominates. This form is useful for presentations because it makes the trade-off explicit in a single integral.

The implemented objective does not use Equation 2.1 directly. Instead, degradation enters through the **terminal** salvation map $\Psi(\mathbf{d})$ in Equation 5.4, while the running integral contains only the revenue rate R. As noted in Section 5.1.1, an equivalent modelling choice would penalize a **running** degradation rate inside the integral rather than terminal state. The two constructions answer slightly different questions:

- $\int_0^T R\, \mathrm{d}t + \Psi(\mathbf{d}(T))$ **(this work):** cash flow over $[0,T]$ plus end-of-horizon residual value; tends to favour gentle operation when Ψ is steep near failure limits.
- $\int_0^T [\alpha R - (1-\alpha)\, \dot{\mathbf{d}}]\mathrm{d}t$**:** explicit trade-off at every instant; can push away from high $\dot{d}$ even when the terminal state is still healthy.

- $\Psi = 0$, T **equal to lifetime (ideal):** total lifetime revenue; requires predicting prices and degradation far ahead.

The salvation-based terminal penalty approximates the lifetime objective from Figure 5.1 on a rolling horizon $T \ll$ stack life. It does not reproduce a time-varying $\alpha(t)$ unless α is identified with exogenous price signals—which was not done in the reported runs.

7.6.3 Objectives in the literature

Recent SOEC dynamic-optimization studies use several distinct scalar objectives [4, 7, 10]:

- **Minimize terminal degradation** (or maximize remaining life margin) at fixed horizon.
- **Maximize integral efficiency** over the operating window.
- **Minimize levelized cost of hydrogen (LCOH)**, coupling production, energy, capital, and replacement.
- **Maximize hydrogen production** or **minimize energy consumption** subject to thermal and degradation constraints [7].

Giridhar et al. report that LCOH-oriented control can outperform strategies that optimize terminal degradation or integral efficiency alone, with relative gains that depend on electricity price. Beyrami et al. embed degradation and stack replacement into plant-scale economic optimization. These works suggest that the qualitative conclusions of this thesis—especially the sensitivity of $u^*(t)$ to delivery constraints and the flat objective landscape in Section 6.3—could shift if the scalar objective were changed.

7.6.4 Implications for the results presented

Several findings are **objective-dependent** rather than universal:

1. **Non-uniqueness under delivery contracts.** Two trajectories with equal $\int R\,\mathrm{d}t + \Psi$ (Figure 6.3, Figure 6.4) might separate if the objective penalized control jumps, minimum inventory, or running degradation rate.
2. **Potentiostatic–isothermal baseline.** Near-thermoneutral operation is optimal for $\int R\,\mathrm{d}t + \Psi$ without delivery forcing. A pure terminal-degradation minimization might accept lower immediate revenue for slower $\dot{d}$; an LCOH objective might favour higher utilisation when electricity is cheap (relevant to the diurnal scenario in Table 6.3).
3. **Salvation sharpness α_Ψ.** The exponent in Equation 5.4 plays a role analogous to $(1 - \alpha)$ in Equation 2.1, but only at $t = T$. Sweeps over α_Ψ or over an explicit α in a running penalty were not carried out here.

Implementing Equation 2.1 with price-linked $\alpha(t)$, or switching to LCOH as in [10], would be a natural extension. The existing NLP and adjoint infrastructure could accommodate a different integrand without changing the spatial or degradation models, provided gradients of the new terms are derived or verified in the same way as in Appendix A5.

7.7 Use of generative AI tools

7.7.1 Notes

Large language models (LLMs) and AI-assisted coding environments were used during this project. This section records that use transparently, in line with institutional expectations for thesis work. The primary tool was Cursor [2], an AI-augmented editor integrated with the project repository.

7.7.1.1 Scope of AI assistance

AI tools were applied to **supporting** tasks rather than to the scientific conclusions themselves. Documented uses include:

- **Software development:** debugging, vectorizing numerical code, structuring modules, and generating repetitive boilerplate (e.g. parallel implementations of similar residual blocks).
- **Verification support:** drafting scripts for finite-difference gradient checks against the adjoint implementation (Appendix A5 and appendix gradient plots).
- **Documentation and prose:** reformulating sentences, drafting parts of the appendix, and generating nomenclature tables (Nomenclature).
- **Literature workflow:** locating and formatting references (all citations were checked against the primary sources before inclusion).

The following remained under direct author responsibility: problem formulation with supervisors (Dynelectro, DTU Energy), choice of degradation mechanisms and parameters, model validation against published data, interpretation of optimization results, and all figures presenting final numerical outcomes.

7.7.2 Verification and limitations

Outputs from generative models were not accepted without review. In particular:

- **Physics and equations** were checked against the cited literature and hand-derived limits (e.g. IV-curve validation in Figure 5.3, ODE replay in the solver-validation section).
- **Gradients** used in the NLP were verified numerically before trusting optimization results.
- **Prose** was edited for technical accuracy, notation consistency with Table 4.1, and alignment with the implemented code.

LLMs can produce plausible but incorrect equations, wrong unit conversions, or citations that do not match the claimed source. They also reflect training data that may not include the latest SOEC literature. For these reasons, AI-generated text and code were treated as drafts, not as authoritative references.

7.8 Conclusions and outlook

7.8.1 Conclusions

1. **Control freedom is scenario-dependent.** Additional temporal freedom in U and T_{in} did not change the optimal operating mode for the unrestricted 200-day baseline: potentiostatic–isothermal operation near the thermoneutral point remained optimal. Freedom became economically relevant only when a fixed hydrogen delivery contract and buffer tank were introduced.

2. **Delivery contracts activate inventory-aware operation.** With contractual offtake, the optimizer alternates between gentle potentiostatic–isothermal periods and galvanostatic periods dictated by tank inventory and penalty prices. Multiple control trajectories can achieve the same objective, so uniqueness should not be assumed without further constraints.
3. **Degradation-aware objectives favour uniform, moderate operation.** Flat temperature profiles and operation away from aggressive current densities slow both modelled degradation mechanisms and align with the salvation term. This provides a consistent link between the physical degradation model and the economic optimum in the cases studied.
4. **The toolchain is fit for purpose at engineering fidelity.** IV validation, ODE replay against reference degradation data, and gradient checks support using the model for qualitative optimal-control studies, subject to the spatial-resolution and mechanism-scope limitations above.
5. **Objective choice shapes what "optimal" means.** The implemented form $\int_0^T R\,\mathrm{d}t + \Psi$ differs from the weighted schematic in Equation 2.1 and from LCOH- or efficiency-based objectives in the literature (Section 7.6). Several reported degeneracies—especially equivalent delivery trajectories—are artefacts of that scalar choice.

7.8.2 Future Work

A1 Deriving PDE's for molecule balances

From [15] eq. [A-3], [A-4] and [A-5]:

$$\frac{PV}{RT}\frac{\partial x_{\mathrm{H_2O}}}{\partial t} = \underbrace{A_{int}N_{in,\mathrm{H_2O}}}_{\text{Inlet}} + \underbrace{A_{cell}N_{a,\mathrm{H_2O}}}_{\text{Cell contribution}} - \underbrace{A_{int}N_{out,\mathrm{H_2O}}}_{\text{Outlet}}$$

$$N_{in/out,\mathrm{H_2O}} = v_{fuel}\,\mu_{in/out,\mathrm{H_2O}}$$

$$N_{a,\mathrm{H_2O}} = \nu_{\mathrm{H_2O}}\frac{i}{2F}$$

The equations are also required in terms of the concentration $\mu_{\mathrm{H_2O}}$, defined by:

$$\mu_{\mathrm{H_2O}} = \frac{n_{\mathrm{H_2O}}}{V}$$

Inserting the above and the ideal gas law, and discretizing, yields

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{A_{cell}}{V}\nu_{\mathrm{H_2O}}\frac{i}{2F} + \frac{A_{int}}{V}v_{fuel}(\mu_{\mathrm{H_2O}}(z-\Delta z) - \mu_{\mathrm{H_2O}}(z))$$

Further, $A_{cell} = w\Delta z$ and $V = \Delta z A_{int}$ are discretized. Inserting and isolating the time derivative gives:

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w\Delta z}{A_{int}\Delta z}\nu_{\mathrm{H_2O}}\frac{i}{2F} + \frac{A_{int}}{A_{int}\Delta z}v_{fuel}(\mu_{\mathrm{H_2O}}(z-\Delta z) - \mu_{\mathrm{H_2O}}(z))$$

$$= \frac{w}{A_{int}}\nu_{\mathrm{H_2O}}\frac{i}{2F} + v_{fuel}\frac{(\mu_{\mathrm{H_2O}}(z-\Delta z) - \mu_{\mathrm{H_2O}}(z))}{\Delta z}$$

Letting $\Delta z \to 0$ yields

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{int}}\nu_{\mathrm{H_2O}}\frac{i}{2F} - v_{fuel}\frac{\partial \mu_{\mathrm{H_2O}}}{\partial z}$$

or since, $N_{\mathrm{H_2O}} = v_{fuel}\mu_{\mathrm{H_2O}}$

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{int}}\nu_{\mathrm{H_2O}}\frac{i}{2F} - \frac{\partial N_{\mathrm{H_2O}}}{\partial z}$$

For x_{H_2} and x_{O_2} the same steps give:

$$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{int}}\nu_i\frac{i}{2F} - \frac{\partial N_i}{\partial z}, \qquad \text{for } i \in \{\mathrm{H_2O}, \mathrm{H_2}, \mathrm{O_2}\}$$

Where $\nu_i = \left(1, -1, -\frac{1}{2}\right)^{\top}$.

A1.1 Unit computation test only for $\mathrm{H_2O}$

The units for the $\mathrm{H_2O}$ equation are verified as follows:

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{int}}\frac{i}{2F} - v_{fuel}\frac{\partial \mu_{\mathrm{H_2O}}}{\partial z}$$

Left side units: - $\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t}$: $[mol \cdot m^{-3} \cdot s^{-1}]$

Right side units: - $\nu_{\mathrm{H_2O}}\frac{w}{A_{int}}\frac{i}{2F}$: - $\nu_{\mathrm{H_2O}}$ is dimensionless. - $\frac{w}{A_{int}}$: $[m]/[m^2] = [m^{-1}]$ - $\frac{i}{2F}$: $[A \cdot m^{-2}]/[C \cdot mol^{-1}]$

$$[m^{-1}] \cdot [A\, m^{-2}]/[C\, mol^{-1}] = [A\, m^{-3}]/[C\, mol^{-1}]$$

Next, since $1\, A = 1\, C/s$:

$$= [C\, s^{-1}\, m^{-3}]/[C\, mol^{-1}] = [mol\, s^{-1}\, m^{-3}]$$

So the total units are $[mol\, m^{-3}\, s^{-1}]$

- $v_{fuel}\frac{\partial \mu_{\mathrm{H_2O}}}{\partial z}$: $[m \cdot s^{-1}] \cdot \frac{[mol \cdot m^{-3}]}{[m]} = [mol \cdot m^{-3} \cdot s^{-1}]$ (since $\partial\mu/\partial z$ has units concentration per length, i.e. $[mol \cdot m^{-3}]/[m] = [mol \cdot m^{-4}]$).

Both terms on the right have units $[mol \cdot m^{-3} \cdot s^{-1}]$, which matches the left side. The unit computation is consistent.

A2 Deriving PDE's for energy balances

Starting with. Setting $\mathcal{S} = \{\mathrm{H_2O}, \mathrm{O_2}, \mathrm{H_2}\}$:

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (h_i N_i) + Q_T$$

Using the assumption $\frac{\partial p}{\partial t} = \frac{\partial p}{\partial z} = 0$, $h \approx c_p T$ and the equation becomes:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (c_p T N_i) + Q_T$$

Neglecting conductivity gives:

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \nabla \cdot (T N_i) + Q_T$$

In 1D the divergence reduces to the derivative:

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \frac{\partial T N_i}{\partial z} + Q_T$$

With $N_i = v_i \mu_i$,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + Q_T$$

A2.1 Getting source term

The source term follows from the energy per second (power) in the specific volume:

First the power, which is the current times the "voltage drop"

$$P = i A_{cell} U$$

Dividing by the volume gives Q_T:

$$Q_T = \frac{P}{V} = \frac{A_{cell}}{V} iU$$

With $A_{cell} = wdz$ and $V = A_{int}dz$,

$$Q_T = \frac{w}{A_{int}} iU$$

Hence the overall equation is

$$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + \frac{w}{A_{int}} iU$$

A2.2 Getting the steady state equation:

For the steady-state equation, the derivative in the sum is expanded and the time derivative is set to zero:

$$0 = -\sum_{i \in \mathcal{S}} v_i c_{p,i} \left(\frac{\partial T}{\partial z} \mu_i + T \frac{\partial \mu_i}{\partial z} \right) + \frac{w}{A_{int}} iU$$

And substitute back in that $h_i = c_{p,i} T$ and the steady state equation for $v \frac{\partial}{\partial z} \mu_i = \frac{w}{A_{int}} \nu_i \frac{i}{n_e F}$ to get that

$$0 = -\sum_{i \in \mathcal{S}} \left(v_i c_{p,i} \frac{\partial T}{\partial z} \mu_i + \frac{w}{A_{int}} \frac{i}{n_e F} \nu_i h_i \right) + \frac{w}{A_{int}} iU$$

Now takin all but the indexed paramters out of the sum:

$$0 = -\frac{\partial T}{\partial z} \sum_{i \in \mathcal{S}} v_i c_{p,i} \mu_i - i \frac{w}{A_{int}} \sum_{i \in \mathcal{S}} \nu_i h_i + i \frac{w}{A_{int}} U$$

Defining the sums as

$$\sum_{i \in S} v_i c_{p,i} \mu_i = C_p, \quad U_{th} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i h_i$$

yields:

$$0 = i \frac{w}{A_{int}} (U - U_{th}) - C_p \frac{\partial T}{\partial z}$$

A3 Computing Specific Finite Volume Equations

A3.1 Equations for moles

Starting from Equation 4.5, Equation 4.6 and Equation 4.7, integration over the volume gives:

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t}\, dV = \int_{\Omega_k} \frac{w}{A_{int}} N_{c,i}\, dV - \int_{\Omega_k} \frac{\partial N_i}{\partial z}\, dV$$

The time derivative can be moved outside the integral, and the concentration integrated over a volume equals the amount of moles in that volume:

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t}\, dV = \partial_t \int_{\Omega_k} \mu_i\, dV = \frac{\partial n_i}{\partial t} = \dot{n}_i$$

Using the approximation in Section 5.2.2 with $V_k = A_{int}\Delta z$,

$$\int_{\Omega_k} \frac{w}{A_{int}} N_{c,i}\, dV \approx \frac{w}{A_{int}} N_{c,i}(z_k) V_k = w N_{c,i}(z_k)\Delta z$$

For the last term, the integral is split so that integration over the area normal to z follows integration over z, giving

$$\begin{aligned}\int_{A_{int}} \int_{z_{k-\frac{1}{2}}}^{z_{k+\frac{1}{2}}} \frac{\partial N_i}{\partial z}\, dz\, dA &\approx \int_{A_{int}} \Big(N_i(z_k) - N_i(z_{k-1})\Big)\, dA \\ &= \Big(N_i(z_k) - N_i(z_{k-1})\Big) A_{int}\end{aligned}$$

Hence for each volume:

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) = \dot{n}_i(z_k) \approx w N_{c,i}\Delta z - A_{int}(N_i(z_k) - N_i(z_{k-1}))$$

A3.2 Equations for heat

Steady state is treated first, then the time derivative.

A3.2.1 Steady state

Starting from Equation 4.18:

$$0 = i\frac{w}{A_{int}}(U - U_{th}) - C_p\frac{\partial T}{\partial z}$$

The same steps as for the mole balances apply:

$$0 = \int_{\Omega_k} i\frac{w}{A_{int}}(U - U_{th}) - \int_{\Omega_k} C_p\frac{\partial T}{\partial z}$$

Following the same steps as the previous subsection:

$$\int_{\Omega_k} C_p\frac{\partial T}{\partial z}\,dV \approx C_p(T(z_k) - T(z_{k-1}))A_{int}$$

And the rest is obvious, then the final equation is:

$$0 \approx iw(U - U_{th})\Delta z - C_p(T(z_k) - T(z_{k-1}))A_{int}$$

A3.2.2 Time derivative

For the time derivative:

$$\int_{\Omega_k} \rho c_p\frac{\partial T}{\partial t} = \int_{\Omega_k} Q_T - v\sum_{i\in\mathcal{S}}\int_{\Omega_k} c_{p,i}\frac{\partial T\mu_i}{\partial z}$$

intergrating the right hand side, wee see that the density becomes the mass in tha volume, and the rest will be the average of that volume:

$$\int_{\Omega_k} \rho c_p\frac{\partial T}{\partial t} \approx m_k c_p\frac{\partial T}{\partial t}(z_k)$$

the source term is treated as for the other equations, $\int_{\Omega_k} Q_T = Q_T(z_k)A_{int}\Delta z$, and for the flux term one species suffices (the others follow analogously):

$$\int_{\Omega_k} c_{p,i}\frac{\partial T\mu_i}{\partial z} \approx c_{p,i}\Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1})\Big)A_{int}$$

The final expression is:

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T(z_k) A_{int} \Delta z - \sum_{i \in \mathcal{S}} v_i c_{p,i} \Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1}) \Big) A_{int}$$

A4 Relavant Gradients for numerical solvers

A4.1 Molar balance

$$\text{Jacobian}_{\mu}[x] = \begin{bmatrix} \frac{1}{\mu^f} & -\frac{1}{\mu^f} & 0 \\ -\frac{1}{\mu^f} & \frac{1}{\mu^f} & 0 \\ 0 & 0 & \frac{\mu_{in,\mathrm{N}_2}}{(\mu_{in,\mathrm{N}_2}+\mu_{\mathrm{O}_2})} \end{bmatrix}$$

Then

$$\nabla_{\mu} f(x) = \left(\text{Jacobian}_{\mu}[x]\right)^{\top} \nabla_x f$$

A4.1.1 Nernst Equation

A4.1.1.1 Gradient for molar fractions: ∇_x

$$\nabla_x \eta_{nernst} = \frac{RT}{2F} \begin{bmatrix} -\frac{1}{x_{\mathrm{H_2O}}} \\ \frac{1}{x_{\mathrm{H}_2}} \\ \frac{1}{2x_{\mathrm{O}_2}} \end{bmatrix}$$

A4.1.1.2 Gradient for temperature: ∇_T

$$\nabla_T \eta_{nernst} = \frac{\nabla_T(\Delta G)}{2F} + \frac{R}{2F} \log \left\{ \frac{x_{\mathrm{H}_2}\sqrt{x_{\mathrm{H}_2}}}{x_{\mathrm{H_2O}}\sqrt{x_0}} \right\}$$

$$\nabla_T(\Delta G) = -0.0062T + -49.119$$

A4.1.2 Activation overpotential

A4.1.2.1 Gradient for molar fractions: ∇_x

$$\nabla_x(\text{Butler-Volmer}_{\zeta}) = \nabla_x(i_0^{\zeta})\Big(\exp\left[\alpha_{\zeta} \tfrac{n_e F}{RT} \eta_{act}^{\zeta}\right] - \exp\left[-(1-\alpha_{\zeta}) \tfrac{n_e F}{RT} \eta_{act}^{\zeta}\right]\Big)$$

$$\nabla_x(i_0^a) = \gamma_a T \exp\left\{-\frac{E_{act}^a}{RT}\right\} \begin{bmatrix} 0 \\ 0 \\ m_{\mathrm{O_2}}\left(\frac{x_{\mathrm{O_2}}}{x_0}\right)^{m_{\mathrm{O_2}}-1} \end{bmatrix}$$

$$\nabla_x(i_0^f) = \lambda_f \gamma_f T \exp\left\{-\frac{E_{act}^f}{RT}\right\} \begin{bmatrix} m_{\mathrm{H_2}}\left(\frac{x_{\mathrm{H_2O}}}{x_0}\right)^{m_{\mathrm{H_2O}}-1}\left(\frac{x_{\mathrm{H_2}}}{x_0}\right)^{m_{\mathrm{H_2}}} \\ m_{\mathrm{H_2O}}\left(\frac{x_{\mathrm{H_2O}}}{x_0}\right)^{m_{\mathrm{H_2O}}}\left(\frac{x_{\mathrm{H_2}}}{x_0}\right)^{m_{\mathrm{H_2}}-1} \\ 0 \end{bmatrix}$$

A4.1.2.2 Gradient for temperature: ∇_T

$$\begin{aligned}\nabla_T(\text{Butler-Volmer}_\zeta) &= \nabla_T(i_0^\zeta)\Big(\exp\big[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{act}^\zeta\big] - \exp\big[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\big]\Big) \\ &+ i_0^\zeta\Big(-\alpha_\zeta \tfrac{n_e F}{RT^2}\eta_{act}^\zeta \exp\big[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{act}^\zeta\big] - (1-\alpha_\zeta)\tfrac{n_e F}{RT^2}\eta_{act}^\zeta \exp\big[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\big]\Big)\end{aligned}$$

$$\nabla_T(i_0^\zeta) = \left(1 - \frac{E_{act}^\zeta}{RT}\right)\lambda_\zeta \gamma_\zeta \exp\left\{-\frac{E_{act}^\zeta}{RT}\right\}\prod_{i\in\mathcal{S}}\left(\frac{x_i}{x_0}\right)^{m_i}$$

A4.1.2.3 Gradient for activation overpotential: $\nabla_{\eta_{act}^\zeta}$

$$\nabla_{\eta_{act}^\zeta}(\text{Butler-Volmer}_\zeta) = i_0^\zeta\Big(\alpha_\zeta \tfrac{n_e F}{RT}\exp\big[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{act}^\zeta\big] + (1-\alpha_\zeta)\tfrac{n_e F}{RT}\exp\big[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{act}^\zeta\big]\Big)$$

A4.2 Degredation

A4.2.1 Ohmic degredation

A4.2.1.1 Gradient for molar fractions: ∇_x

$$\nabla_x R_{ohm} = \mathbf{0}$$

A4.2.1.2 Gradient for temperature: ∇_T

$$\nabla_T R_{ohm} = \left(1 - \frac{E_{a,ohm}}{RT}\right)\frac{1}{B_{ohm}}\exp\left\{\frac{E_{a,ohm}}{RT}\right\} + \sum_{d\in\mathcal{D}_{ohm}}\nabla_T R_d$$

$$\nabla_T R_{\text{Ni-mi}} = d_{\text{Ni-mi}}\frac{T - E_{\text{act,elec}}/R}{\sigma_{\text{elec,o}} T \exp\{-\frac{E_{\text{act,elec}}}{RT}\}}$$

A4.3 Temperature equations

A4.3.1 Specific heat

A4.3.1.1 Gradient for molar fractions: ∇_μ

$$\nabla_\mu C_p = \begin{bmatrix} c_{p,\mathrm{H_2O}} \\ c_{p,\mathrm{H_2}} \\ c_{p,\mathrm{O_2}} \end{bmatrix}$$

A4.3.1.2 Gradient for temperature: ∇_T

$$\nabla_T C_p = \sum_{i \in \mathcal{S}} \mu_i \nabla_T c_{p,i}$$

A4.3.2 Thermaneutral voltage

A4.3.2.1 Gradient for molar fractions: ∇_μ

$$\nabla_\mu U_{th} = \mathbf{0}$$

A4.3.2.2 Gradient for temperature: ∇_T

$$\nabla_T U_{th} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i \nabla_T h_i$$

A5 NLP Gradients

This appendix records the gradients used by the nonlinear programming (NLP) formulation discussed in Equation 5.1. The discrete chain rule for the stage model is stated first, then the partial derivatives that enter it, then the linear systems that yield sensitivities of the spatial finite-volume model and of the slow trajectory in time.

A5.1 Discrete chain rule for the stage model

Between shooting stages, the integrated objective contribution ϕ and the degradation variables d satisfy a discrete sensitivity chain. For a scalar stage index i and a control component u_k,

$$\frac{\mathrm{d}}{\mathrm{d}u_k}\phi_{i+1} = \nabla_{u_i}\dot{\phi} \cdot \nabla_{u_k} u_i + \nabla_{d_i}\dot{\phi}\,\frac{\mathrm{d}}{\mathrm{d}u_k} d_i + \frac{\mathrm{d}}{\mathrm{d}u_k}\phi_i,$$

with the analogous relation for the degradation state,

$$\frac{\mathrm{d}}{\mathrm{d}u_k} d_{i+1} = (\nabla_{u_i}\dot{d}_i)\,\nabla_{u_k} u_i + (\nabla_{d_i}\dot{d}_i)\,\frac{\mathrm{d}}{\mathrm{d}u_k} d_i + \cdots,$$

where the trailing terms collect any additional dependence of the stage map on u_k (for example through implicit stage equations if d_{i+1} is defined implicitly).

The partial gradients of the instantaneous rates are

$$\begin{aligned}
\nabla_u\dot{\phi} &= c_{\mathrm{H_2}}\nabla_u \frac{\partial n_{\mathrm{H_2}}}{\partial t} - c_P(u\nabla_u I - I),\\
\nabla_d\dot{\phi} &= c_{\mathrm{H_2}}\nabla_d \frac{\partial n_{\mathrm{H_2}}}{\partial t} - c_P u \nabla_d I - \left[c_{i,\text{utilization}}\frac{1}{d_{i,\text{critical}}}\right]_{i\in\mathcal{D}_{\text{fail}}},\\
\nabla_\phi\dot{\phi} &= 0,\\
\nabla_u\dot{d} &= c_2\nabla_u i + c_3\nabla_u T + c_4(\nabla_u i\, T + i\nabla_u T),\\
\nabla_d\dot{d} &= c_2\nabla_d i + c_3\nabla_d T + c_4(\nabla_d i\, T + i\nabla_d T),\\
\nabla_\phi\dot{d} &= 0.
\end{aligned}$$

Note that $\nabla_\zeta I = \frac{A_{\text{cell}}}{N_{\text{cell}}}\sum_{j=1}^{N_{\text{cell}}} \nabla_\zeta i_j$ for $\zeta \in \{u, d, \phi, \dots\}$.

The remaining ingredients are the gradients of the current density with respect to the cell voltage and with respect to the local degradation state; those are given in the next two sections. Further electrochemical partials are listed in Appendix A4.

A5.2 Gradient with respect to cell voltage: $\nabla_u i$

The current is expressed in terms of fuel- and air-side activation branches in the same notation as Appendix A4. With scalars $\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)$ and $\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)$ denoting $\partial i/\partial \eta^f_{\text{act}}$ and $\partial i/\partial \eta^a_{\text{act}}$ at the operating point,

$$\nabla_u i = \frac{1}{R_{\text{ohm}} + \dfrac{1}{\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)} + \dfrac{1}{\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)}}.$$

A5.3 Gradient with respect to degradation: $\nabla_d i$

Similarly, with $\nabla_d R_{\text{ohm}}$ the partial of the ohmic resistance with respect to the local degradation variable,

$$\nabla_d i = -i \frac{\nabla_d R_{\text{ohm}}}{R_{\text{ohm}} + \dfrac{1}{\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)} + \dfrac{1}{\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)}}.$$

A5.4 Sensitivity of the spatial FVM discretization

At each time level the converged spatial state satisfies a residual of flux form. With cell index k along the stack, let $Q(u_k)$ denote the cellwise source residual and $F(u_k)$ the numerical flux between neighbours. The source term is scaled by the cell thickness Δz and the flux difference by the interface area A_{int}, so the semi-discrete steady relation is

$$0 = \Delta z\, Q(u_k) - A_{\text{int}}\left(F(u_k) - F(u_{k-1})\right).$$

Differentiating with respect to a parameter p (a control coefficient, a boundary value, etc.) and writing $\partial u_k/\partial p$ for the sensitivity of the converged state gives the familiar block recursion

$$\begin{aligned}\left(\Delta z \frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}} \frac{\partial F}{\partial u_k}(u_k)\right)\frac{\partial u_k}{\partial p} + A_{\text{int}} \frac{\partial F}{\partial u_{k-1}}(u_{k-1})\frac{\partial u_{k-1}}{\partial p} \\ = A_{\text{int}}\left(\frac{\partial F}{\partial p}(u_k) - \frac{\partial F}{\partial p}(u_{k-1})\right) - \Delta z \frac{\partial Q}{\partial p}(u_k).\end{aligned}$$

Stacking all cells, this is the linear system

$$J_{\text{fvm}} \frac{\partial \mathbf{u}}{\partial p} = b^{(p)}_{\text{fvm}},$$

with the block-lower-bidiagonal Jacobian and right-hand side

$$J_{\text{fvm}} = \begin{bmatrix} \Delta_1 & & & \\ A_{\text{int}} \dfrac{\partial F}{\partial u_1}(u_1) & \Delta_2 & & \\ & \ddots & \ddots & \\ & & A_{\text{int}} \dfrac{\partial F}{\partial u_{N-1}}(u_{N-1}) & \Delta_N \end{bmatrix},$$

$$\text{with } \Delta_k = \Delta z \frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}} \frac{\partial F}{\partial u_k}(u_k),$$

$$b_{\text{fvm}}^{(p)} = \begin{bmatrix} A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_1) - \frac{\partial F}{\partial p}(u_{\text{in}}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_1) \\ A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_2) - \frac{\partial F}{\partial p}(u_1) \right) - \Delta z \frac{\partial Q}{\partial p}(u_2) \\ \vdots \\ A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_N) - \frac{\partial F}{\partial p}(u_{N-1}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_N) \end{bmatrix}$$

The same block structure matches the flux assembly used in Appendix A3.

A5.5 Sensitivity of the ODE system.

For this section the problem is written in general form:

$$\begin{aligned} \max_u \Big\{ &\text{Objective} = \int_0^T \Phi(X, d, u)\, dt + \Psi(X, d, u) \Big\} \\ \text{s.t.} \quad & 0 = R_{\text{fvm}}(X, d, u) \\ & \dot{d} = f(X, d, u) \end{aligned}$$

The method of adjoint gradients [16], [8] is used, with the following discretization ($t_0 = 0 < t_1 \cdots t_i \cdots t_N = T$):

$$X(t_i) \approx X_i, \quad d(t_i) \approx d_i, \quad u(t_i) \approx u_i.$$

$$\begin{aligned} R_{\text{fvm},i} &= R_{\text{fvm}}(X_{i+1}, d_{i+1}, u_i), \\ D_i &= d_{i+1} - f(X_{i+1}, d_{i+1}, u_i)\Delta t - d_i \\ \Phi_i &= \Phi(X_{i+1}, d_{i+1}, u_i)\Delta t \approx \int_{t_i}^{t_{i+1}} \Phi(X, d, u) dt \end{aligned}$$

The combined residual for the FVM constraints and the ODE defect is then:

$$R_i = \begin{bmatrix} D_i \\ R_{\text{fvm},i} \end{bmatrix}$$

The state variables are combined as

$$w_i = \begin{bmatrix} X_i \\ d_i \end{bmatrix}$$

The Jacobian of the residual with respect to the internal and control variables is

$$\frac{\partial R_i}{\partial w_i} = \begin{bmatrix} \mathbf{0} & \frac{\partial D_i}{\partial d_i} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \frac{\partial R_i}{\partial w_{i+1}} = \begin{bmatrix} \frac{\partial D_i}{\partial X_{i+1}} & \frac{\partial D_i}{\partial d_{i+1}} \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}} \end{bmatrix} \quad \frac{\partial R_i}{\partial u_i} = \begin{bmatrix} \frac{\partial D_i}{\partial u_i} \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i} \end{bmatrix}$$

$$\frac{\partial \Phi_i}{\partial w_{i+1}} = \begin{bmatrix} \frac{\partial \Phi_i}{\partial X_{i+1}} \\ \frac{\partial \Phi_i}{\partial d_{i+1}} \end{bmatrix} \quad \frac{\partial \Phi_i}{\partial u_i}$$

For implementation these blocks are spelled out in more detail:

$$\frac{\partial R_i}{\partial w_i} = \begin{bmatrix} \mathbf{0} & -\mathbf{I} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$$\frac{\partial R_i}{\partial w_{i+1}} = \begin{bmatrix} -\Delta t \frac{\partial f}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \mathbf{I} - \Delta t \frac{\partial f}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix}$$

$$\frac{\partial R_i}{\partial u_i} = \begin{bmatrix} -\Delta t \frac{\partial f}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix}$$

The blocks $\partial R_{\text{fvm},i}/\partial X_{i+1}$ coincide with J_{fvm} from the spatial sensitivity section (same flux assembly). At each time level:

$$\frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} = J_{\text{fvm}} \quad \frac{\partial R_{\text{fvm},i}}{\partial u_i} = \begin{bmatrix} | & | \\ b_{\text{fvm}}^{(u_1^{(U)})} & b_{\text{fvm}}^{(u_i^{(T_0)})} \\ | & | \end{bmatrix} \quad \frac{\partial R_{\text{fvm},i}}{\partial d_i} = b_{\text{fvm}}^{(d)}$$

With these blocks, the gradient of the objective with respect to u_i is.

$$\nabla_{u_i} \text{Objective} = \frac{\partial \Phi_i}{\partial u_i} + \left(\frac{\partial R_i}{\partial u_i} \right)^\top \lambda_{i+1} \quad \text{for} \quad i = 1, \dots, N$$

The adjoint variables $\Lambda = (\lambda_k)_k$ are found by solving the system:

$$\left(\frac{\partial R_{N-1}}{\partial w_N} \right)^\top \lambda_N = -\frac{\partial \Phi_{N-1}}{\partial w_N} - \frac{\partial \Psi}{\partial w_N},$$

$$\left(\frac{\partial R_{i-1}}{\partial w_i} \right)^\top \lambda_i + \left(\frac{\partial R_i}{\partial w_i} \right)^\top \lambda_{i+1} = -\frac{\partial \Phi_{i-1}}{\partial w_i}, \qquad i = 1, \dots, N-1.$$

Equivalently, one large block system can be assembled for Λ.

$$\begin{bmatrix} \frac{\partial R_0}{\partial w_1} & & & & \\ \frac{\partial R_1}{\partial w_1} & \frac{\partial R_1}{\partial w_2} & & & \\ & \ddots & \ddots & & \\ & & & \frac{\partial R_{N-2}}{\partial w_{N-1}} & \\ & & & \frac{\partial R_{N-1}}{\partial w_{N-1}} & \frac{\partial R_{N-1}}{\partial w_N} \end{bmatrix}^\top \Lambda = - \begin{bmatrix} \frac{\partial \Phi_0}{\partial w_1} \\ \frac{\partial \Phi_1}{\partial w_2} \\ \vdots \\ \frac{\partial \Phi_{N-2}}{\partial w_{N-1}} \\ \frac{\partial \Phi_{N-1}}{\partial w_N} + \frac{\partial \Psi}{\partial w_N} \end{bmatrix}$$

A6 Squared continuity penalty

Piecewise-constant controls on N_t segments can introduce artificial jumps between adjacent intervals. When both voltage and inlet temperature are optimized segment-wise, the optimizer may exploit those jumps unless they are discouraged. The implementation adds an optional **soft penalty** to the running objective, proportional to the sum of squared differences between neighbouring segment values.

For segment index $k = 0, \dots, N_t - 2$, let U_k and $T_{\mathrm{in},k}$ denote the piecewise-constant controls on segment k. With non-negative weights w_U and w_T (CLI: `--continuity-weight-u`, `--continuity-weight-T`), the penalty is

$$P_{\mathrm{cont}} = w_U \sum_{k=0}^{N_t-2} (U_{k+1} - U_k)^2 + w_T \sum_{k=0}^{N_t-2} (T_{\mathrm{in},k+1} - T_{\mathrm{in},k})^2.$$

It is added to the economic running cost before integration over the horizon. Setting $w_U = w_T = 0$ disables the term (the default in Table 6.1).

When the stack is operated **potentiostatically** ($U_k \equiv U$), the voltage is a single decision variable rather than a length-N_t profile, so every adjacent voltage difference vanishes and the w_U term contributes nothing regardless of its numerical value. Only the inlet-temperature sum is active in that mode.

The penalty is smooth and cheap to evaluate; its gradient with respect to the control knots is implemented analytically in the NLP (`code/src/scripts/nlp/continuity.py`). Reported decompositions P_{cont}, P_U, and $P_{T_{\mathrm{in}}}$ in saved run metadata refer to the two sums in Appendix A6.

A7 Gradient verification plots

This appendix collects all current diagnostic figures generated in `code/figures/testers/gradients_test/` (42 plots at the time of writing). Most panels compare analytical derivatives to finite-difference checks from the standalone gradient test scripts.

A7.1 Complete current plot set

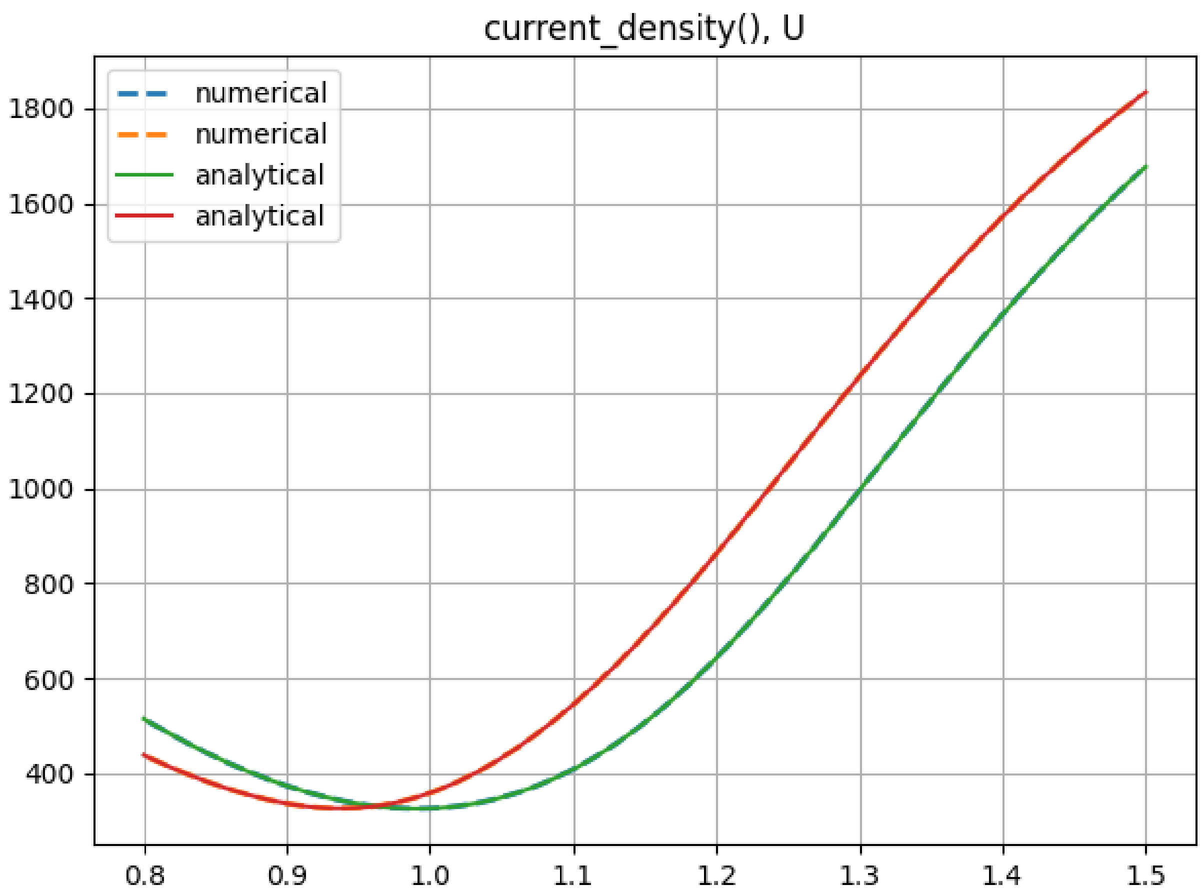


Figure A7.1: Gradient verification output: `current_density_U.png`.

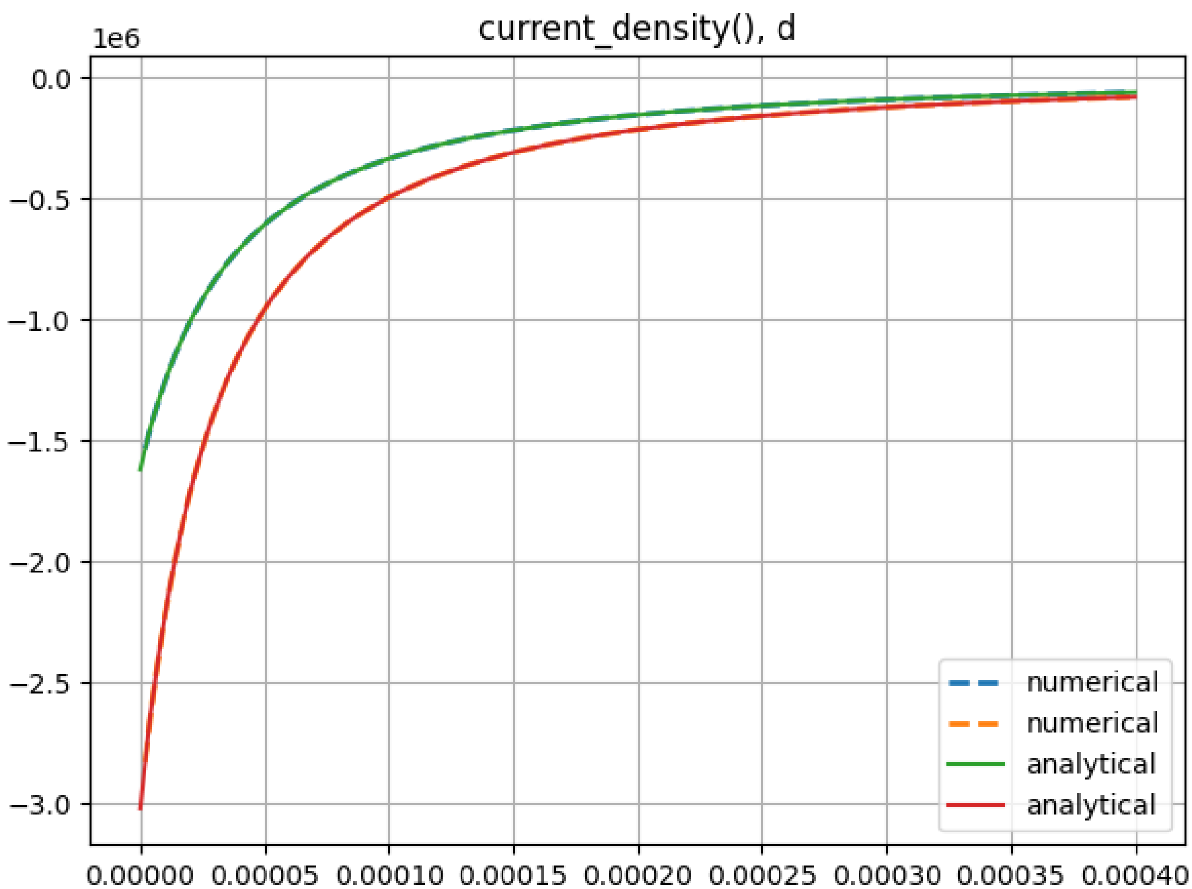


Figure A7.2: Gradient verification output: `current_density_d.png`.

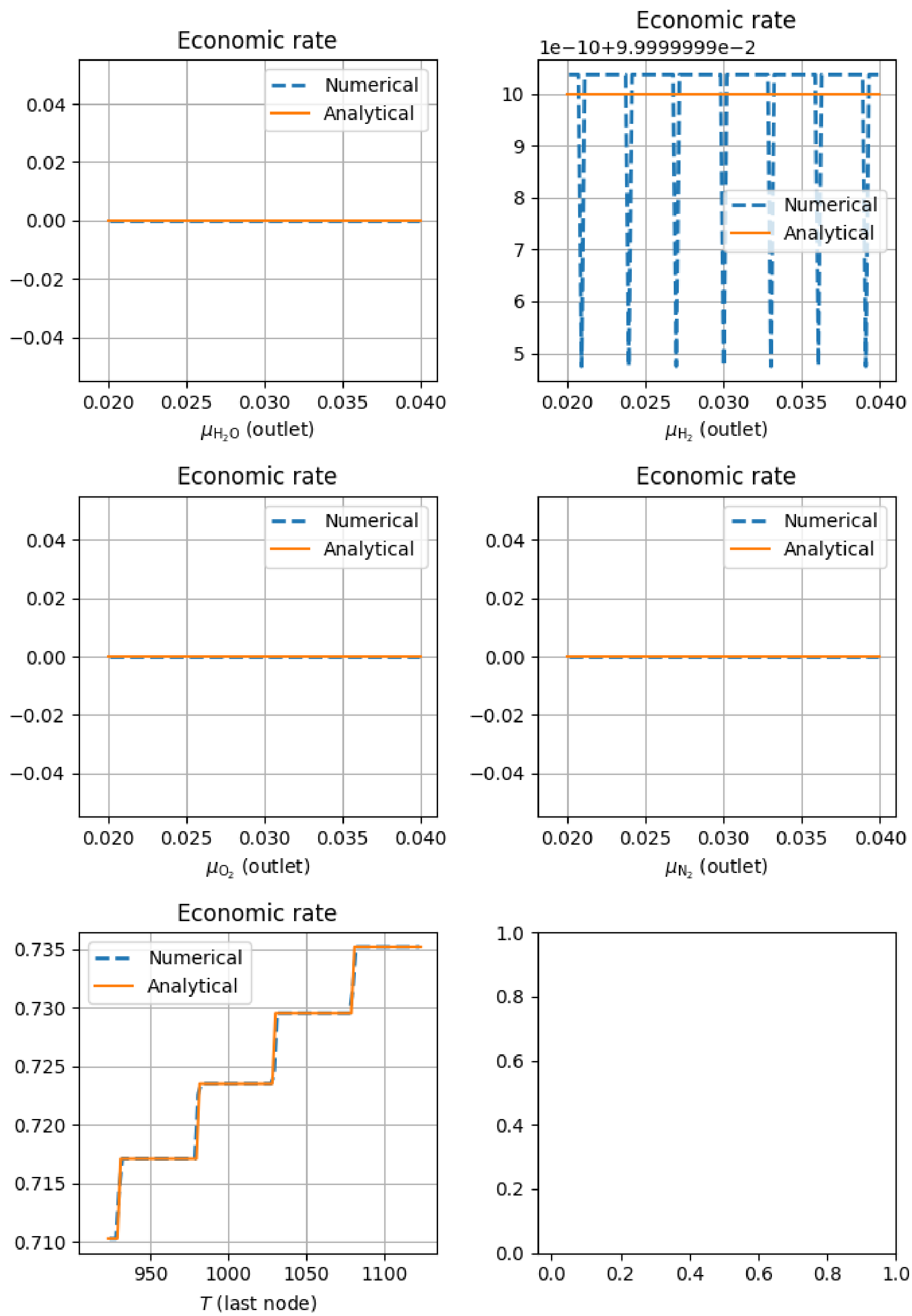


Figure A7.3: Gradient verification output: `economic_rate.png`.

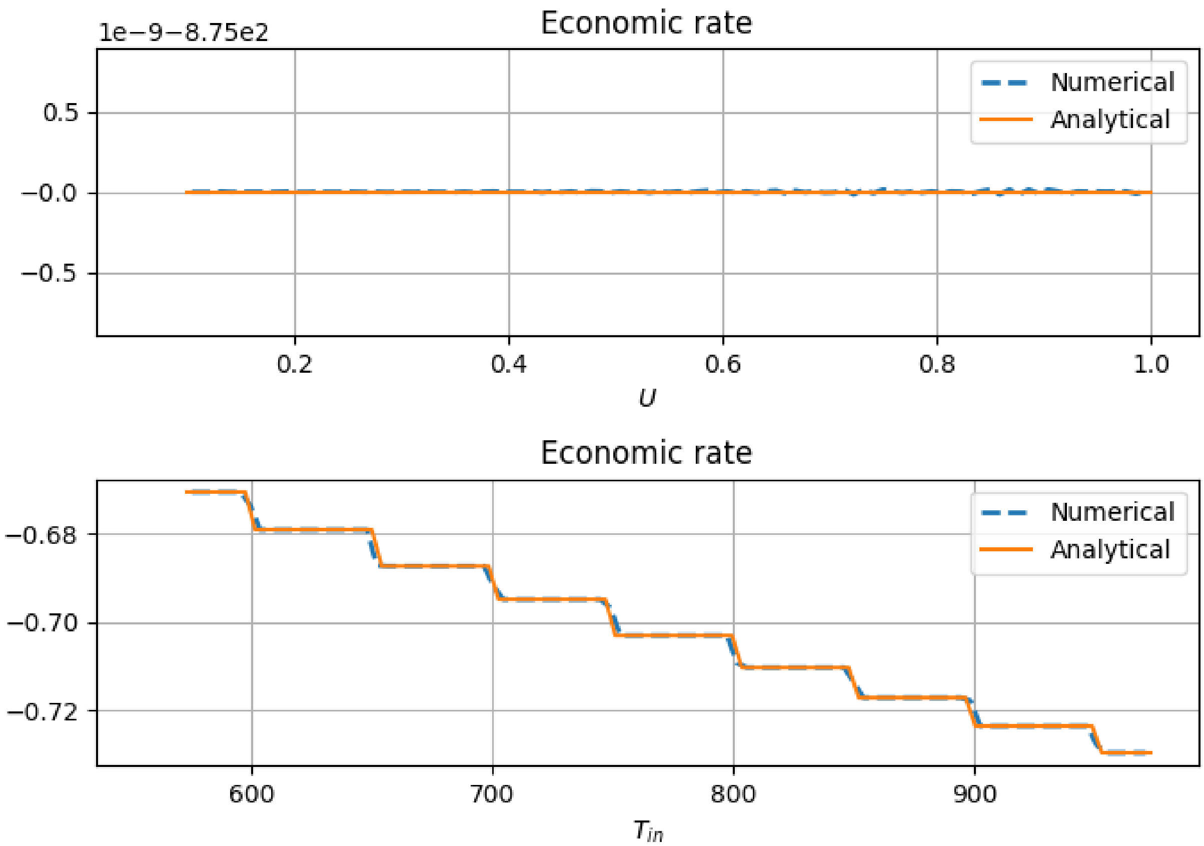


Figure A7.4: Gradient verification output: `economic_rate_dp.png`.

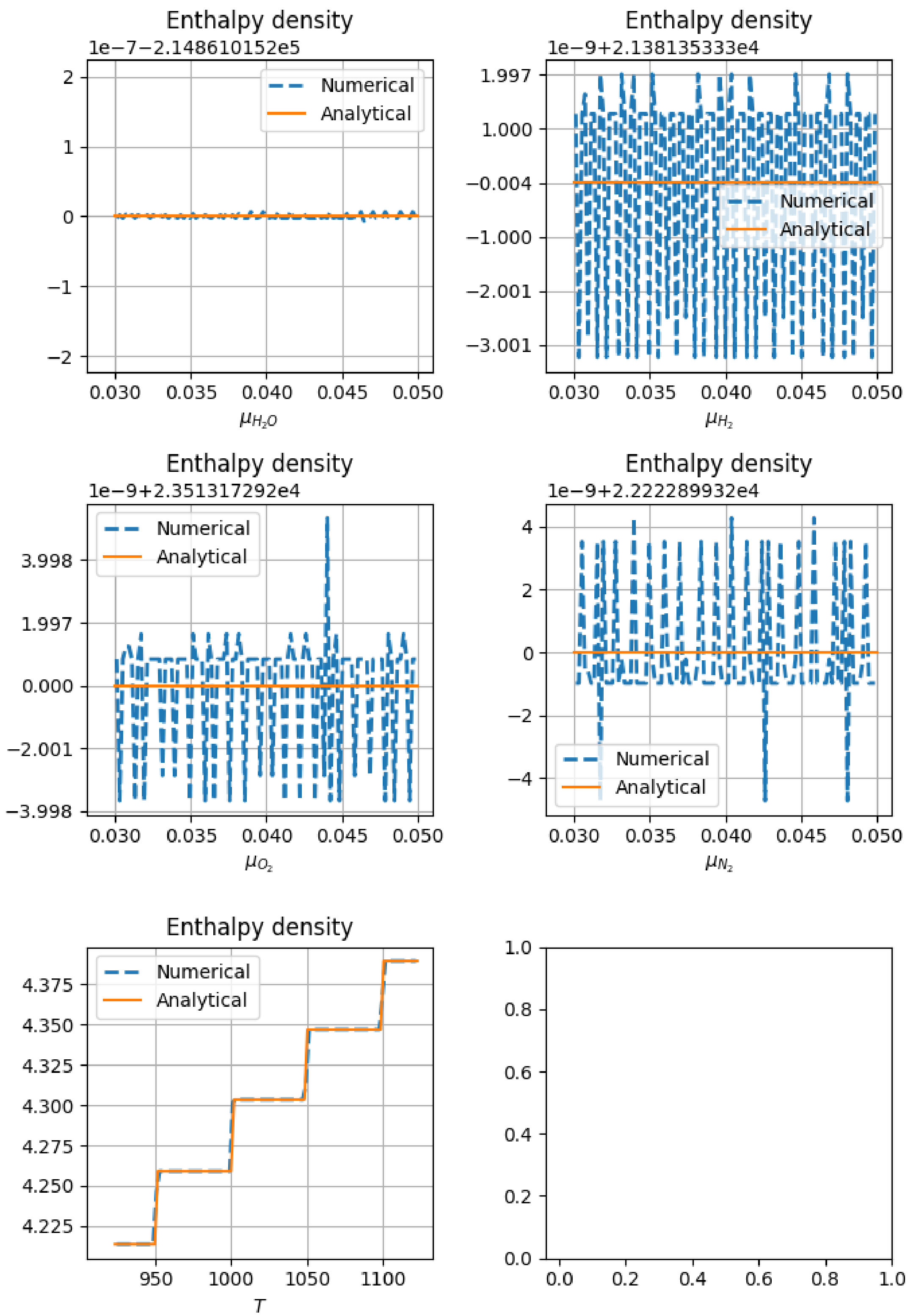


Figure A7.5: Gradient verification output: `enthalpy_density.png`.

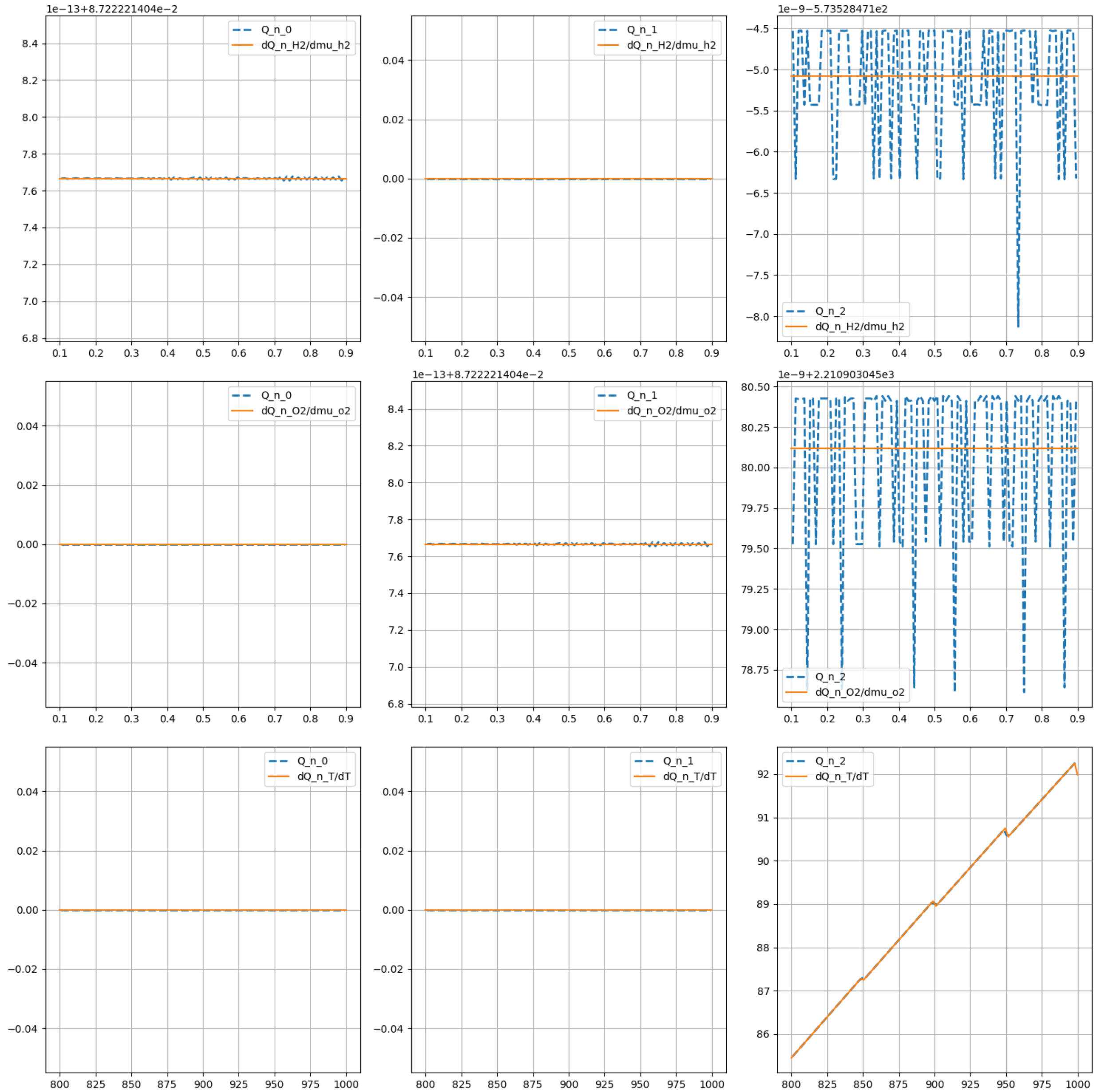


Figure A7.6: Gradient verification output: `flux_term_jacobian.png`.

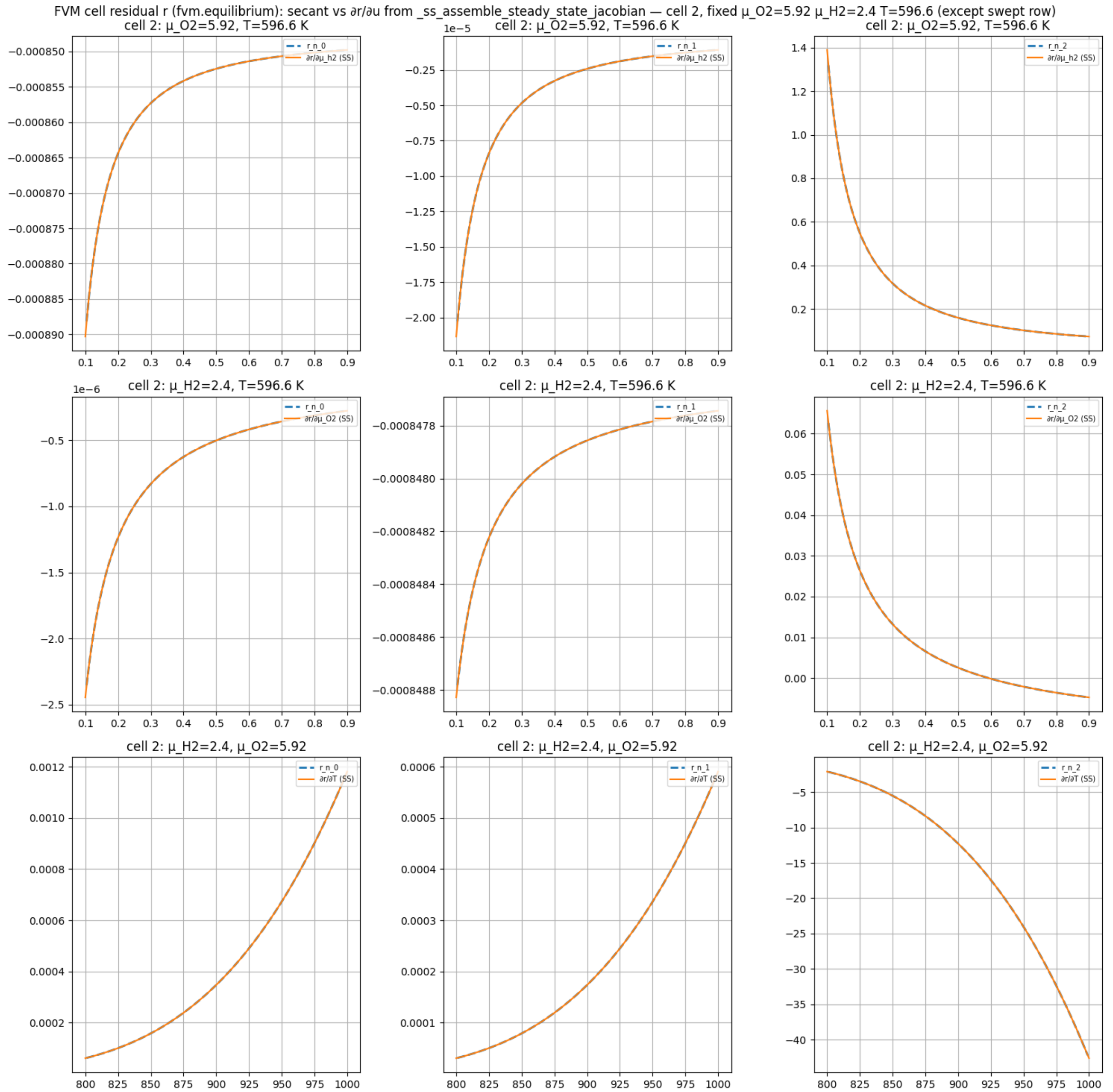


Figure A7.7: Gradient verification output: `fvm_residual_ss_jacobian_vs_fd.png`.

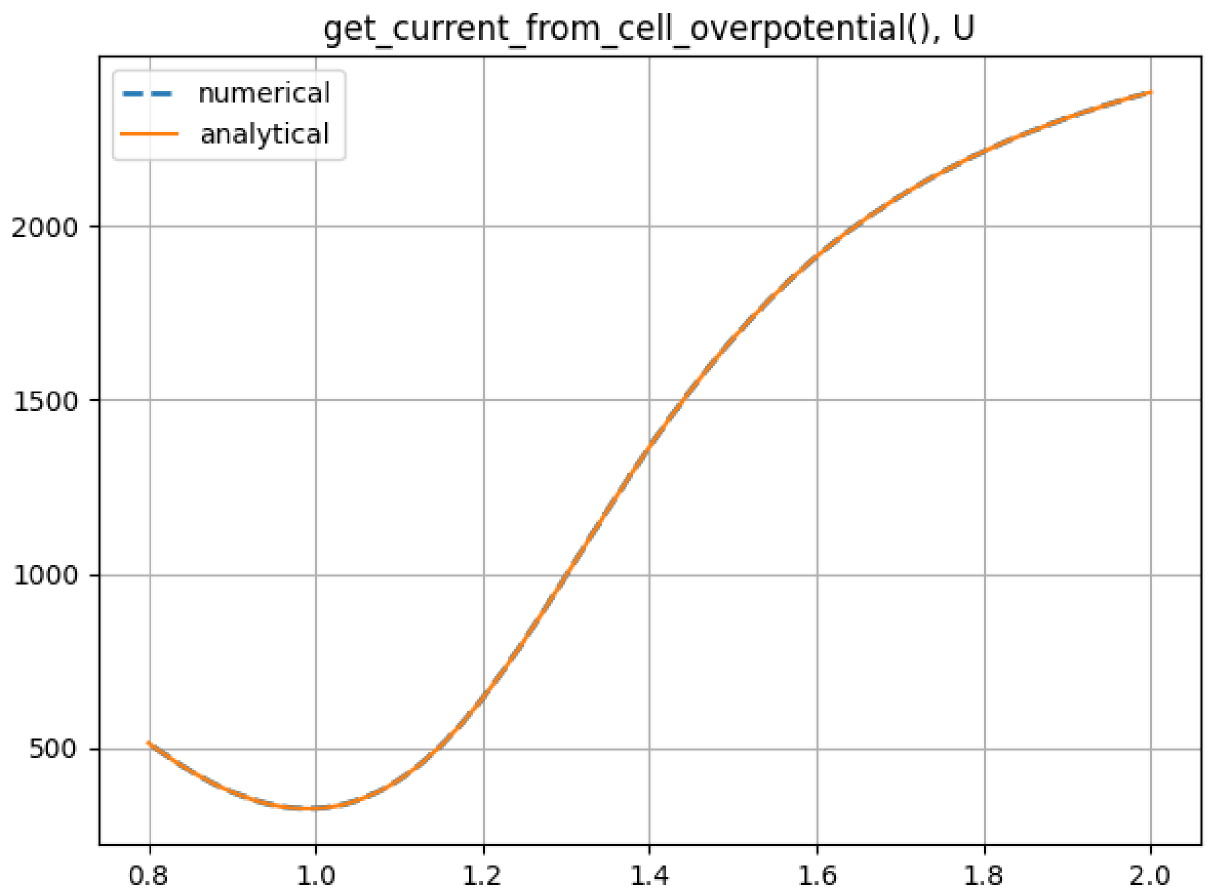


Figure A7.8: Gradient verification output: `get_current_from_cell_overpotential_U.png`.

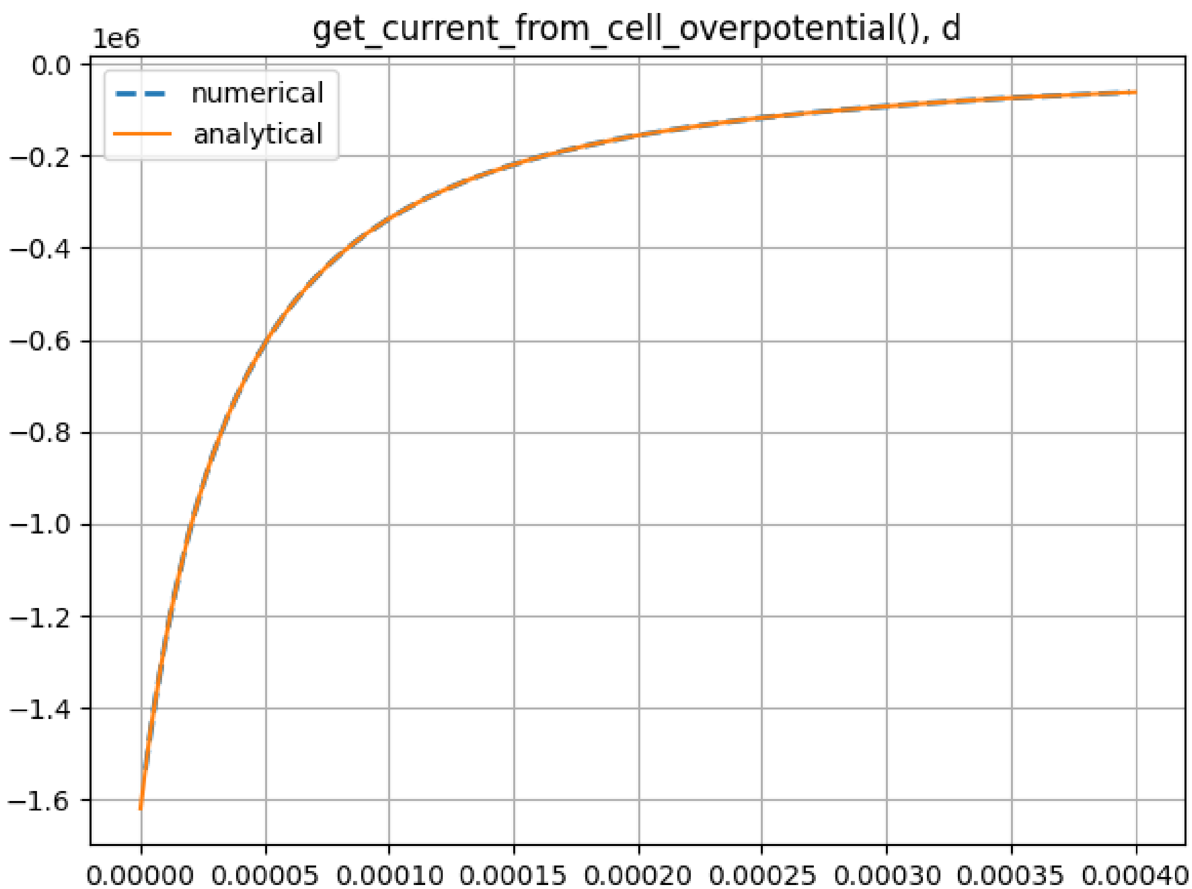


Figure A7.9: Gradient verification output: `get_current_from_cell_overpotential_d.png`.

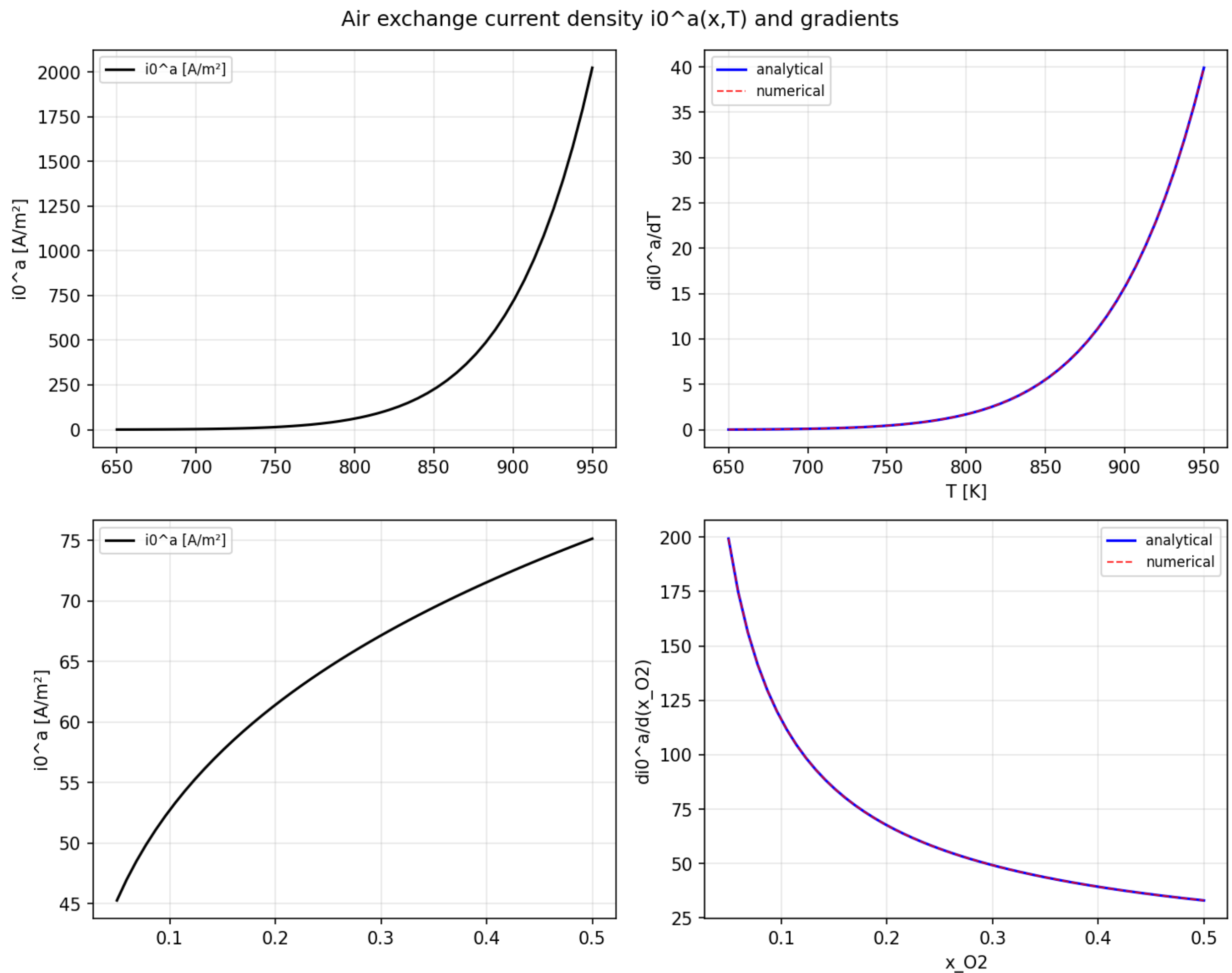


Figure A7.10: Gradient verification output: `gradient_test_air_i0.png`.

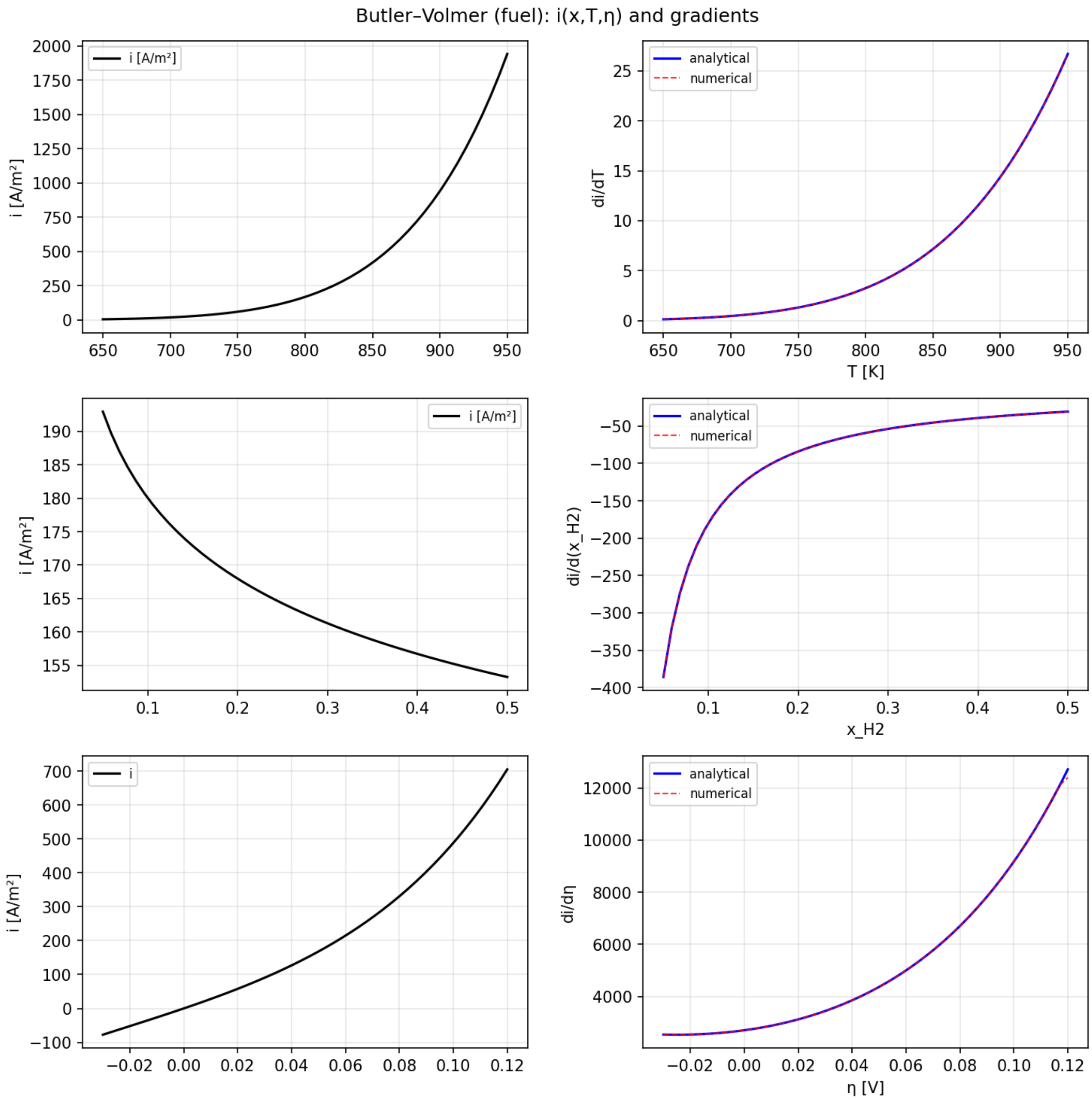


Figure A7.11: Gradient verification output: `gradient_test_butler_volmer.png`.

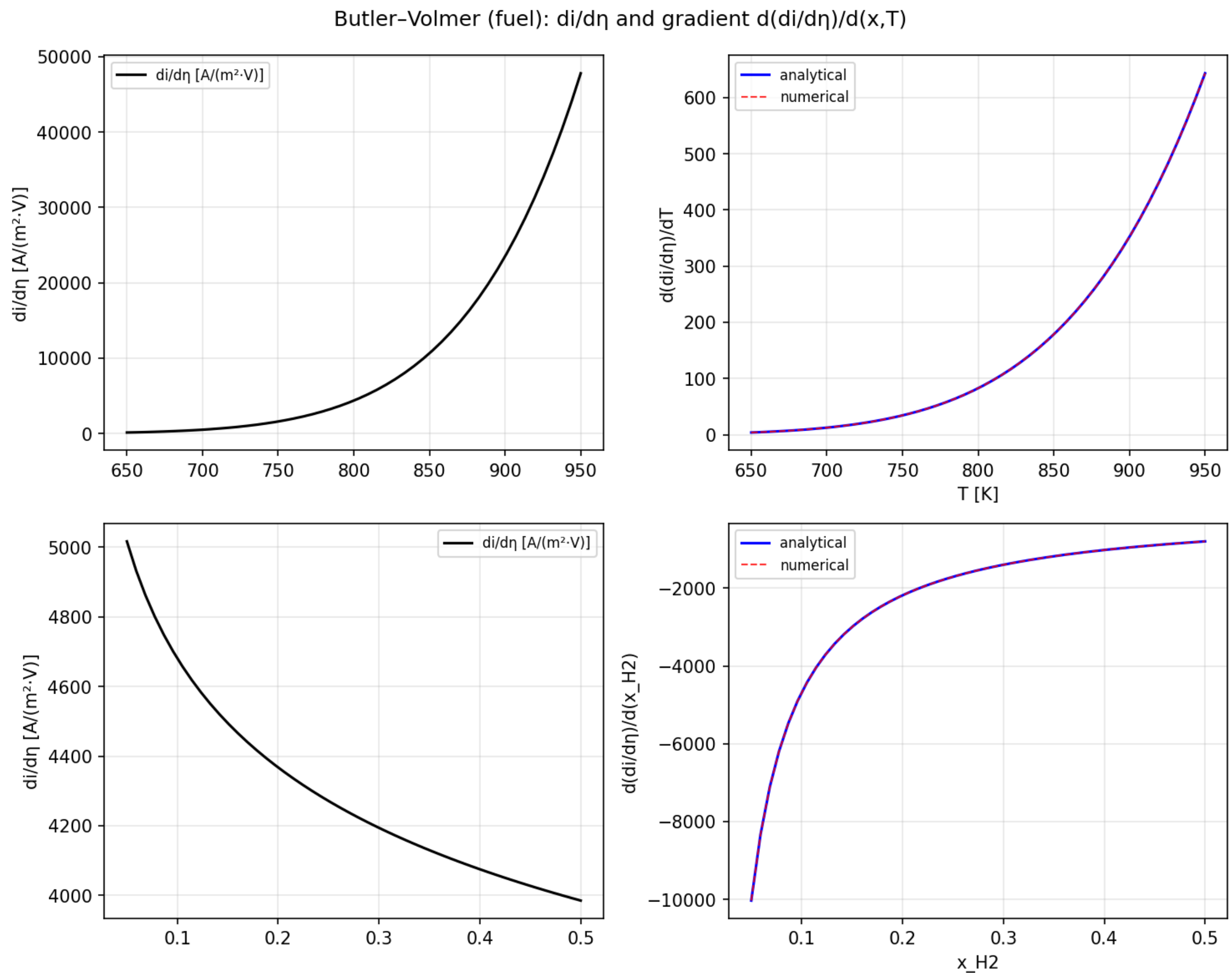


Figure A7.12: Gradient verification output: `gradient_test_butler_volmer_deta_gradient.png`.

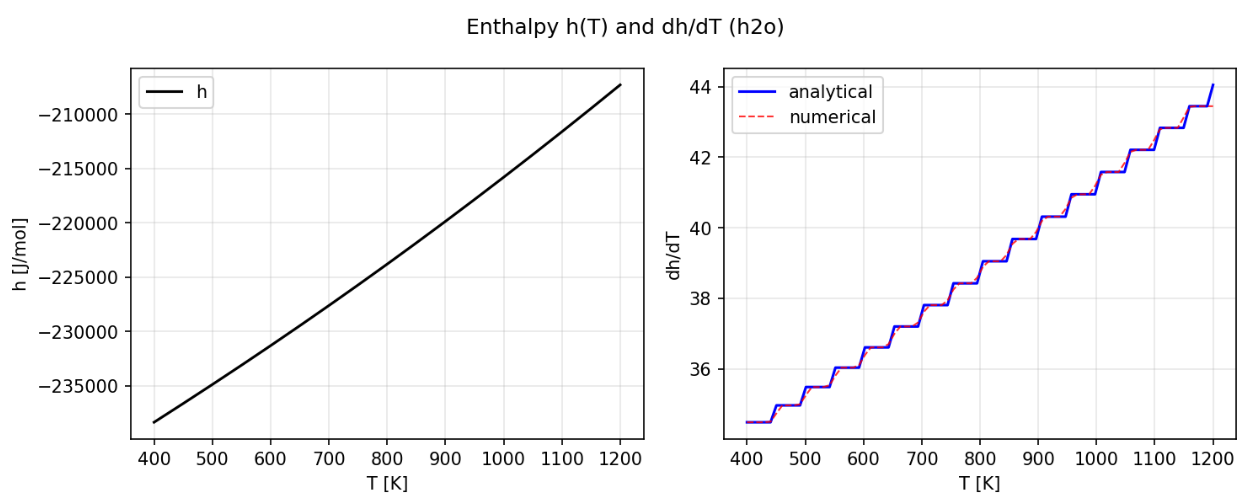


Figure A7.13: Gradient verification output: `gradient_test_enthalpy.png`.

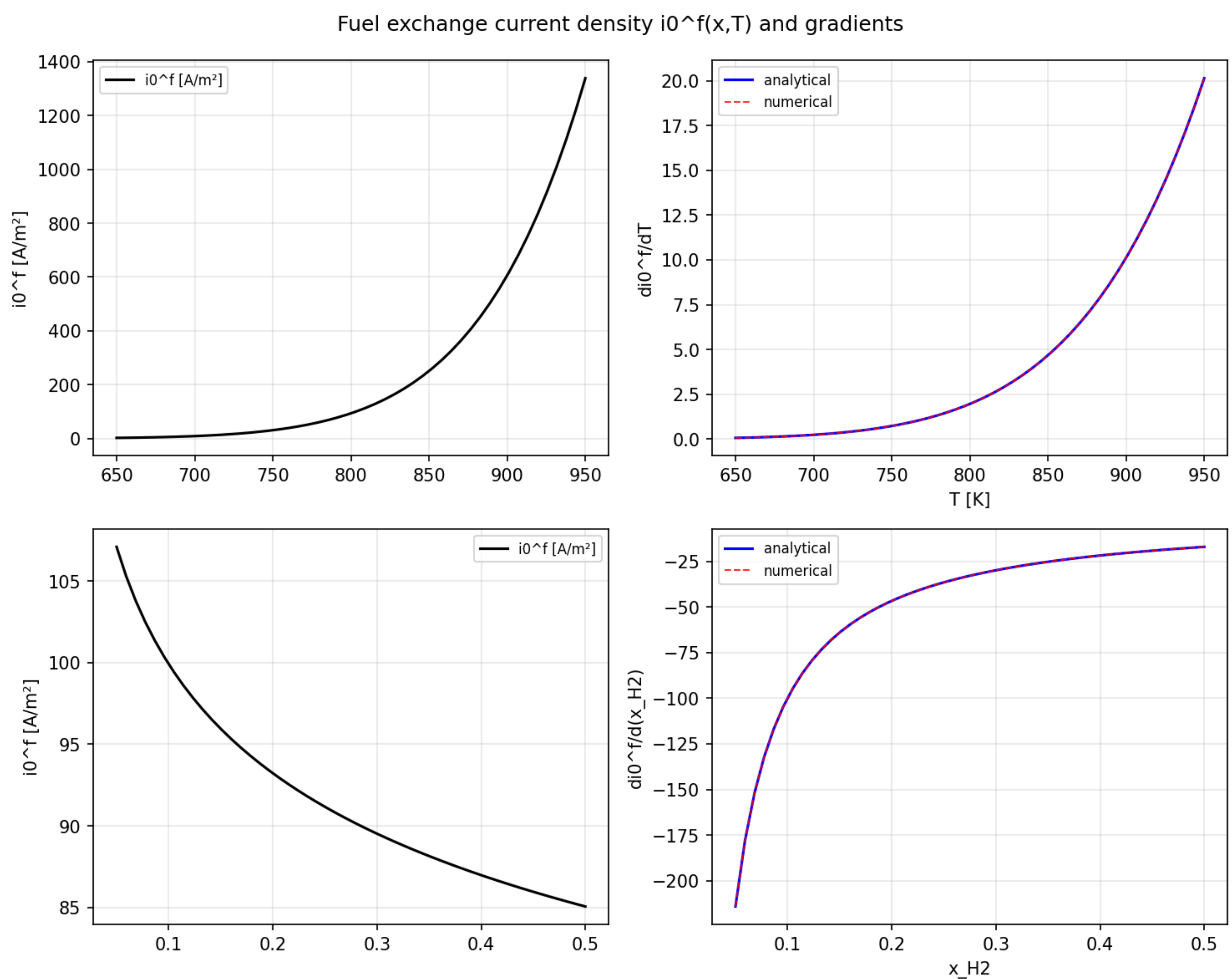


Figure A7.14: Gradient verification output: `gradient_test_fuel_i0.png`.

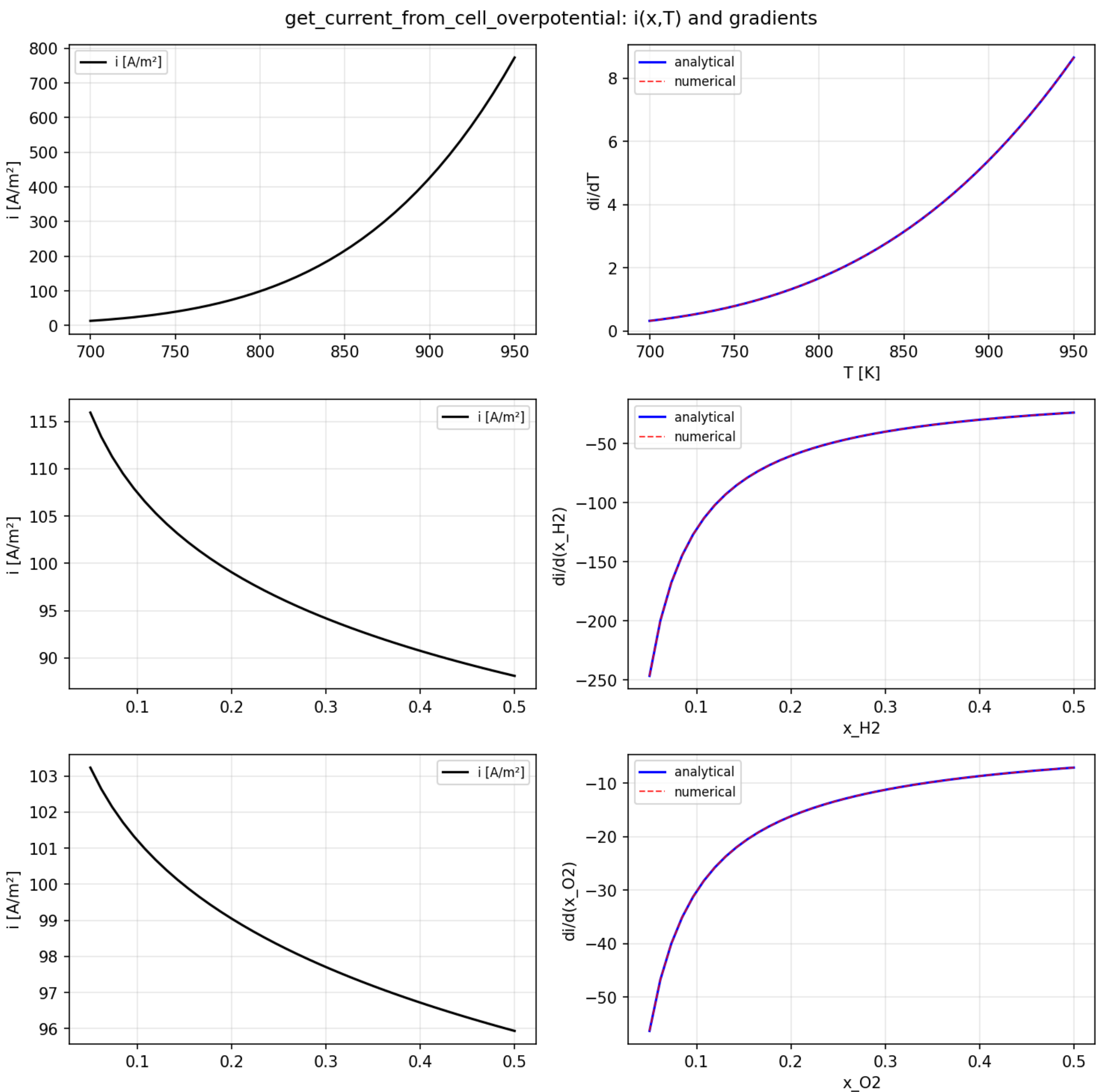


Figure A7.15: Gradient verification output: `gradient_test_get_current.png`.

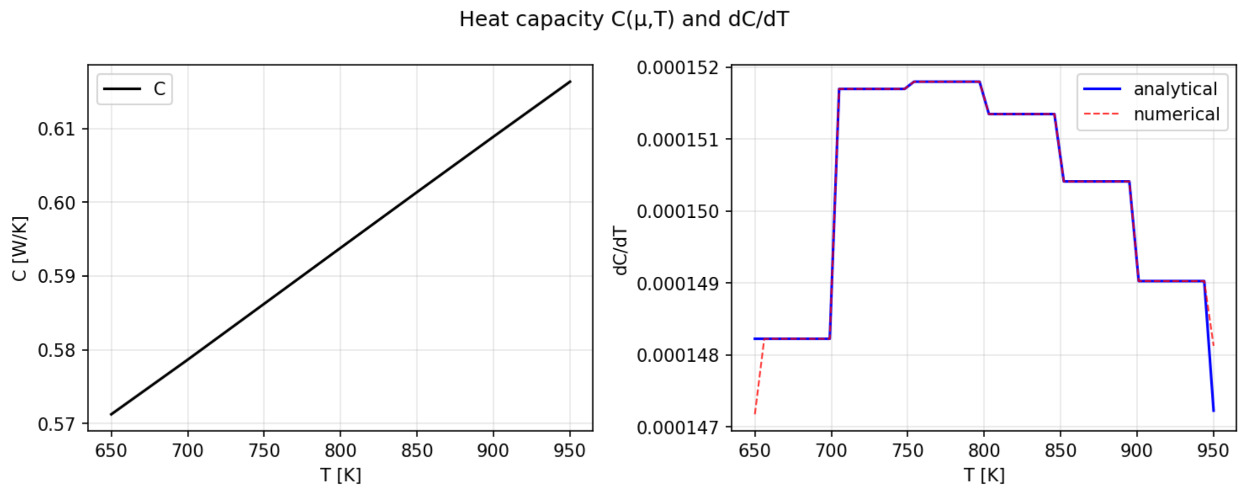


Figure A7.16: Gradient verification output: `gradient_test_heat_capacity.png`.

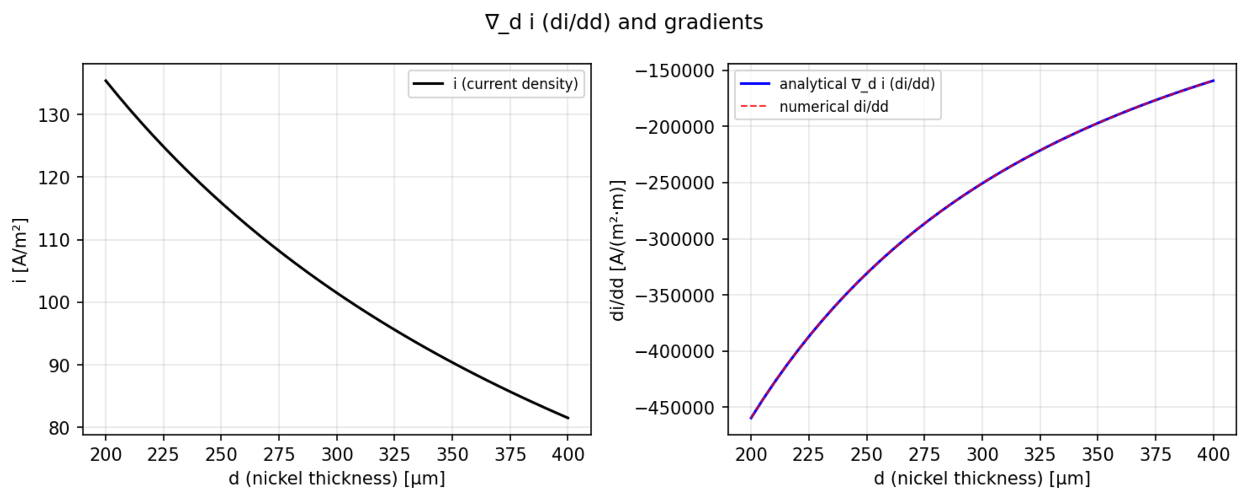


Figure A7.17: Gradient verification output: `gradient_test_nabla_d.png`.

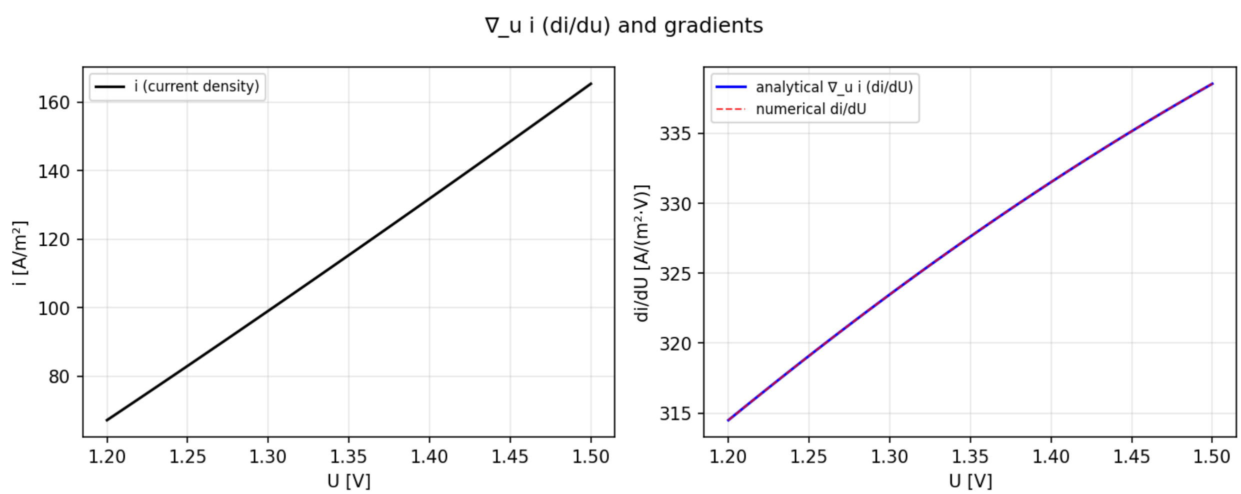


Figure A7.18: Gradient verification output: `gradient_test_nabla_u.png`.

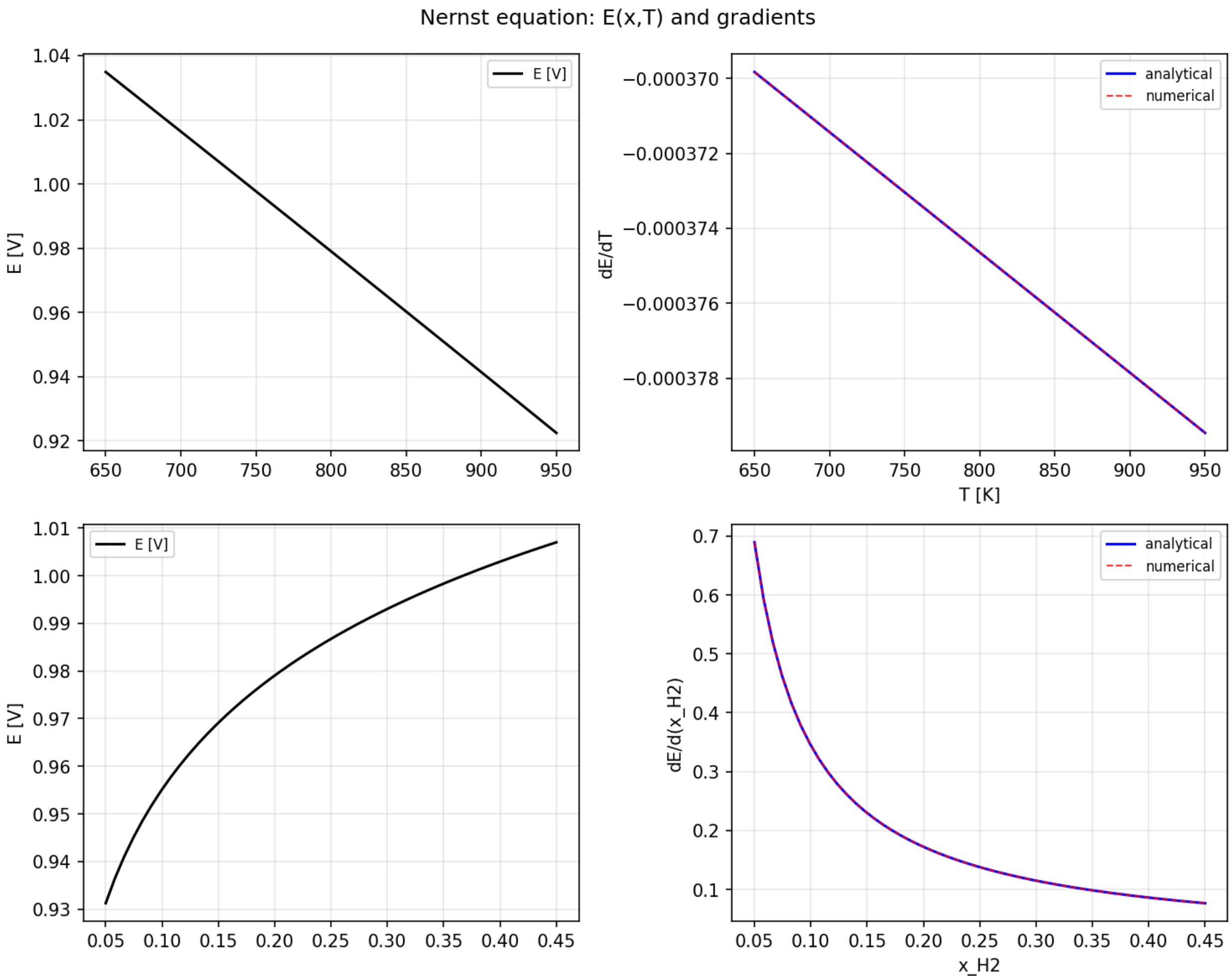


Figure A7.19: Gradient verification output: `gradient_test_nernst.png`.

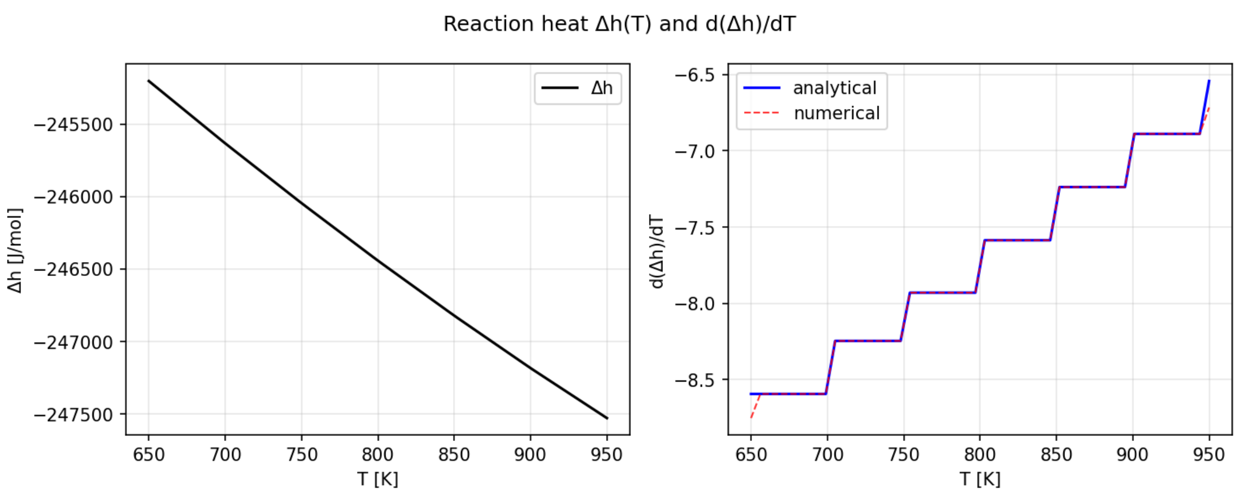


Figure A7.20: Gradient verification output: `gradient_test_reaction_heat.png`.

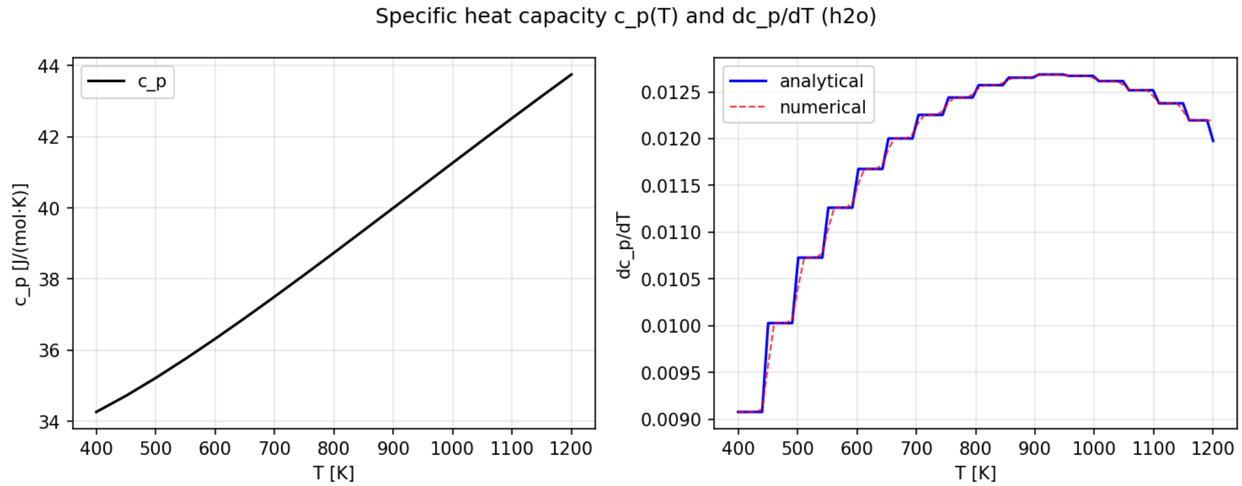


Figure A7.21: Gradient verification output: `gradient_test_specific_heat_capacity.png`.

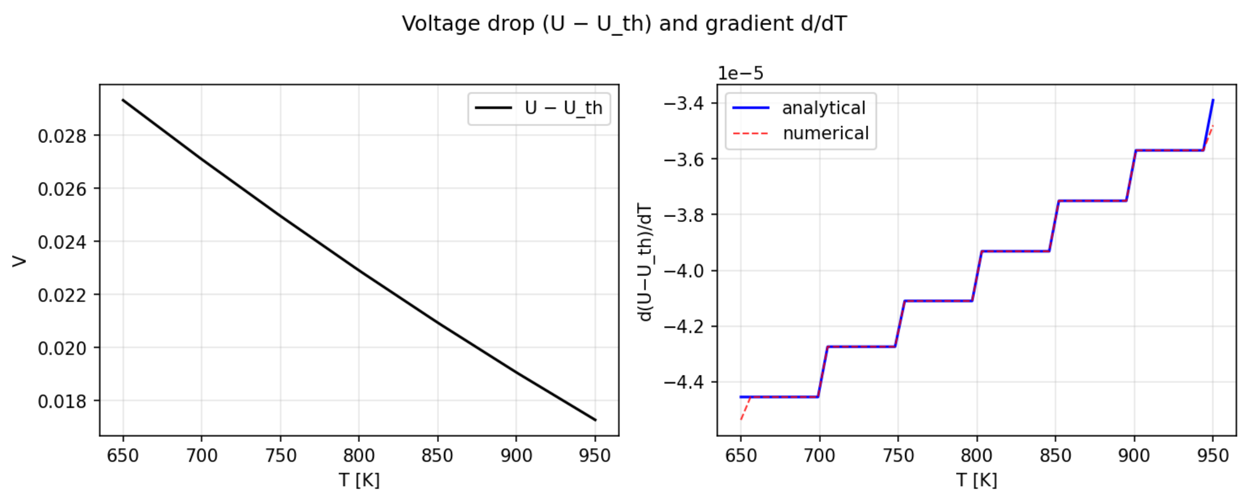


Figure A7.22: Gradient verification output: `gradient_test_voltage_drop_gradient.png`.

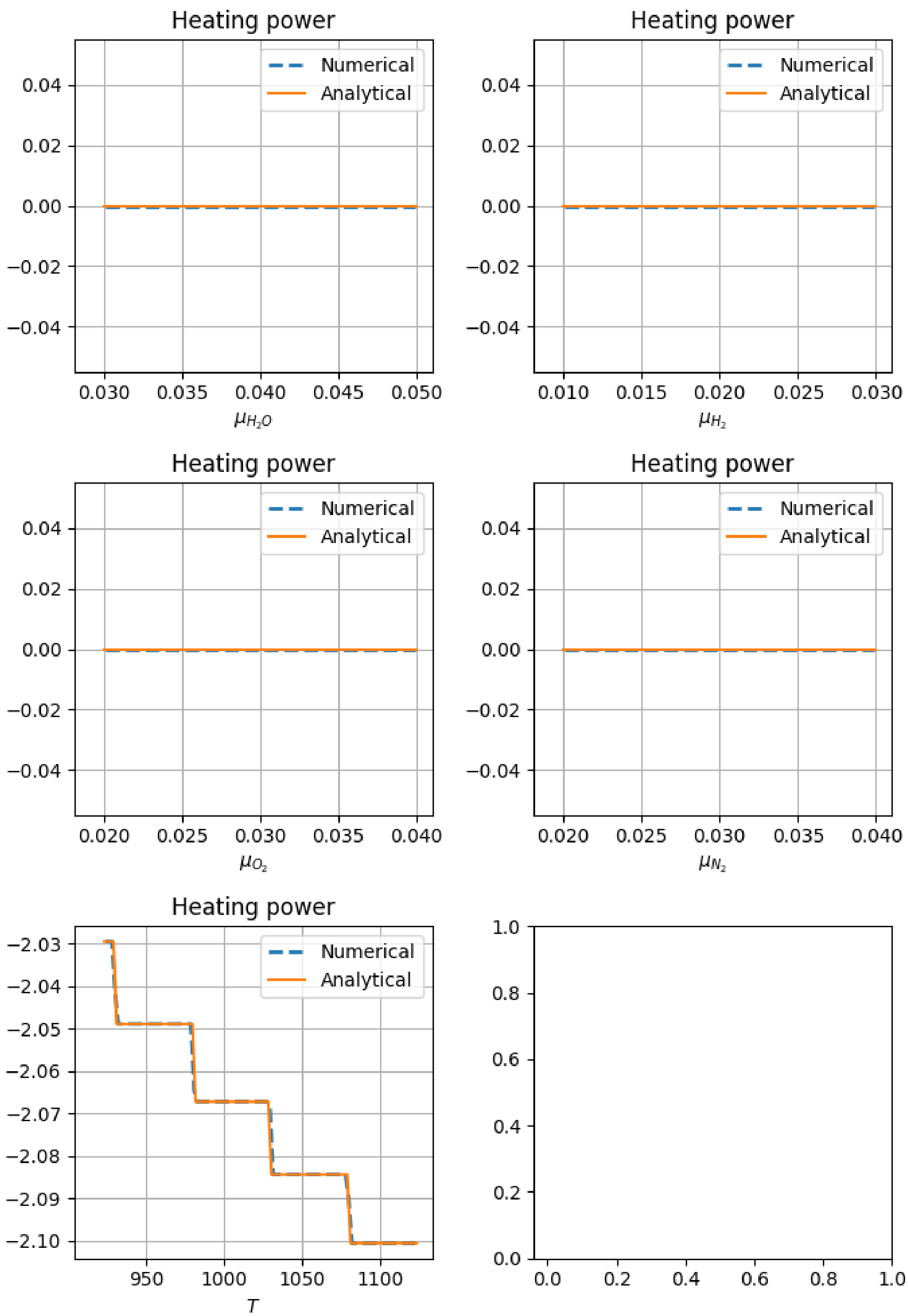


Figure A7.23: Gradient verification output: `heating_power.png`.

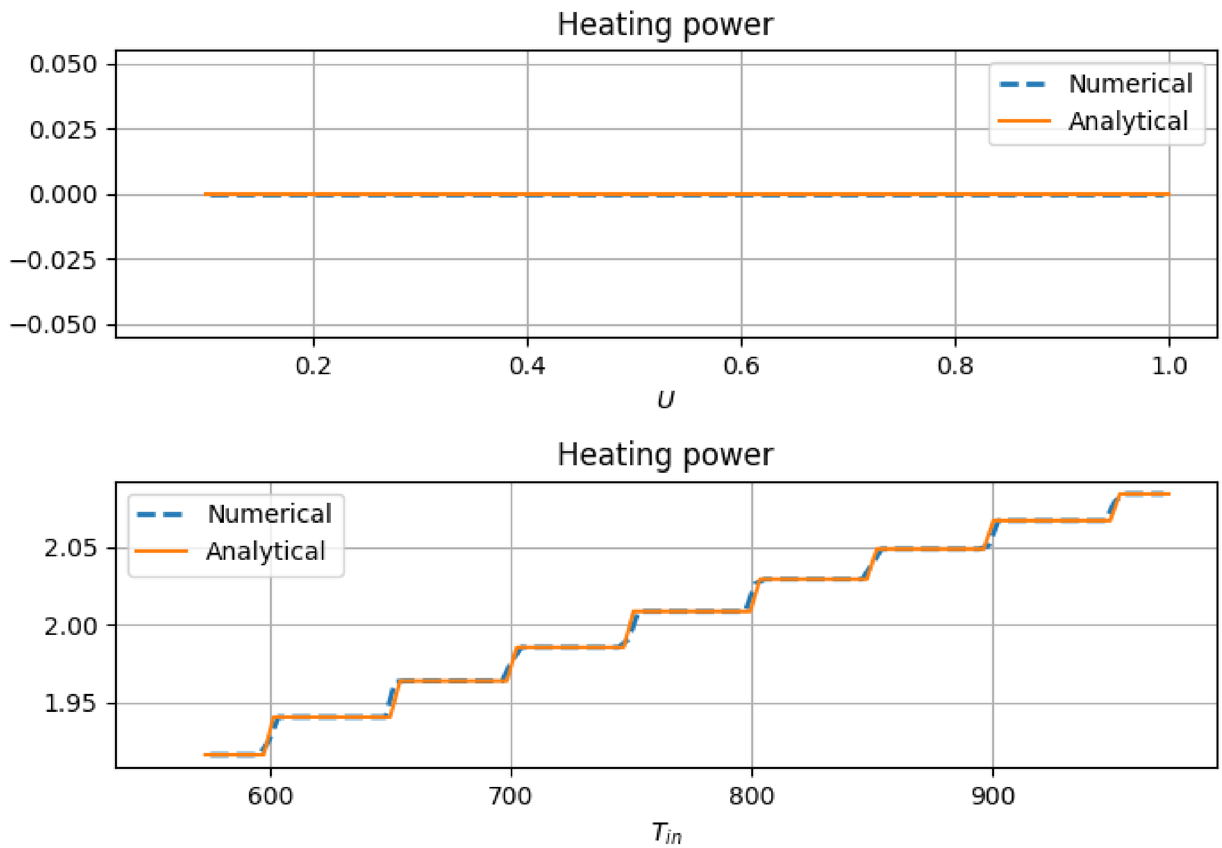


Figure A7.24: Gradient verification output: `heating_power_dp.png`.

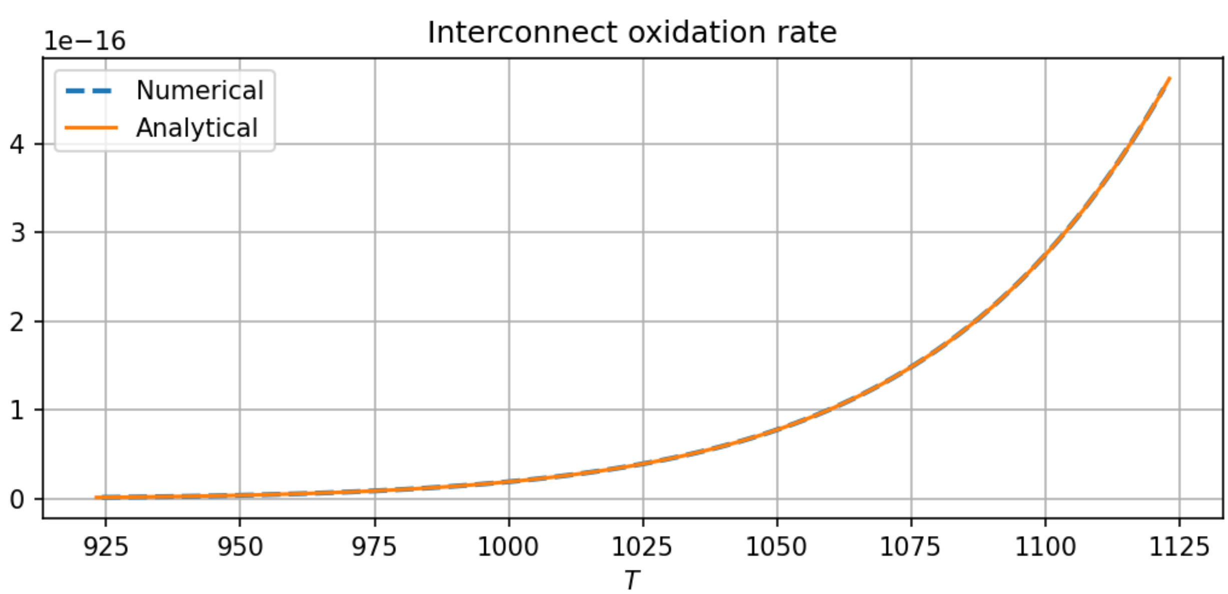


Figure A7.25: Gradient verification output: `interconnect_oxidation_rate.png`.

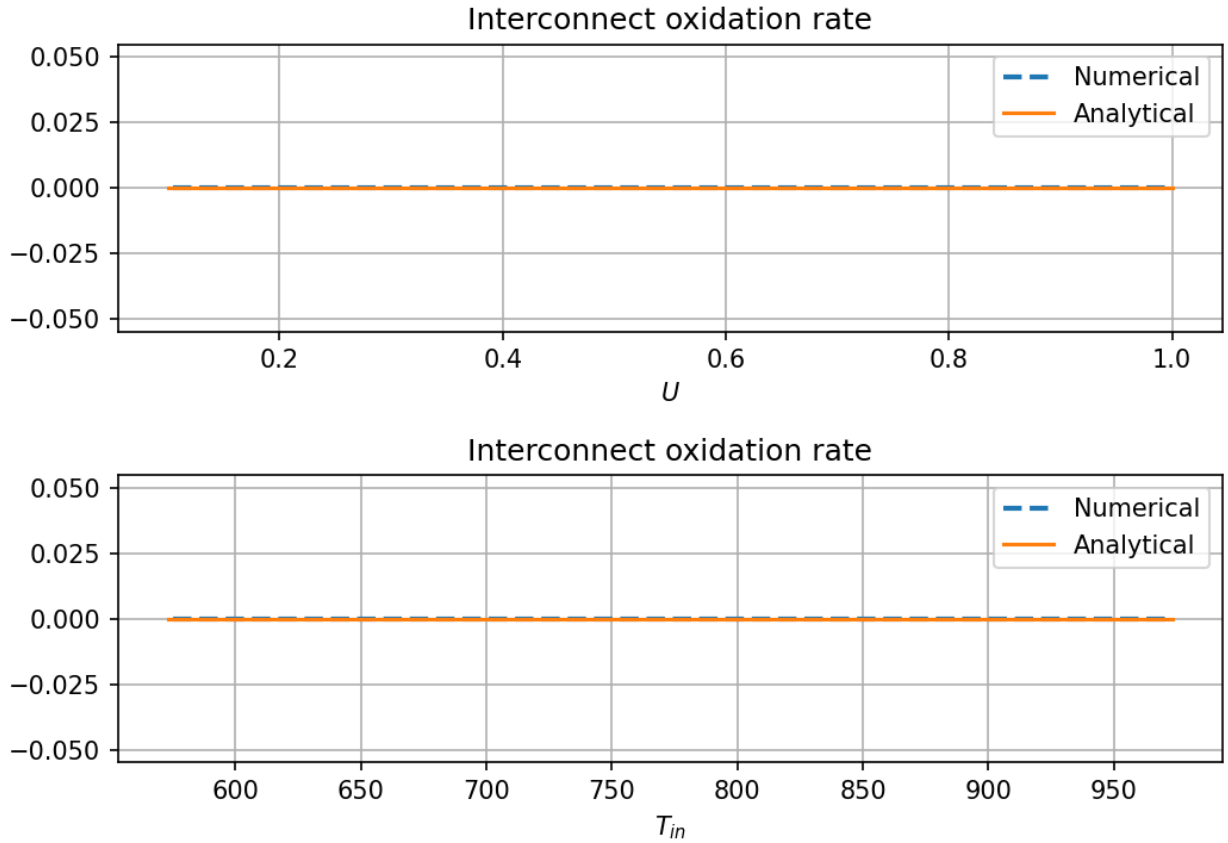


Figure A7.26: Gradient verification output: `interconnect_oxidation_rate_dp.png`.

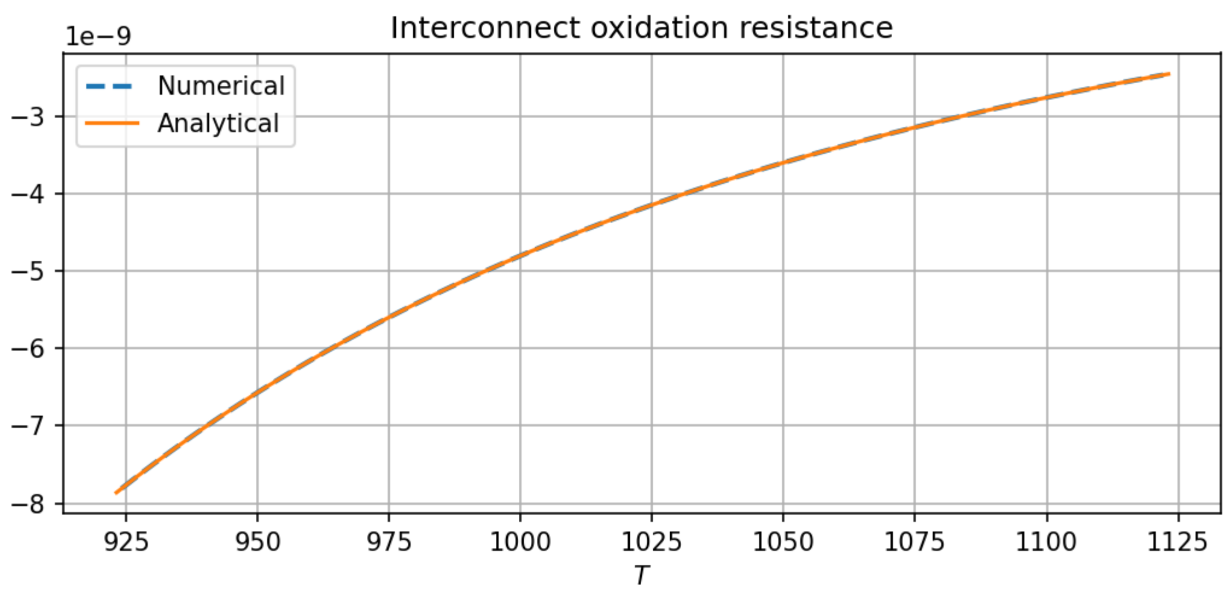


Figure A7.27: Gradient verification output: `interconnect_oxidation_resistance.png`.

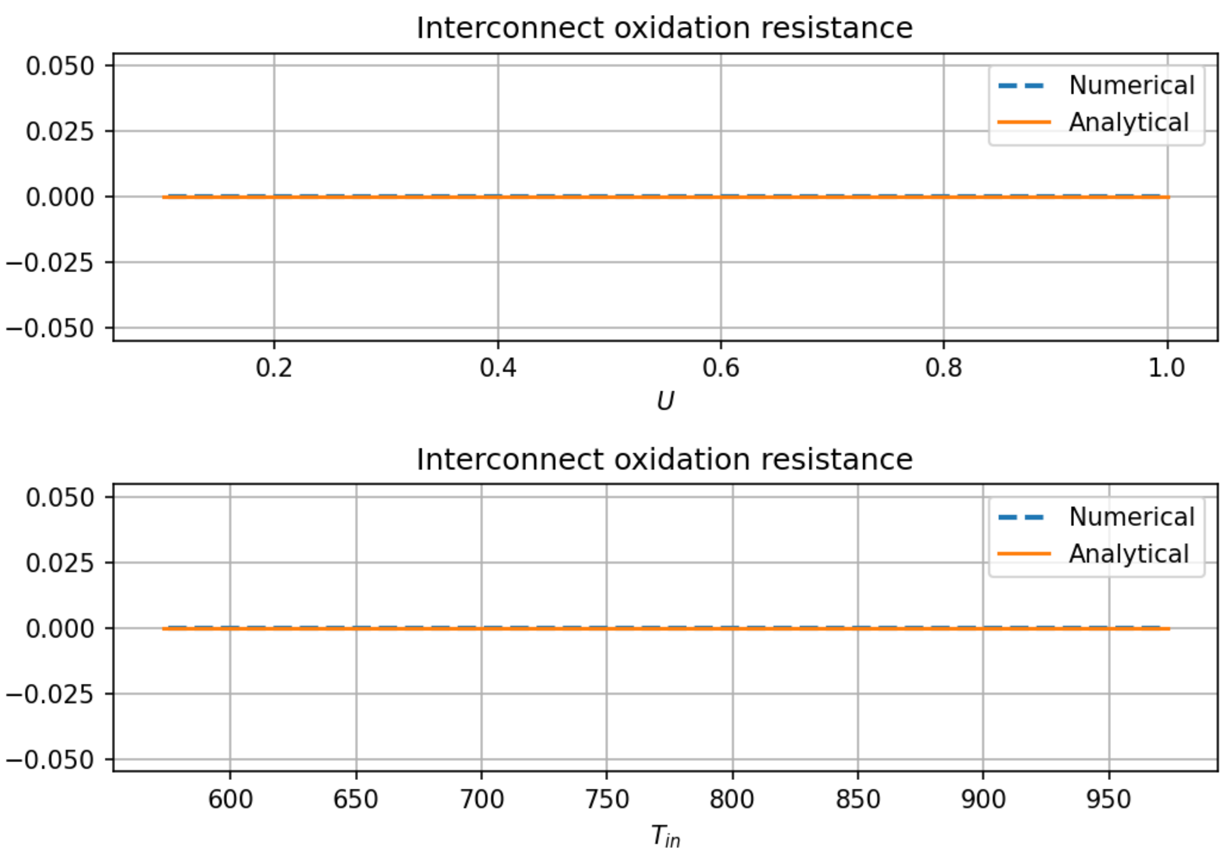


Figure A7.28: Gradient verification output: `interconnect_oxidation_resistance_dp.png`.

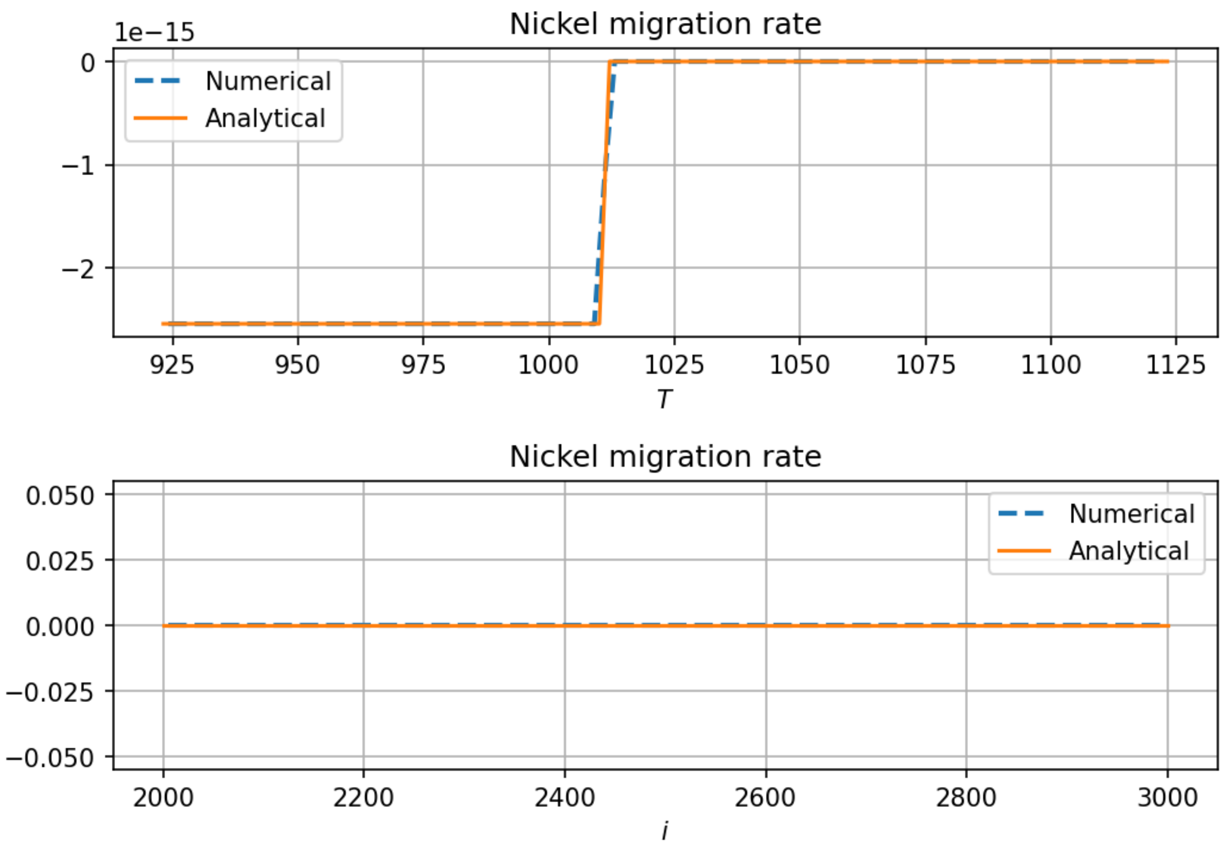


Figure A7.29: Gradient verification output: `nickel_migration_rate.png`.

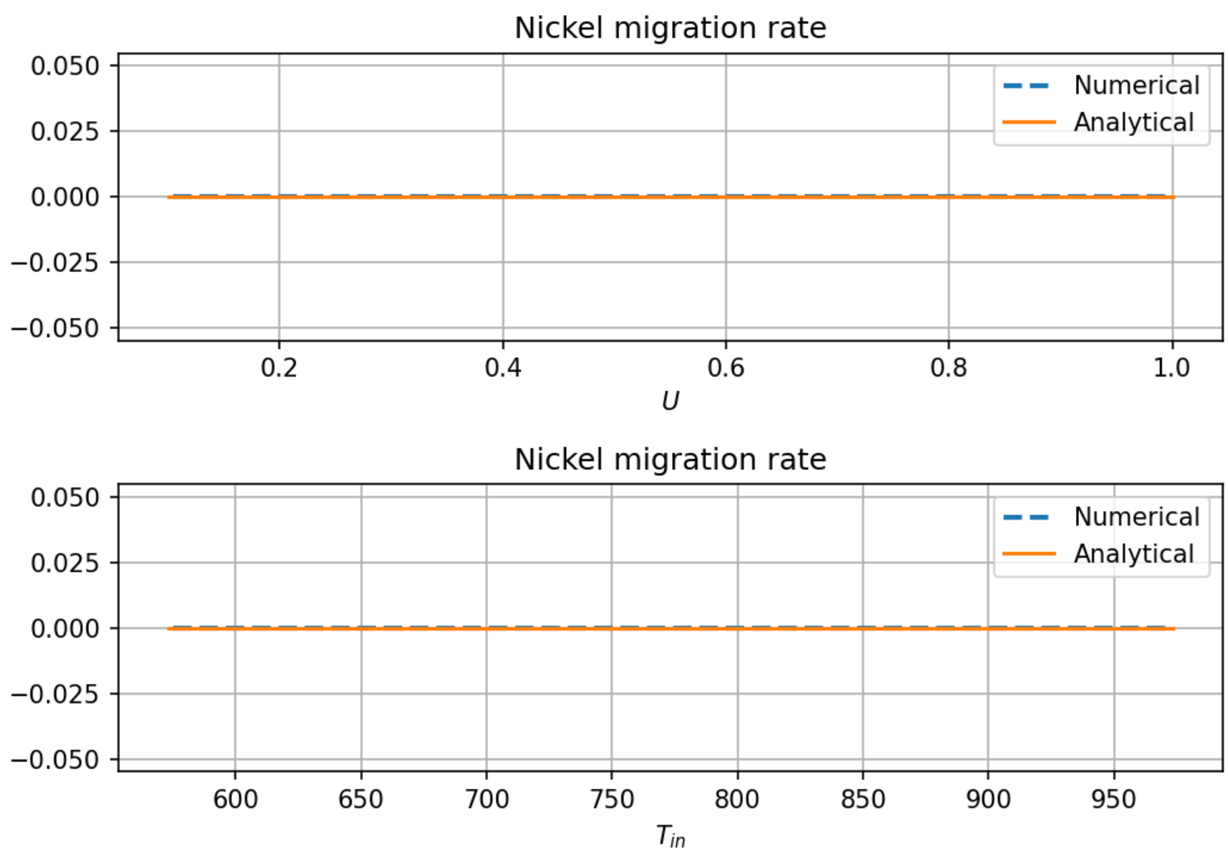


Figure A7.30: Gradient verification output: `nickel_migration_rate_dp.png`.

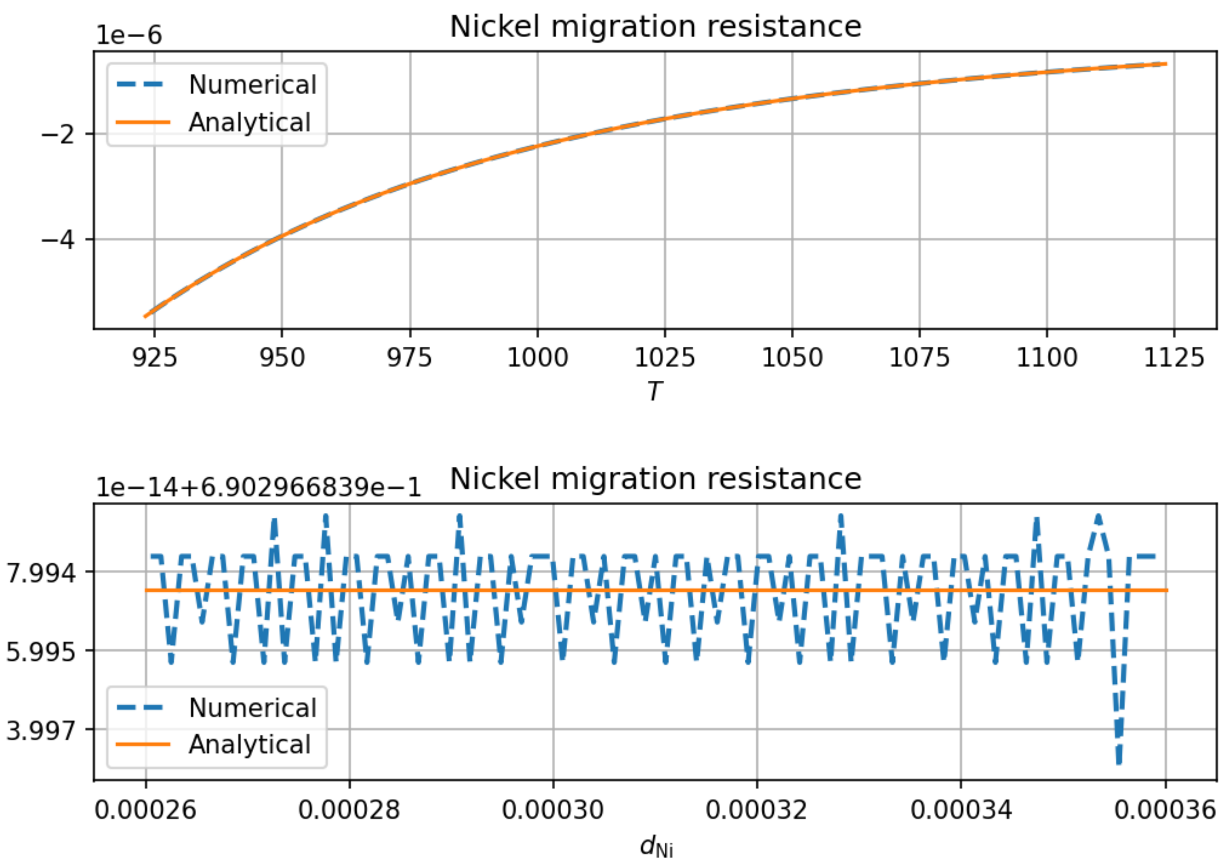


Figure A7.31: Gradient verification output: `nickel_migration_resistance.png`.

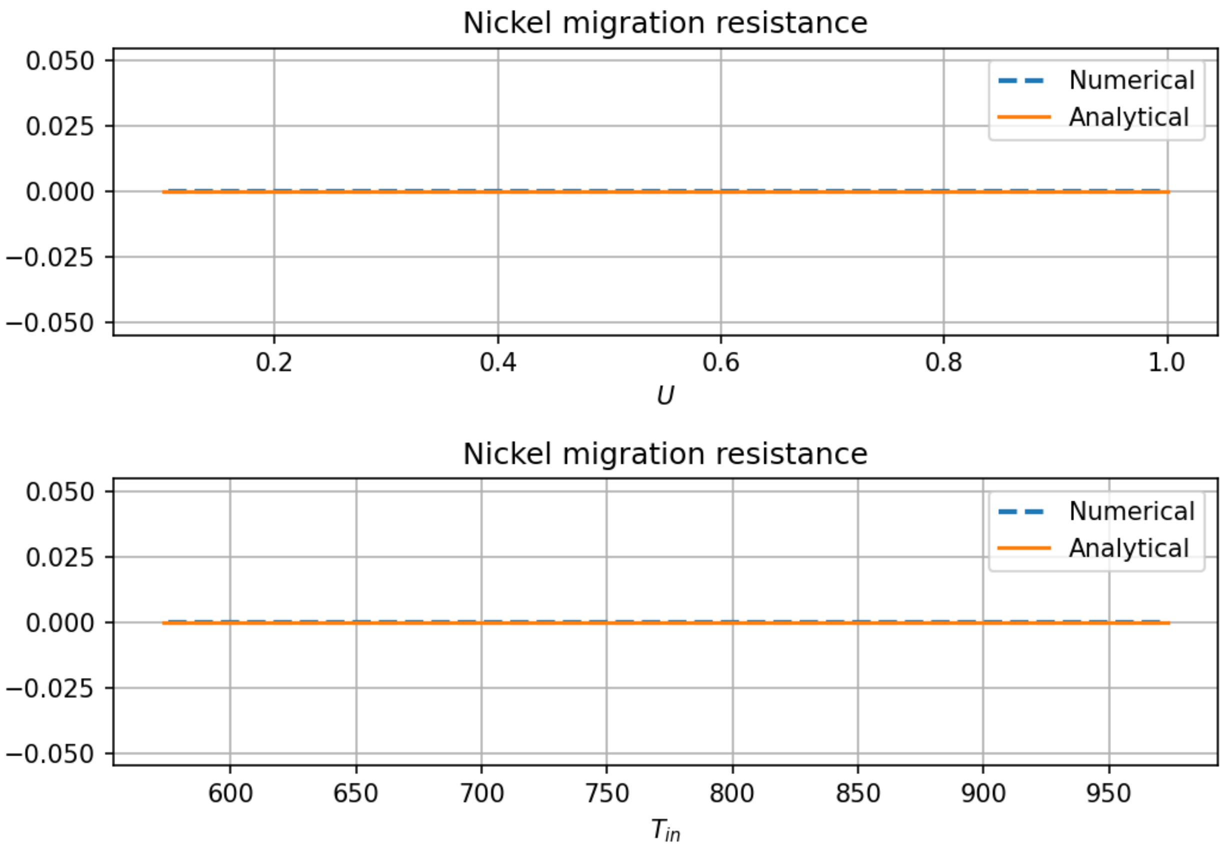


Figure A7.32: Gradient verification output: `nickel_migration_resistance_dp.png`.

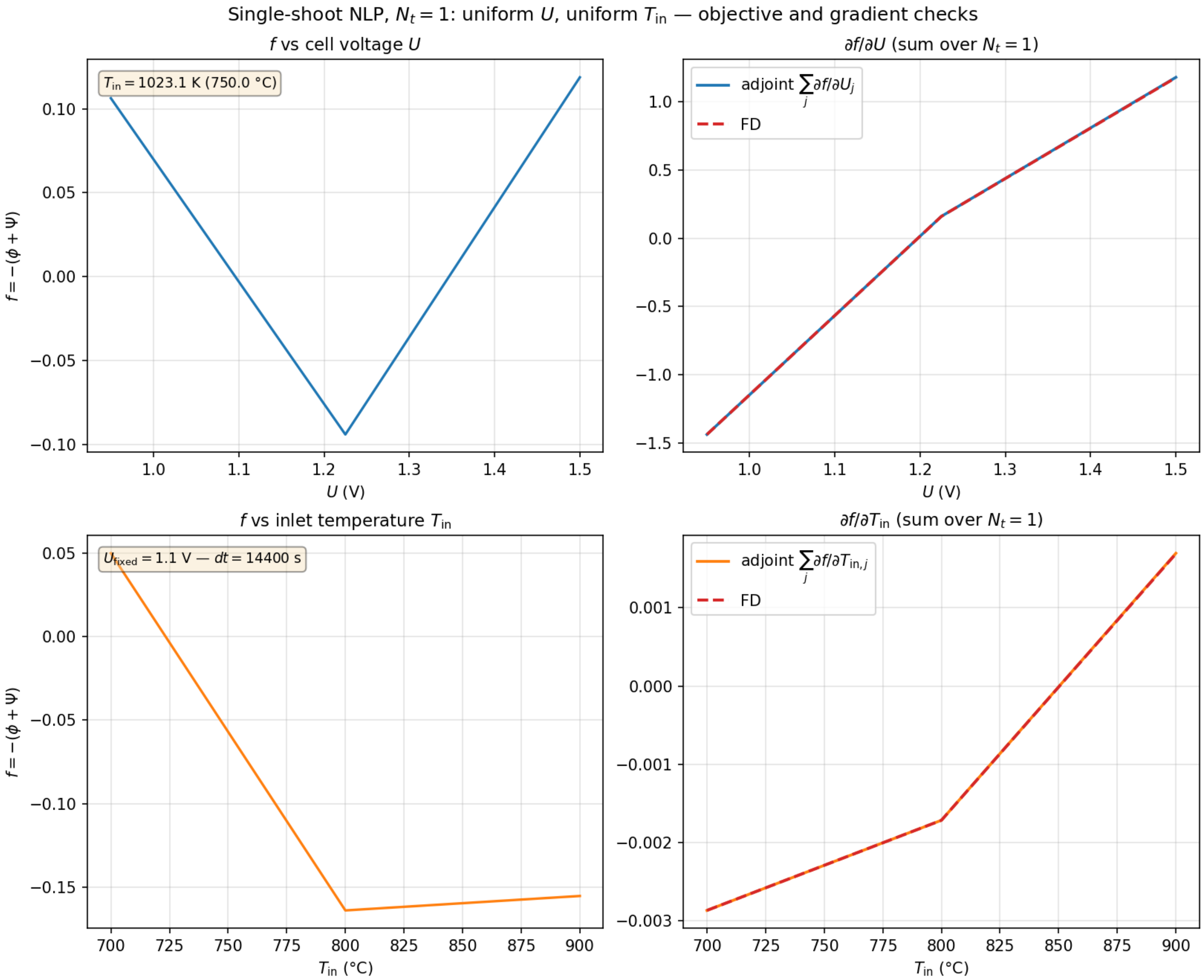


Figure A7.33: Gradient verification output: `nlp_gradients_NT1.png`.

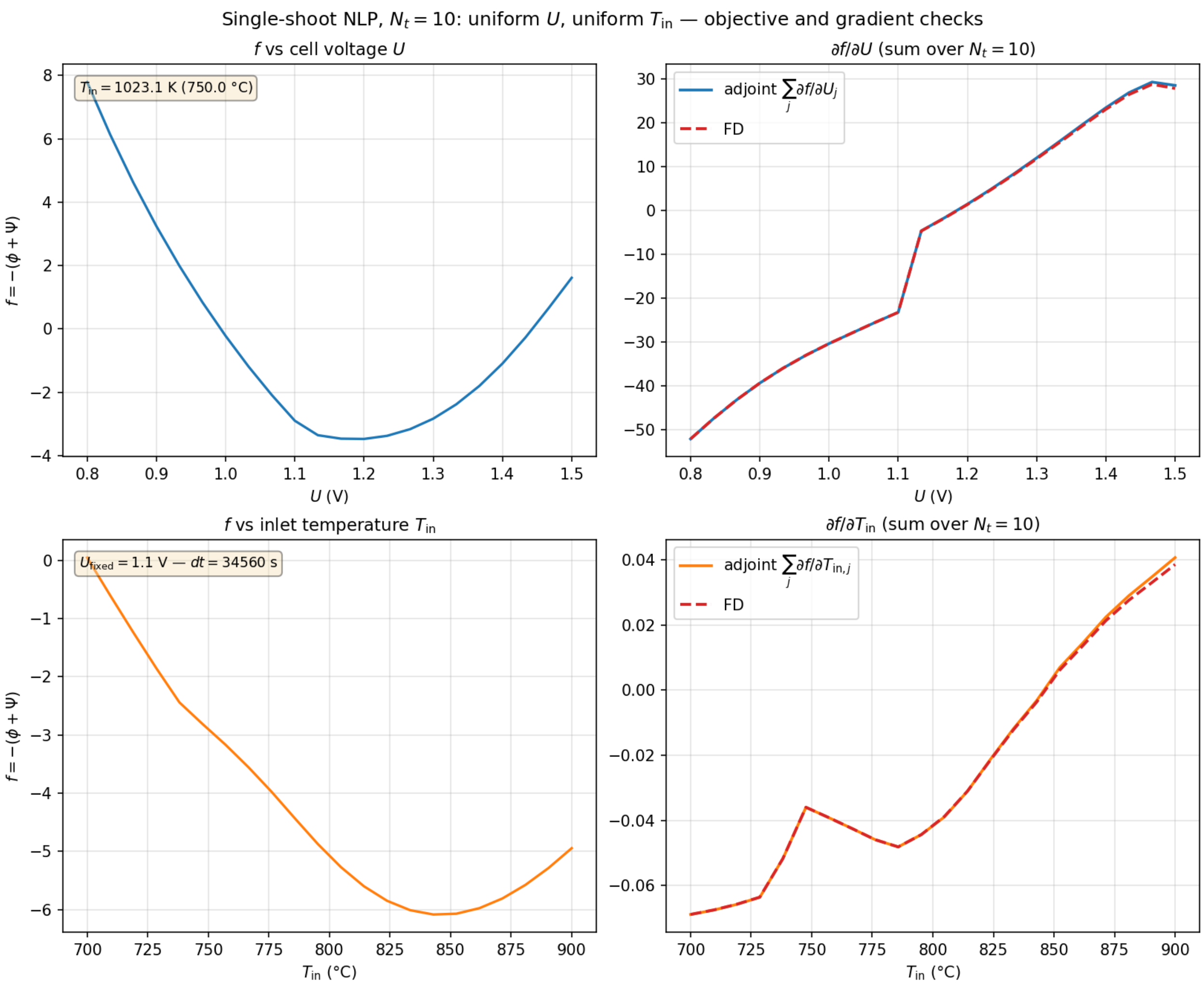


Figure A7.34: Gradient verification output: `nlp_gradients_NT10.png`.

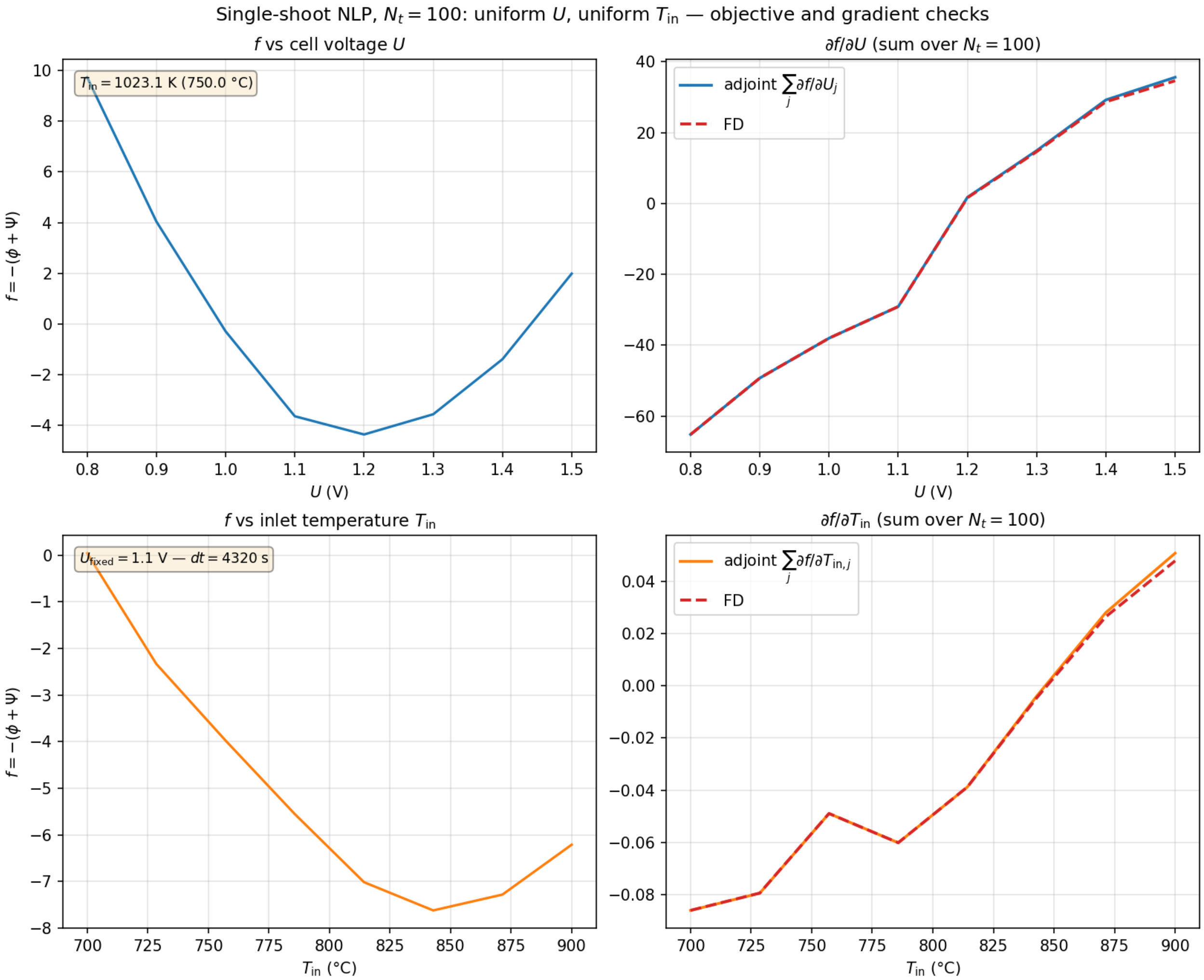


Figure A7.35: Gradient verification output: `nlp_gradients_NT100.png`.

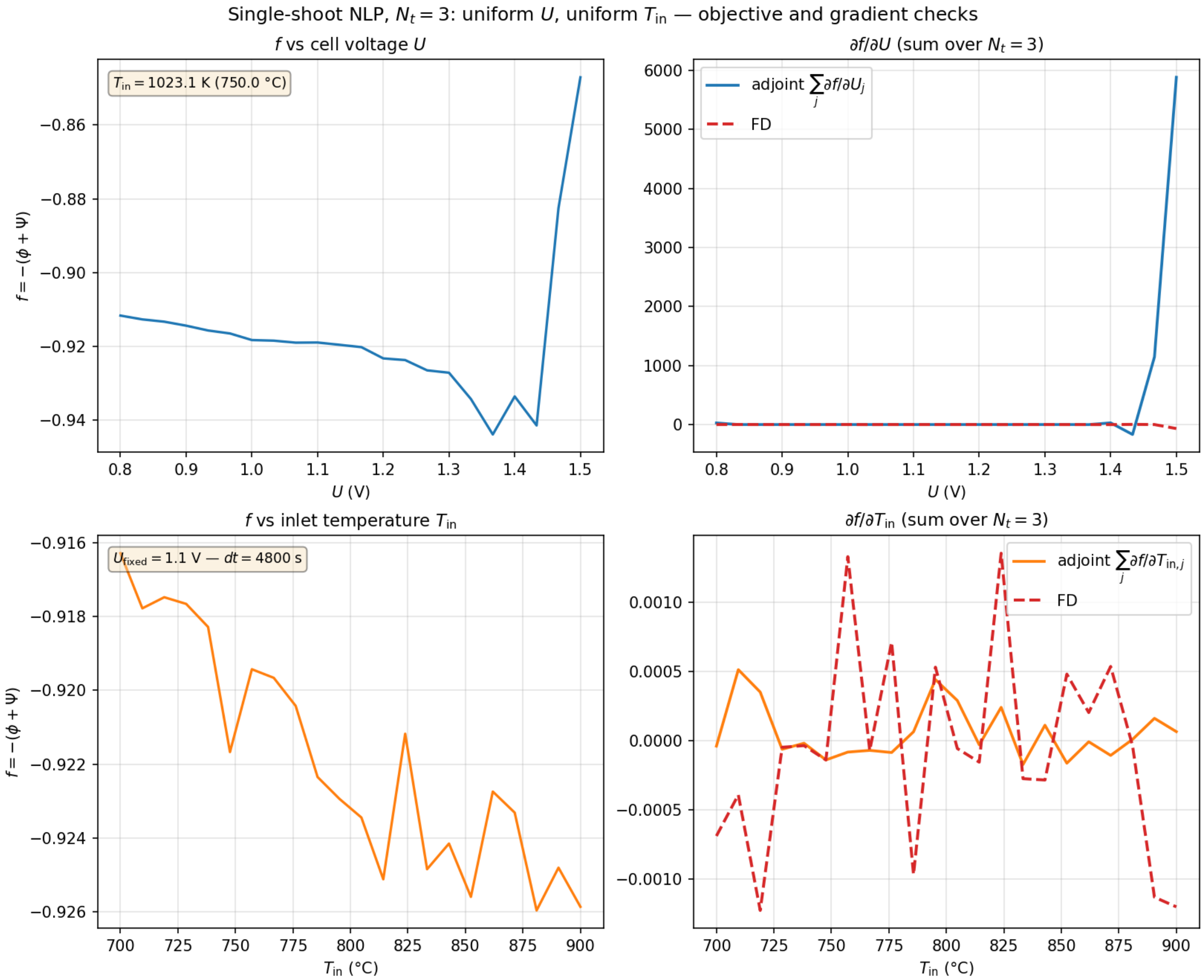


Figure A7.36: Gradient verification output: `nlp_gradients_NT3.png`.

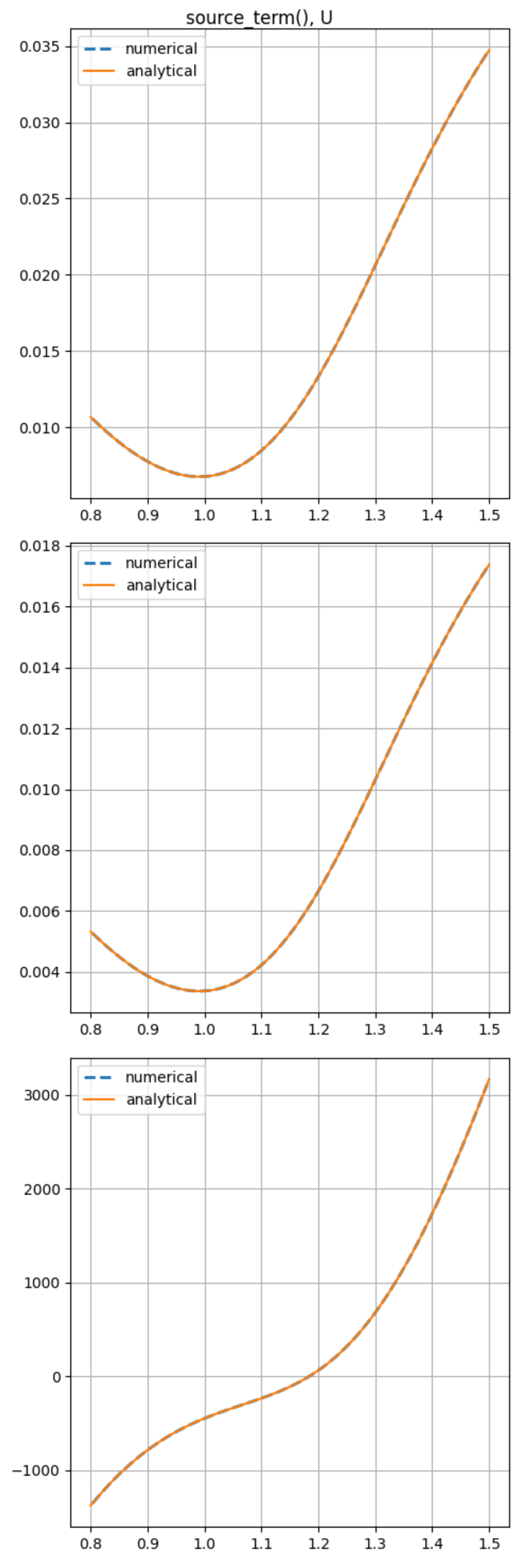


Figure A7.37: Gradient verification output: `source_term_U.png`.

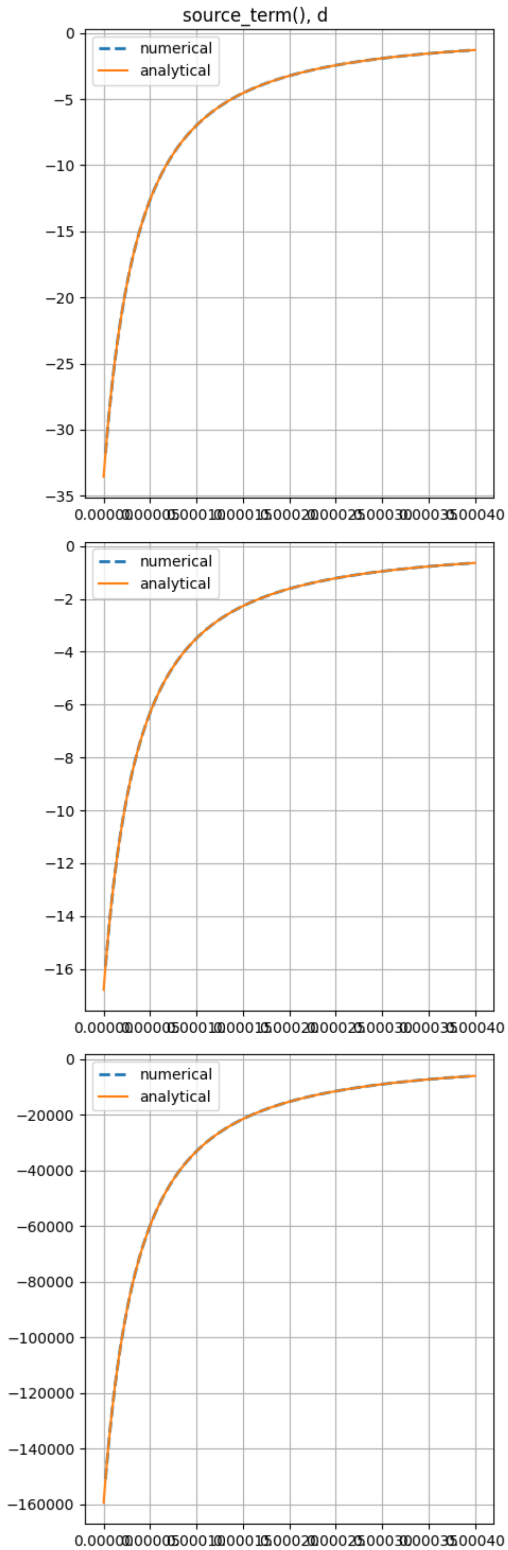


Figure A7.38: Gradient verification output: `source_term_dd.png`.

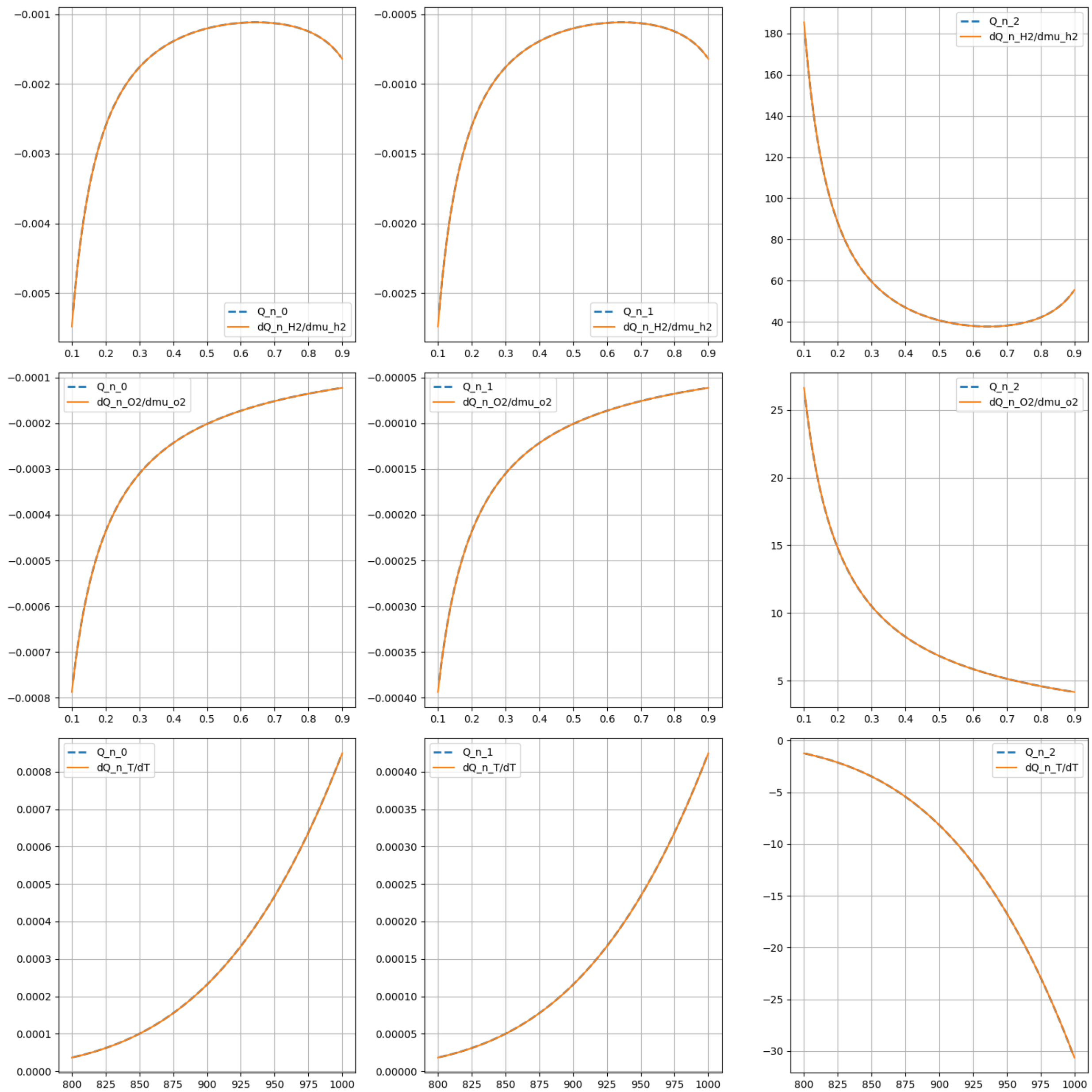


Figure A7.39: Gradient verification output: `source_term_jacobian.png`.

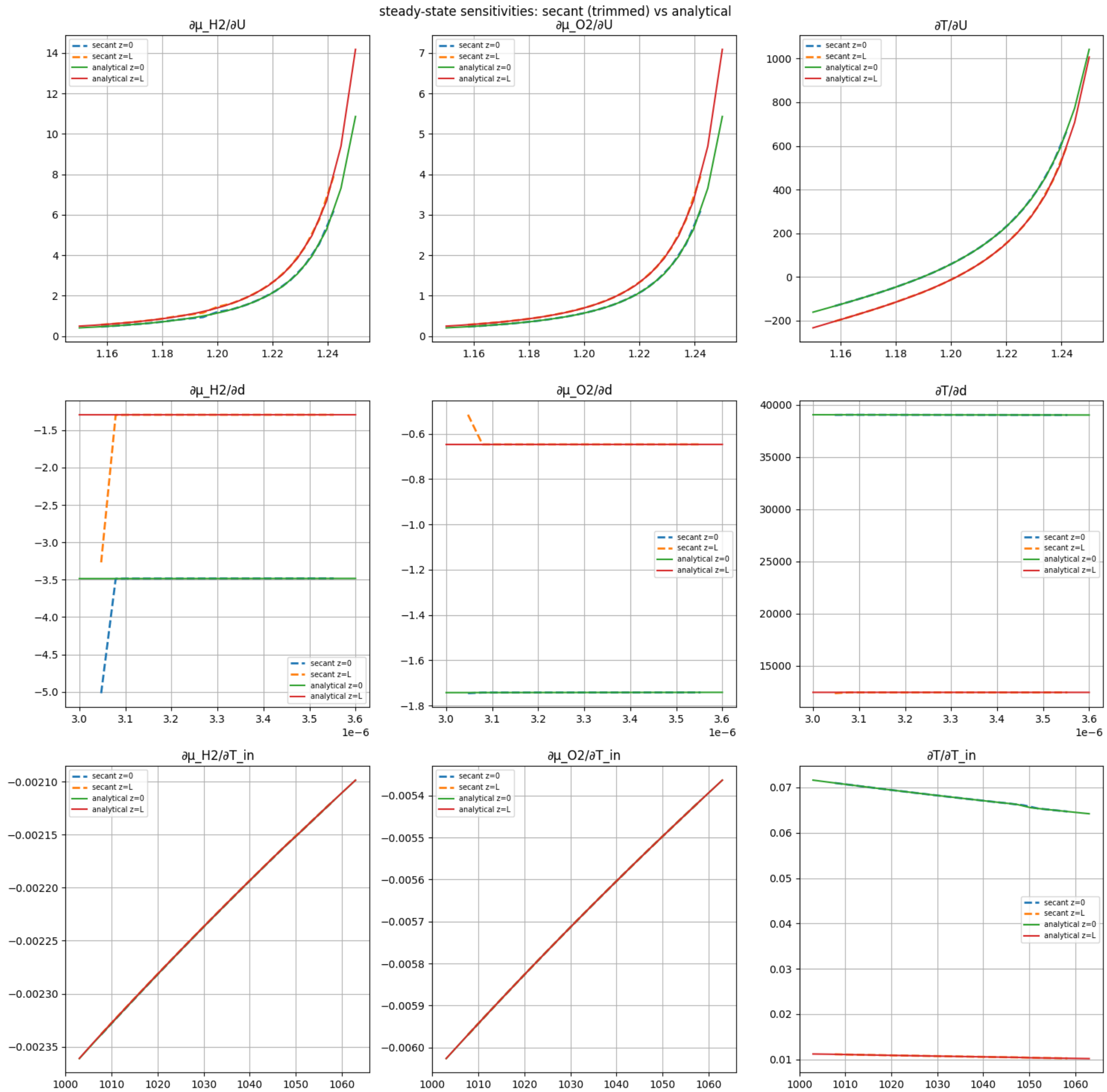


Figure A7.40: Gradient verification output: `steady_state_dmu_3x3.png`.

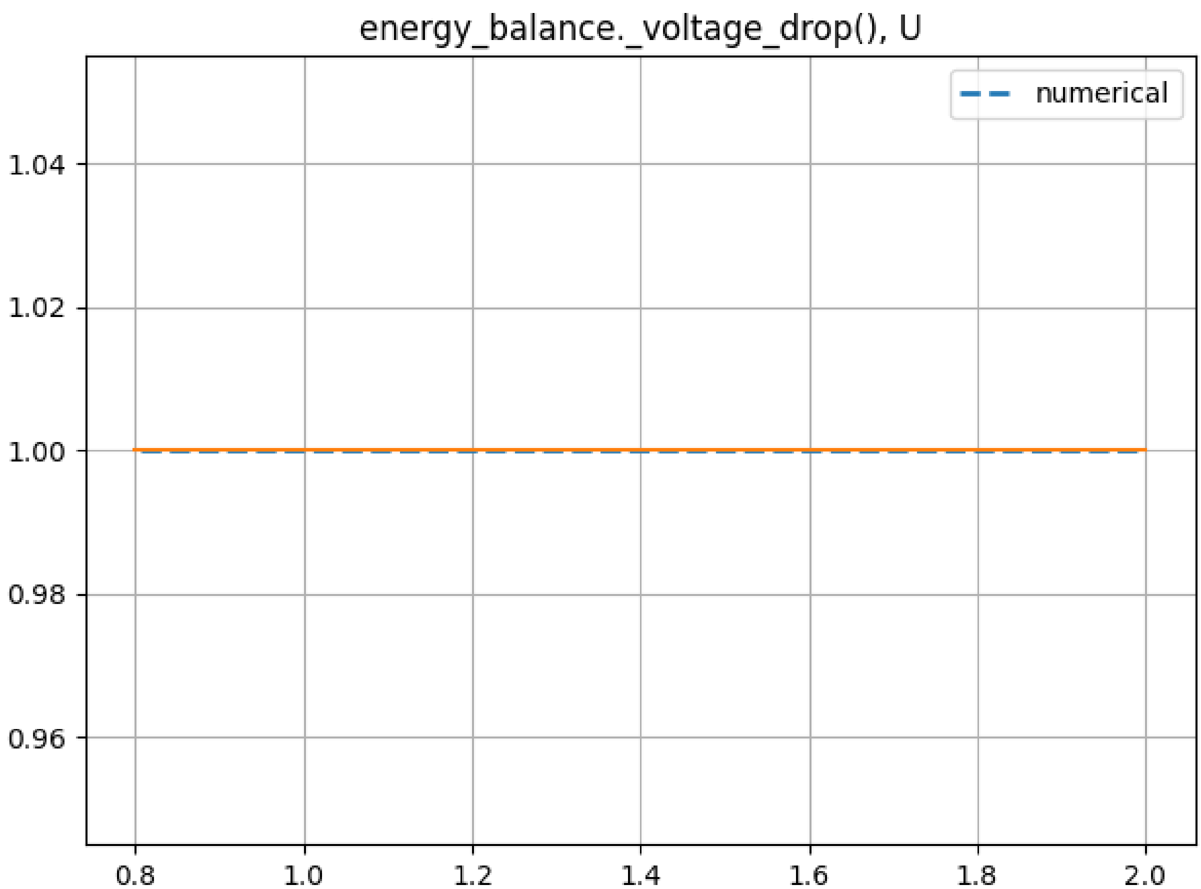


Figure A7.41: Gradient verification output: `voltage_drop_U.png`.

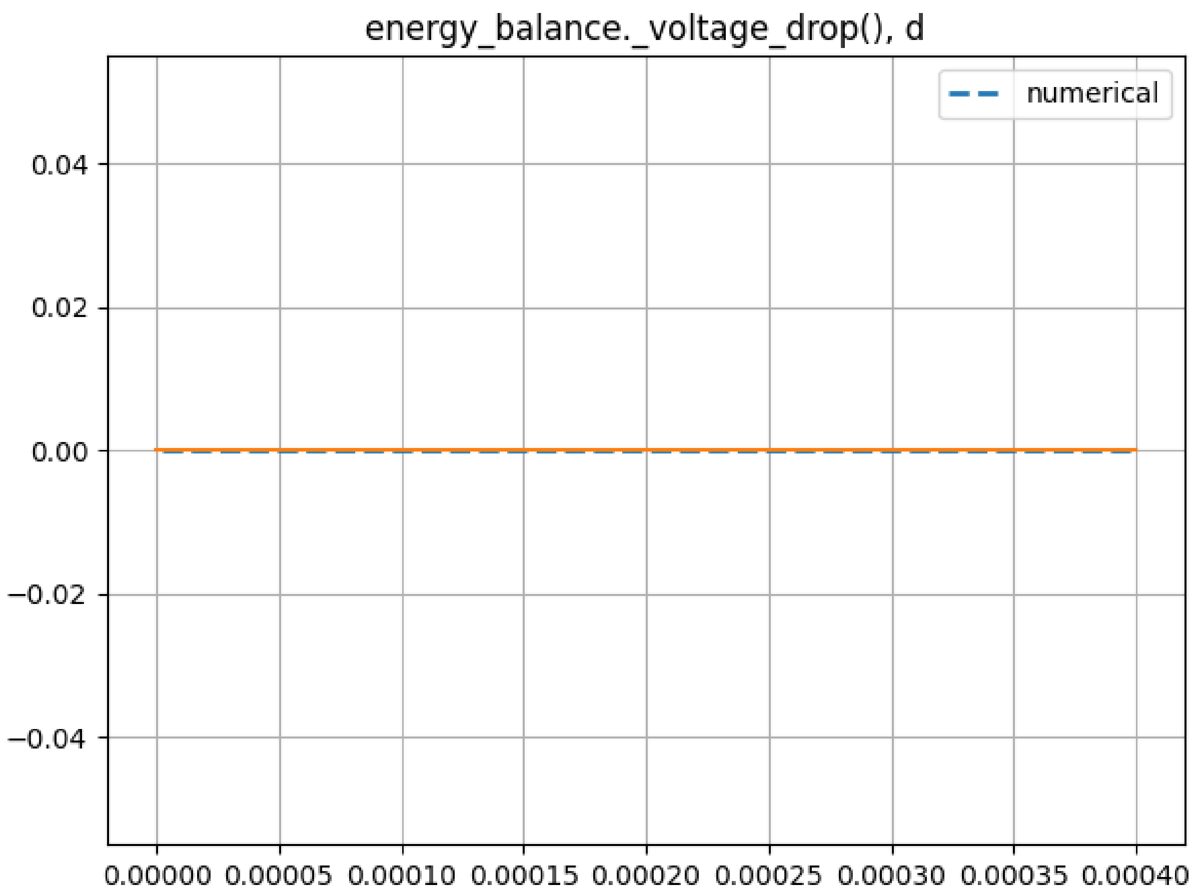


Figure A7.42: Gradient verification output: `voltage_drop_d.png`.

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